Calculus III Notes

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1 3-Dimensional Space

The 3-D coordinate system is often denoted by $\mathbb{R}^3$. Likewise, the 2-D coordinate system is denoted by $\mathbb{R}^2$, and the 1-D coordinate system is denoted by $\mathbb{R}$.

1.1 Equations of Lines

Vector form

If $\vec{a}$ and $\vec{v}$ are parallel vectors, then $\vec{a} = t\vec{v}$ for some scalar t.
Now if we have a vector $\vec{r}$ as follows

$$\vec{r} = \vec{r_0} + \vec{a}$$

Then we can write

$$\vec{r} = \vec{r_0} + t\vec{v} = \langle x_0, y_0, z_0 \rangle + t\langle a, b, c \rangle$$

This is called the **vector form of the equation of a line**.

Parametric form

We can rewrite the vector form as

$$\vec{r} = \langle x_0, y_0, z_0 \rangle + t\langle a, b, c \rangle$$
$$\langle x, y, z \rangle = \langle x_0 + ta, y_0 + tb, z_0 + tc \rangle$$

In other words

$$x = x_0 + ta$$
$$y = y_0 + tb$$
$$z = z_0 + tc$$

This set of equations is called the **parametric form of the equation of a line**.

Symmetric Equations of a Line

If we assume that a, b, and c are non-zero numbers, then we can solve each of the parametric equations for t. This gives us

$$\frac{x - x_0}{a} = \frac{y - y_0}{b} = \frac{z - z_0}{c}$$

Example 1.1: Find the Equations of lines:
1. Through the points $(7, -3, 1)$ **and** $(-2, 1, 4)$
2. Through the point $(1, -5, 0)$ **and parallel to the line given by** $\vec{r}(t) = \langle 8 - 3t, -10 + 9t, -1 - t \rangle$
3. Through the point $(-7, 2, 4)$ **and orthogonal to both** $\vec{v} = \langle 0, -9, 1 \rangle$ **and** $\vec{w} = 3\hat{i} + \hat{j} - 4\hat{k}$

1.
Direction vector $\vec{d} = \langle -2 - 7, 1 + 3, 4 - 1 \rangle = \langle -9, 4, 3 \rangle$
Now, the vector form of the line is

$$\vec{r} = \langle 7, -3, 1 \rangle + t\langle -9, 4, 3 \rangle$$

The parametric form is

$$x = 7 - 9t, \quad y = -3 + 4t, \quad z = 1 + 3t$$

The symmetric form is

$$\frac{x-7}{-9} = \frac{y+3}{4} = \frac{z-1}{3}$$

2.
The direction vector is $\vec{d} = \langle -3, 9, -1 \rangle$
Hence, the vector form of the line is

$$\vec{r} = \langle 1, -5, 0 \rangle + t\langle -3, 9, -1 \rangle$$

The parametric form is

$$x = 1 - 3t, \quad y = -5 + 9t, \quad z = -t$$

And the symmetric form is

$$\frac{x-1}{-3} = \frac{y+5}{9} = -z$$

3.
Direction vector

$$\vec{d} = \vec{v} \times \vec{w} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 0 & -9 & 1 \\ 3 & 1 & -4 \end{vmatrix} = \langle 35, 3, 27 \rangle$$

Hence, the vector form of the line is

$$\vec{r} = \langle -7, 2, 4 \rangle + t\langle 35, 3, 27 \rangle$$

The parametric form is

$$x = -7 + 35t, \quad y = 2 + 3t, \quad z = 4 + 27t$$

The symmetric form is

$$\frac{x+7}{35} = \frac{y-2}{3} = \frac{z-4}{27}$$

Example 1.2: Determine if the two lines are parallel, orthogonal, or neither:
1. The line given by $\vec{r}(t) = \langle 4-7t, -10+5t, 21-4t \rangle$ and the line given by $\vec{r}(t) = \langle -2+3t, 7+5t, 5+t \rangle$
2. The line given by $x = 29, y = -3-6t, z = 12-t$ and the line given by $\vec{r}(t) = \langle 12-14t, 2+7t, -10+3t \rangle$
1.
The direction vectors are

$$\vec{d_1} = \langle -7, 5, -4 \rangle, \quad \vec{d_2} = \langle 3, 5, 1 \rangle$$

To check if they are parallel, we can check:

$$\frac{-7}{3} \neq \frac{5}{5} \neq \frac{-4}{1}$$

which means they are not parallel.
To check if they are orthogonal, we can check:

$$\vec{d_1} \cdot \vec{d_2} = -7(3) + 5(5) + (-4)(1) = -21 + 25 - 4 = 0$$

Hence, they are orthogonal.
2.
The direction vectors are

$$\vec{d_1} = \langle 0, -6, -1 \rangle, \quad \vec{d_2} = \langle -14, 7, 3 \rangle$$

To check if they are parallel, we can check:

$$\frac{0}{-14} \neq \frac{-6}{7} \neq \frac{-1}{3}$$

which means they are not parallel.
To check if they are orthogonal, we can check:

$$\vec{d_1} \cdot \vec{d_2} = 0(-14) + (-6)(7) + (-1)(3) = -42 - 3 = -45 \neq 0$$

Hence, they are neither parallel nor orthogonal.

Example 1.3: Determine the intersection point of the two lines or show that they don't not intersect:
1. The line passing through the point $(0, -9, -1)$ **and** $(1, 6, -3)$ **and the line given by** $\vec{r}(t) = \langle -9 - 4t, 10 + 6t, 1 - 2t \rangle$
2. The line given by $x = 1 + 6t, y = -1 - 3t, z = 4 + 12t$ **and the line given by** $x = 4 + t, y = -10 - 8t, z = 3 - 5t$
1.
The direction vector of the first line is

$$\vec{d_1} = \langle 1 - 0, 6 + 9, -3 + 1 \rangle = \langle 1, 15, -2 \rangle$$

We can write the parametric equations of the first line as:

$$x = s, y = -9 + 15s, z = -1 - 2s$$

And the parametric equations of the second line as:

$$x = -9 - 4t, y = 10 + 6t, z = 1 - 2t$$

Setting them equal to each other we get,

$$\begin{aligned} s &= -9 - 4t \\ -9 + 15s &= 10 + 6t \\ -1 - 2s &= 1 - 2t \end{aligned}$$

Solving the first two equations, we get

$$t = -\frac{7}{3}, \quad s = \frac{1}{3}$$

Now, verifying the third equation, we get

$$\begin{aligned} -1 - 2\left(\frac{1}{3}\right) &= 1 - 2\left(-\frac{7}{3}\right) \\ -1 - \frac{2}{3} &= 1 + \frac{14}{3} \\ -\frac{5}{3} &\neq \frac{17}{3} \end{aligned}$$

Since the third equation is not satisfied, the two lines do not intersect.
2.
The lines are given in parametric form.
Setting them equal to each other we get,

$$\begin{aligned} 1 + 6s &= 4 + t \\ -1 - 3s &= -10 - 8t \\ 4 + 12s &= 3 - 5t \end{aligned}$$

Solving the first two equations, we get

$$s = \frac{1}{3}, \quad t = -1$$

Now, verifying the third equation, we get

$$\begin{aligned} 4 + 12\left(\frac{1}{3}\right) &= 3 - 5(-1) \\ 8 &= 8 \end{aligned}$$

That means, the lines intersect. Substituting the values in the parametric equation, we get

$$\begin{aligned} x &= 1 + 6\left(\frac{1}{3}\right) = 3 \\ y &= -1 - 3\left(\frac{1}{3}\right) = -2 \\ z &= 4 + 12\left(\frac{1}{3}\right) = 8 \end{aligned}$$

Hence, the intersection point is $(3, -2, 8)$.

Example 1.4: Which of the three coordinate planes does the line given by $x = 16t, y = -4 - 9t, z = 34$ intersect?

To intersect the xy-plane, we need $z = 0$. But here $z = 34$ is constant. Hence, the line does not intersect the xy-plane.
To intersect the yz-plane, we need $x = 0$. Hence,

$$16t = 0 \implies t = 0$$

And the intersection point is $(0, -4 - 9 \times 0, 34)$ or $(0, -4, 34)$.
To intersect the xz-plane, we need $y = 0$. Hence,

$$-4 - 9t = 0 \implies t = -\frac{4}{9}$$

And the intersection point is $\left(16\left(-\frac{4}{9}\right), 0, 34\right)$ or $\left(-\frac{64}{9}, 0, 34\right)$.

1.2 Equations of Planes

Vector form

Let's assume $\vec{r_0} = \langle x_0, y_0, z_0 \rangle$ and $\vec{r} = \langle x, y, z \rangle$ are two position vectors and $\vec{r} - \vec{r_0}$ is a vector in the plane.
If $\vec{n} = \langle a, b, c \rangle$ is a normal to the plane (which means it's orthogonal to the vector $\vec{r} - \vec{r_0}$), then we can write

$$\vec{n} \cdot (\vec{r} - \vec{r_0}) = 0 \quad \implies \quad \vec{n} \cdot \vec{r} = \vec{n} \cdot \vec{r_0}$$

This is called the **vector form of the equation of a plane**.

Scalar form

If we expand the vector equation in the following way,

$$\begin{aligned} \vec{n} \cdot (\vec{r} - \vec{r_0}) &= 0 \\ \langle a, b, c \rangle \cdot \langle x - x_0, y - y_0, z - z_0 \rangle &= 0 \end{aligned}$$

Computing the dot product, we get

$$a(x - x_0) + b(y - y_0) + c(z - z_0) = 0$$

This is called the **scalar form of the equation of a plane**.
This equation can also be written as

$$ax + by + cz = d$$

where $d = ax_0 + by_0 + cz_0$.

Example 1.5: Find the equation of the plane:
1. Through the point $(6, -3, 1)$**,** $(5, -4, 1)$**, and** $(3, -4, 0)$
2. The plane containing the point $(1, -5, 8)$ **and orthogonal to the line given by** $x = -3 + 15t, y = 14 - t, z = 9 - 3t$
3. The plane containing the point $(-8, 3, 7)$ **and parallel to the plane given by** $4x + 8y - 2z = 45$
4. The plane containing the two lines given by $\vec{r}(t) = \langle 7 + 5t, 2 + t, 6t \rangle$ **and** $\vec{r}(t) = \langle 7 - 6t, 2 - 2t, 10t \rangle$

1.
The given points are

$$A(6, -3, 1), B(5, -4, 1), C(3, -4, 0)$$

Two vectors in the plane are

$$\begin{aligned} \vec{AB} &= \langle 5 - 6, -4 + 3, 1 - 1 \rangle = \langle -1, -1, 0 \rangle \\ \vec{BC} &= \langle 3 - 5, -4 + 4, 0 - 1 \rangle = \langle -2, 0, -1 \rangle \end{aligned}$$

Normal vector on the place:

$$\vec{n} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ -1 & -1 & 0 \\ -2 & 0 & -1 \end{vmatrix} = \hat{i} - \hat{j} - 2\hat{k}$$

Now, using the point A, we can write the equation of the plane as

$$\begin{aligned}(x-6)-(y+3)-2(z-1)&=0\\ x-y-2z&=7\end{aligned}$$

2.
The normal vector is

$$\vec{n}=\langle 15,-1,-3\rangle$$

Using the point $(1,-5,8)$, the equation of the plane is

$$\begin{aligned}15(x-1)-(y+5)-3(z-8)&=0\\ 15x-y-3z&=15+5-24\\ 15x-y-3z+4&=0\end{aligned}$$

3.
The normal vector is

$$\vec{n}=\langle 4,8,-2\rangle$$

Using the point $(-8,3,7)$, the equation of the plane is

$$\begin{aligned}4(x+8)+8(y-3)-2(z-7)&=0\\ 4x+8y-2z&=-32+24+14\\ 4x+8y-2z+22&=0\end{aligned}$$

4.
The direction vectors of the two lines are

$$\vec{d_1}=\langle 5,1,6\rangle,\quad \vec{d_2}=\langle -6,-2,10\rangle$$

The normal vector is

$$\vec{n}=\vec{d_1}\times\vec{d_2}=\begin{vmatrix}\hat{i}&\hat{j}&\hat{k}\\ 5&1&6\\ -6&-2&10\end{vmatrix}=\langle 22,-86,-4\rangle$$

Using the point $A(7,2,0)$, the equation of the plane is

$$\begin{aligned}22(x-7)-86(y-2)-4(z-0)&=0\\ 22x-86y-4z-154+172&=0\\ 22x-86y-4z+18&=0\end{aligned}$$

Example 1.6: Determine if the two planes are parallel, orthogonal, or neither: The plane given by $3x+9y+7z=-1$ and the plane containing the points $(1,-1,9),(4,-1,2),(-2,3,4)$

The normal vector of the first plane is

$$\vec{n_1}=\langle 3,9,7\rangle$$

Let the points be

$$A(1,-1,9), B(4,-1,2), C(-2,3,4)$$

Two vectors in the plane are

$$\vec{AB} = \langle 4-1, -1+1, 2-9 \rangle = \langle 3, 0, -7 \rangle$$
$$\vec{AC} = \langle -2-1, 3+1, 4-9 \rangle = \langle -3, 4, -5 \rangle$$

The normal vector of the second plane is

$$\vec{n_2} = \vec{AB} \times \vec{AC} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 3 & 0 & -7 \\ -3 & 4 & -5 \end{vmatrix} = \langle 28, 36, 12 \rangle = \langle 7, 9, 3 \rangle$$

To check if they are parallel, we can check:

$$\frac{3}{7} \neq \frac{9}{9} \neq \frac{7}{3}$$

which means they are not parallel.
To check if they are orthogonal, we can check:

$$\vec{n_1} \cdot \vec{n_2} = 3(7) + 9(9) + 7(3) = 21 + 81 + 21 = 123 \neq 0$$

Hence, they are neither parallel nor orthogonal.

Example 1.7: Find the intersection of the plane given by $4x + y + 10z = -2$ and the plane given by $-8x + 2y + 3z = -8$

The two planes are

$$4x + y + 10z = -2$$
$$-8x + 2y + 3z = -8$$

Multiplying the first equation by 2 and adding it to the second equation, we get

$$4y + 23z = -12 \implies y = -3 - \frac{23}{4}z$$

Substituting the value of y in the first equation, we get

$$4x - 3 - \frac{23}{4}z + 10z = -2 \implies x = \frac{1}{4} - \frac{17}{16}z$$

Let $z = t$ (a parameter). Then we get

$$x = \frac{1}{4} - \frac{17}{16}t$$
$$y = -3 - \frac{23}{4}t$$
$$z = t$$

This is the parametric form of the line of intersection.
We can also write it in vector form as

$$\vec{r} = \left\langle \frac{1}{4}, -3, 0 \right\rangle + t\left\langle -\frac{17}{16}, -\frac{23}{4}, 1 \right\rangle$$

1.3 Quadratic Surfaces

General form

The general form of a quadratic surface is

$$Ax^2 + By^2 + Cz^2 + Dxy + Exz + Fyz + Gx + Hy + Iz + J = 0$$

where $A, B, C, D, E, F, G, H, I, J$ are constants.

Ellipsoid

The general equation of an ellipsoid is

$$\frac{(x-h)^2}{a^2} + \frac{(y-k)^2}{b^2} + \frac{(z-l)^2}{c^2} = 1$$

where (h, k, l) is the center of the ellipsoid and a, b, c are the semi-axis lengths.
If $a = b = c$, we get a sphere.

Cone

The general equation of a cone that opens along the z-axis is

$$\frac{(x-h)^2}{a^2} + \frac{(y-k)^2}{b^2} = \frac{(z-l)^2}{c^2}$$

where (h, k, l) is the center of the cone and a, b, c are the semi-axis lengths.

Cylinder

The general equation of a cylinder that opens along the z-axis is

$$\frac{(x-h)^2}{a^2} + \frac{(y-k)^2}{b^2} = 1$$

where (h, k) is the center of the cylinder and a, b are the semi-axis lengths.
If $a = b$, we get a circular cylinder.

Hyperboloid of One Sheet

The general equation of a hyperboloid of one sheet is

$$\frac{(x-h)^2}{a^2} + \frac{(y-k)^2}{b^2} - \frac{(z-l)^2}{c^2} = 1$$

where (h, k, l) is the center of the hyperboloid and a, b, c are the semi-axis lengths.

Hyperboloid of Two Sheets

The general equation of a hyperboloid of two sheets is

$$-\frac{(x-h)^2}{a^2} - \frac{(y-k)^2}{b^2} + \frac{(z-l)^2}{c^2} = 1$$

where (h, k, l) is the center of the hyperboloid and a, b, c are the semi-axis lengths.

Elliptic Paraboloid

The general equation of an elliptic paraboloid is

$$\frac{(x-h)^2}{a^2} + \frac{(y-k)^2}{b^2} = \frac{z-l}{c}$$

where (h, k, l) is the center of the paraboloid and a, b are the semi-axis lengths.

Hyperbolic Paraboloid

The general equation of a hyperbolic paraboloid is

$$\frac{(x-h)^2}{a^2} - \frac{(y-k)^2}{b^2} = \frac{z-l}{c}$$

where (h, k, l) is the center of the paraboloid and a, b are the semi-axis lengths.

1.4 Calculus with Vector Functions

Let

$$\vec{r}(t) = \langle f(t), g(t), h(t)\rangle$$

Note:-

$$\lim_{t\to a} \vec{r}(t) = \langle \lim_{t\to a} f(t), \lim_{t\to a} g(t), \lim_{t\to a} h(t)\rangle$$

Note:-

$$\frac{\mathrm{d}}{\mathrm{d}t}(\vec{u}+\vec{v}) = \vec{u}' + \vec{v}'$$

$$\frac{\mathrm{d}}{\mathrm{d}t}(c\vec{u}) = c\vec{u}'$$

$$\frac{\mathrm{d}}{\mathrm{d}t}(\vec{u}\cdot\vec{v}) = \vec{u}'\cdot\vec{v} + \vec{u}\cdot\vec{v}'$$

$$\frac{\mathrm{d}}{\mathrm{d}t}(\vec{u}\times\vec{v}) = \vec{u}'\times\vec{v} + \vec{u}\times\vec{v}'$$

$$\frac{\mathrm{d}}{\mathrm{d}t}\big(\vec{u}f(t)\big) = f'(t)\vec{u}'(f(t))$$

Note:-

$$\int \vec{r}(t)\,\mathrm{d}t = \left\langle \int f(t)\,\mathrm{d}t\,, \int g(t)\,\mathrm{d}t\,, \int h(t)\,\mathrm{d}t \right\rangle$$

$$\int_a^b \vec{r}(t)\,\mathrm{d}t = \left\langle \int_a^b f(t)\,\mathrm{d}t + \int_a^b g(t)\,\mathrm{d}t + \int_a^b h(t)\,\mathrm{d}t \right\rangle$$

1.5 Tangent, Normal, and Binormal Vectors

Unit Tangent vector

Given the vector function $\vec{r}(t)$, we call $\vec{r}'(t)$ the **tangent vector** The unit tangent vector is given by

$$\vec{T}(t) = \frac{\vec{r}'(t)}{\|\vec{r}'(t)\|}$$

Unit Normal vector

If $\vec{T}(t)$ is the unit tangent vector, then the **unit normal vector** is given by

$$\vec{N}(t) = \frac{\vec{T}'(t)}{\|\vec{T}'(t)\|}$$

Note:-

If $\vec{r}(t)$ is a vector such that $\|\vec{r}(t)\| = c$ for all t, then $\vec{r}'(t)$ is orthogonal to $\vec{r}(t)$

Binormal vector

The **binormal vector** is given by

$$\vec{B}(t) = \vec{T}(t) \times \vec{N}(t)$$

The binormal vector is orthogonal to both the tangent and normal vectors.

1.6 Arc Length with Vector Functions

Note:-

The arc length of a vector function $\vec{r}(t)$ from $t = a$ to $t = b$ is given by

$$L = \int_a^b \|\vec{r}'(t)\| \, \mathrm{d}t$$

Or,

$$L = \int_a^b \sqrt{\left(\frac{\mathrm{d}f}{\mathrm{d}t}\right)^2 + \left(\frac{\mathrm{d}g}{\mathrm{d}t}\right)^2 + \left(\frac{\mathrm{d}h}{\mathrm{d}t}\right)^2} \, \mathrm{d}t$$

1.7 Curvature

Curvature of a curve in 3-D space

The curvature of a curve in 3-D space is given by

$$\kappa = \frac{\|\vec{T}'(t)\|}{\|\vec{r}'(t)\|}$$

where $\vec{T}(t)$ is the unit tangent vector and $\vec{r}(t)$ is the position vector.
This can also be written as

$$\kappa = \frac{\|\vec{r}'(t) \times \vec{r}''(t)\|}{\|\vec{r}'(t)\|^3}$$

2 Partial Derivatives

2.1 First Order Partial Derivatives

Definition 2.1.1: First Order Partial Derivative

The **first order partial derivative** of a function $f(x, y)$ is the derivative of f with respect to one variable while treating the other variable as a constant. The partial derivative of f wrt x is denoted by:

$$f_x(x, y) = \lim_{h \to 0} \frac{f(x+h, y) - f(x, y)}{h}$$

And the partial derivative of f wrt y is denoted by:

$$f_y(x, y) = \lim_{h \to 0} \frac{f(x, y+h) - f(x, y)}{h}$$

They can also be written in the following notations:

$$f_x(x, y) = f_x = \frac{\partial f}{\partial x} = \frac{\partial}{\partial x}\big(f(x, y)\big) = D_x f$$

$$f_y(x, y) = f_y = \frac{\partial f}{\partial y} = \frac{\partial}{\partial y}\big(f(x, y)\big) = D_y f$$

2.2 Interpretations of Partial Derivatives

Much like the first derivative of a function of one variable, the first order partial derivatives of a function of multiple variables can be interpreted as the slope of the tangent line to the surface defined by the function at a point.

Slopes of Traces

Partial derivatives are the slopes of traces. The partial derivative $f_x(a, b)$ is the slope of the trace of $f(x, y)$ for the plane $y = b$ at the point (a, b). Likewise, the partial derivative $f_y(x, y)$ is the slope of the trace of $f(x, y)$ for the plane $x = a$ at the point (a, b)

Example 2.1: Determine if $f(x, y) = \frac{x^2}{y^3}$ is increasing or decreasing at $(2, 5)$, if:
(a) we allow x to vary and hold y fixed,
(b) we allow y to vary and hold x fixed.

(a) To find the partial derivative with respect to x, we treat y as a constant:

$$f_x(x, y) = \frac{2x}{y^3} \qquad \Longrightarrow \qquad f_x(2, 5) = \frac{4}{125} > 0$$

This means that f is increasing in the x direction at the point $(2, 5)$.
(b) To find the partial derivative with respect to y, we treat x as a constant:

$$f_y(x, y) = -\frac{3x^2}{y^4} \qquad \Longrightarrow \qquad f_y(2, 5) = -\frac{12}{625} < 0$$

This means that f is decreasing in the y direction at the point $(2, 5)$.

Partial derivatives can also be interpreted as the slope of the tangent plane to the surface defined by the function at a point. The tangent plane is a linear approximation of the surface at that point.

> **Example 2.2: Find the slopes of the traces to $z = 10 - 4x^2 - y^2$ at the point $(1, 2)$.**
> The partial derivative with respect to x is:
>
> $$f_x(x, y) = -8x \qquad \Longrightarrow \qquad f_x(1, 2) = -8$$
>
> The partial derivative with respect to y is:
>
> $$f_y(x, y) = -2y \qquad \Longrightarrow \qquad f_y(1, 2) = -4$$
>
> Thus, the slope of the trace in the x direction at $(1, 2)$ is -8, and the slope of the trace in the y direction at $(1, 2)$ is -4.

We can also use partial derivatives to find the equations of the tangent lines to the traces of a surface at a point.

> **Example 2.3: Find the vector equations of the tangent lines to the traces to $z = 10 - 4x^2 - y^2$ at the point $(1, 2)$**
> The point on the trace is
>
> $$(1, 2, f(1, 2)) = (1, 2, 10 - 4(1)^2 - (2)^2) = (1, 2, 2)$$
>
> Hence, the equation of the tangent line to the trace for the plane $y = 2$ is:
>
> $$\vec{r_x}(t) = \langle 1, 2, 2\rangle + t\langle 1, 0, -8\rangle = \langle 1 + t, 2, 2 - 8t\rangle$$
>
> And the equation of the tangent line to the trace for the plane $x = 1$ is:
>
> $$\vec{r_y}(t) = \langle 1, 2, 2\rangle + t\langle 0, 1, -4\rangle = \langle 1, 2 + t, 2 - 4t\rangle$$

> **Example 2.4: Find the vector equations of the tangent lines to the traces for $f(x, y) = \sin x \cos y$ at $\left(\frac{\pi}{3}, \frac{-\pi}{4}\right)$**
> The point on the trace is
>
> $$\left(\frac{\pi}{3}, -\frac{\pi}{4}, f\left(\frac{\pi}{3}, -\frac{\pi}{4}\right)\right) = \left(\frac{\pi}{3}, -\frac{\pi}{4}, \sin\left(\frac{\pi}{3}\right)\cos\left(-\frac{\pi}{4}\right)\right) = \left(\frac{\pi}{3}, -\frac{\pi}{4}, \frac{\sqrt{6}}{4}\right)$$
>
> Hence, the equation of the tangent line to the trace for the plane $y = -\frac{\pi}{4}$ is:
>
> $$\begin{aligned} \vec{r_x}(t) &= \left\langle \frac{\pi}{3}, -\frac{\pi}{4}, \frac{\sqrt{6}}{4}\right\rangle + t\left\langle 1, 0, f_x(x, y)\right\rangle \\ &= \left\langle \frac{\pi}{3}, -\frac{\pi}{4}, \frac{\sqrt{6}}{4}\right\rangle + t\left\langle 1, 0, \cos\left(\frac{\pi}{3}\right)\cos\left(-\frac{\pi}{4}\right)\right\rangle \\ &= \left\langle \frac{\pi}{3}, -\frac{\pi}{4}, \frac{\sqrt{6}}{4}\right\rangle + t\left\langle 1, 0, \frac{1}{2\sqrt{2}}\right\rangle \end{aligned}$$

And the equation of the tangent line to the trace for the plane $x = \frac{\pi}{3}$ is:

$$\begin{aligned}
\vec{r_y}(t) &= \left\langle \frac{\pi}{3}, -\frac{\pi}{4}, \frac{\sqrt{6}}{4} \right\rangle + t\left\langle 0, 1, f_y(x,y) \right\rangle \\
&= \left\langle \frac{\pi}{3}, -\frac{\pi}{4}, \frac{\sqrt{6}}{4} \right\rangle + t\left\langle 0, 1, -\sin\left(\frac{\pi}{3}\right)\sin\left(-\frac{\pi}{4}\right) \right\rangle \\
&= \left\langle \frac{\pi}{3}, -\frac{\pi}{4}, \frac{\sqrt{6}}{4} \right\rangle + t\left\langle 0, 1, \frac{\sqrt{6}}{4} \right\rangle
\end{aligned}$$

2.3 Higher Order Partial Derivatives

Second Order Partial Derivatives

The **Second order partial derivatives** of a function $f(x,y)$ are the partial derivatives of the first order partial derivatives. The second order partial derivatives are denoted by:

$$(f_x)_x = f_{xx} = \frac{\partial}{\partial x}\left(\frac{\partial f}{\partial x}\right) = \frac{\partial^2 f}{\partial x^2}$$

$$(f_x)_y = f_{xy} = \frac{\partial}{\partial y}\left(\frac{\partial f}{\partial x}\right) = \frac{\partial^2 f}{\partial y^2}x$$

$$(f_y)_x = f_{yx} = \frac{\partial}{\partial x}\left(\frac{\partial f}{\partial y}\right) = \frac{\partial^2 f}{\partial x^2}y$$

$$(f_y)_y = f_{yy} = \frac{\partial}{\partial y}\left(\frac{\partial f}{\partial y}\right) = \frac{\partial^2 f}{\partial y^2}$$

Clairaut's Theorem

If the second order partial derivatives f_{xy} and f_{yx} are continuous at a point, then they are equal at that point:

$$f_{xy} = f_{yx}$$

Like second order derivatives, there are higher order partial derivatives as well. The third order partial derivatives are denoted by:

$$f_{xxx} = \frac{\partial^3 f}{\partial x^3}$$

$$f_{xyx} = (f_{xy})_x = \frac{\partial}{\partial x}\left(\frac{\partial^2 f}{\partial y\,\partial x}\right) = \frac{\partial^3 f}{\partial x\,\partial y\,\partial x}$$

$$f_{yxx} = (f_{yx})_x = \frac{\partial}{\partial x}\left(\frac{\partial^2 f}{\partial x\,\partial y}\right) = \frac{\partial^2 f}{\partial x^2\,\partial y}$$

This also applies to functions of more than two variables. For example,

$$f_{xz}(x,y,z) = f_{zx}(x,y,z)$$

Extension of Clairaut's Theorem

In general, we can extend Clairaut's theorem to any function and mixed partial derivatives.

That means:

$$f_{ssrtsrr} = f_{trsrssr} = f_{rrssst} = \dots$$

Example 2.5: Find all the second order partial derivatives of function $Q(u,v,w) = u^4 \sin w^2 - \frac{2v}{u^4} + \ln\big(v^2 w\big)$

To find the second order partial derivatives, we first find the first order partial derivatives:

$$Q_u = 4u^3 \sin w^2 + \frac{8v}{u^5}$$
$$Q_v = \frac{-2}{u^4} + \frac{2vw}{v^2 w} = -\frac{2}{u^4} + \frac{2}{v}$$
$$Q_w = 2u^4 w \cos w^2 + \frac{1}{w}$$

Now we can find the second order partial derivatives:

$$Q_{uu} = 12u^2 \sin w^2 - \frac{40v}{u^6}$$
$$Q_{uv} = Q_{vu} = \frac{8}{u^5}$$
$$Q_{uw} = 8u^3 w \cos w^2$$
$$Q_{vv} = -\frac{2}{v^2}$$
$$Q_{vw} = Q_{wv} = 0$$
$$Q_{ww} = 2u^4 \cos w^2 - 4u^4 w^2 \sin w^2 - \frac{1}{w^2}$$

Example 2.6: Given $w = \ln\left(\frac{xy}{z}\right) + 8x^4 y^3 \sqrt{z}, find \frac{\partial^5 w}{\partial x\, \partial z^2\, \partial y\, \partial x}$

Using Clairaut's theorem,

$$\frac{\partial^5 w}{\partial x\, \partial z^2\, \partial y\, \partial x} = \frac{\partial^5 w}{\partial x^2\, \partial y\, \partial z^2}$$

Now,

$$\frac{\partial w}{\partial x} = \frac{y}{z} \cdot \frac{z}{xy} + 32x^3 y^3 \sqrt{z} = \frac{1}{x} + 32x^3 y^3 \sqrt{z}$$
$$\frac{\partial^2 w}{\partial x^2} = -\frac{1}{x^2} + 96x^2 y^3 \sqrt{z}$$
$$\frac{\partial^3 w}{\partial x^2\, \partial y} = 288x^2 y^2 \sqrt{z}$$
$$\frac{\partial^4 w}{\partial x^2\, \partial y\, \partial z} = \frac{144x^2 y^2}{\sqrt{z}}$$
$$\frac{\partial^5 w}{\partial x^2\, \partial y\, \partial z^2} = -72x^2 y^2 z^{-3/2}$$

Example 2.7: Given $f(x,y) = \frac{x^6}{1+6y} - \cos\big(x^2\big) + 6e^x \sin(y)$**, find** f_{xxyxyx}

$$
\begin{aligned}
f_x &= \frac{6x^5}{1+6y} + 2x\sin\left(x^2\right) + 6e^x\sin(y)\\
f_{xx} &= \frac{30x^4}{1+6y} + 2\sin\left(x^2\right) + 4x^2\cos\left(x^2\right) + 6e^x\sin(y)\\
f_{xxx} &= \frac{120x^3}{1+6y} + 12x\cos\left(x^2\right) - 8x^3\sin\left(x^2\right) + 6e^x\sin(y)\\
f_{xxxx} &= \frac{360x^2}{1+6y} + 12\cos\left(x^2\right) - 48x^2\sin\left(x^2\right) - 16x^4\cos\left(x^2\right) + 6e^x\sin(y)\\
f_{xxxxy} &= -\frac{2160x^2}{(1+6y)^2} + 6e^x\cos(y)\\
f_{xxxxyy} &= \frac{25920x^2}{(1+6y)^3} - 6e^x\sin(y)
\end{aligned}
$$

2.4 Differentials

Differentials

The **differential** of a function $f(x,y)$ is a linear approximation of the change in the function at a point.
The differential of f is denoted by:

$$\mathrm{d}f = f_x\,\mathrm{d}x + f_y\,\mathrm{d}y$$

where dx and dy are small changes in x and y, respectively.
For a given function $w = g(x,y,z)$, the differential is given by:

$$\mathrm{d}w = g_x\,\mathrm{d}x + g_y\,\mathrm{d}y + g_z\,\mathrm{d}z$$

Example 2.8: Compute the differential for $u = \frac{t^3r^6}{s^2}$

$$\mathrm{d}u = \frac{3t^2r^6}{s^2}\,\mathrm{d}t + \frac{6t^3r^5}{s^2}\,\mathrm{d}r - \frac{2t^3r^6}{s^3}\,\mathrm{d}s$$

2.5 Chain Rule

Case 1: If $z = f(x,y)$, $x = g(t)$, and $y = h(t)$, then the chain rule states that:

$$\frac{\mathrm{d}z}{\mathrm{d}t} = \frac{\partial f}{\partial x}\frac{\mathrm{d}x}{\mathrm{d}t} + \frac{\partial f}{\partial y}\frac{\mathrm{d}y}{\mathrm{d}t}$$

$$\text{Or,}\quad \frac{\mathrm{d}z}{\mathrm{d}t} = f_x\frac{\mathrm{d}x}{\mathrm{d}t} + f_y\frac{\mathrm{d}y}{\mathrm{d}t}$$

Example 2.9: Compute $\frac{\mathrm{d}z}{\mathrm{d}t}$ for $z = xe^{xy}$, $x = t^2$, $y = t^{-1}$

$$\begin{aligned}
\frac{\mathrm{d}z}{\mathrm{d}t} &= \frac{\partial f}{\partial x}\frac{\mathrm{d}x}{\mathrm{d}t} + \frac{\partial f}{\partial y}\frac{\mathrm{d}y}{\mathrm{d}t} \\
&= (e^{xy} + yxe^{xy})(2t) + x^2e^{xy}(-t^{-2}) \\
&= 2t(e^{xy} + xye^{xy}) - x^2e^{xy}t^{-2} \\
&= 2t(e^t + te^t) - t^4e^tt^{-2} \\
&= 2te^t + t^2e^t
\end{aligned}$$

Case 2: If $z = f(x, y)$, $x = g(s, t)$, and $y = h(s, t)$, then the chain rule states that:

$$\frac{\partial z}{\partial s} = \frac{\partial f}{\partial x}\frac{\partial x}{\partial s} + \frac{\partial f}{\partial y}\frac{\partial y}{\partial s}$$

$$\text{Or,} \quad \frac{\partial z}{\partial s} = f_x\frac{\partial x}{\partial s} + f_y\frac{\partial y}{\partial s}$$

$$\frac{\partial z}{\partial t} = \frac{\partial f}{\partial x}\frac{\partial x}{\partial t} + \frac{\partial f}{\partial y}\frac{\partial y}{\partial t}$$

$$\text{Or,} \quad \frac{\partial z}{\partial t} = f_x\frac{\partial x}{\partial t} + f_y\frac{\partial y}{\partial t}$$

Example 2.10: Find $\frac{\partial z}{\partial s}$ and $\frac{\partial z}{\partial t}$ for $z = e^{2r}\sin(3\theta)$, $r = st - t^2$, $\theta = \sqrt{s^2 + t^2}$

$$\begin{aligned}
\frac{\partial z}{\partial s} &= \frac{\partial f}{\partial r}\frac{\partial r}{\partial s} + \frac{\partial f}{\partial \theta}\frac{\partial \theta}{\partial s} \\
&= (2e^{2r}\sin(3\theta))(t + 0) + (3e^{2r}\cos(3\theta))\left(\frac{s}{\sqrt{s^2 + t^2}}\right) \\
&= 2te^{2r}\sin(3\theta) + 3e^{2r}\cos(3\theta)\frac{s}{\sqrt{s^2 + t^2}} \\
&= 2te^{2(st-t^2)}\sin\left(3\sqrt{s^2 + t^2}\right) + 3e^{2(st-t^2)}\cos\left(3\sqrt{s^2 + t^2}\right)\frac{s}{\sqrt{s^2 + t^2}}
\end{aligned}$$

And,

$$\begin{aligned}
\frac{\partial z}{\partial t} &= \frac{\partial f}{\partial r}\frac{\partial r}{\partial t} + \frac{\partial f}{\partial \theta}\frac{\partial \theta}{\partial t} \\
&= (2e^{2r}\sin(3\theta))(s - 2t) + (3e^{2r}\cos(3\theta))\left(\frac{t}{\sqrt{s^2 + t^2}}\right) \\
&= 2(s - 2t)e^{2r}\sin(3\theta) + 3e^{2r}\cos(3\theta)\frac{t}{\sqrt{s^2 + t^2}} \\
&= 2(s - 2t)e^{2(st-t^2)}\sin\left(3\sqrt{s^2 + t^2}\right) + 3e^{2(st-t^2)}\cos\left(3\sqrt{s^2 + t^2}\right)\frac{t}{\sqrt{s^2 + t^2}}
\end{aligned}$$

Chain Rule

Given the following conditions:

(i) $z = f(x_1, x_2, \dots, x_n)$ is a function of n variables,

(ii) Each variable $x_i(t_1, t_2, \dots, t_m)$ is a function of m variables, Then for any variable t_i ($i = 1, 2, \dots, m$), we have the following chain rule:

$$\boxed{\frac{\partial z}{\partial t_i} = \frac{\partial z}{\partial x_1}\frac{\partial x_1}{\partial t_i} + \frac{\partial z}{\partial x_2}\frac{\partial x_2}{\partial t_i} + \cdots + \frac{\partial z}{\partial x_n}\frac{\partial x_n}{\partial t_i}}$$

Example 2.11: Compute $\frac{\partial^2 f}{\partial \theta^2}$ for $f(x,y)$ if $x = r\cos\theta$ and $y = r\sin\theta$

$$\begin{aligned}
\frac{\partial f}{\partial \theta} &= \frac{\partial f}{\partial x}\frac{\partial x}{\partial \theta} + \frac{\partial f}{\partial y}\frac{\partial y}{\partial \theta} \\
&= -r\sin\theta\frac{\partial f}{\partial x} + r\cos\theta\frac{\partial f}{\partial y}
\end{aligned}$$

Now, we know the second derivative is

$$\frac{\partial^2 f}{\partial \theta^2} = \frac{\partial}{\partial \theta}\left(\frac{\partial f}{\partial \theta}\right) = \frac{\partial}{\partial \theta}\left(-r\sin\theta\frac{\partial f}{\partial x} + r\cos\theta\frac{\partial f}{\partial y}\right)$$

Now, we can separately compute $\frac{\partial}{\partial \theta}\left(\frac{\partial f}{\partial x}\right)$ and $\frac{\partial}{\partial \theta}\left(\frac{\partial f}{\partial y}\right)$:

$$\begin{aligned}
\frac{\partial}{\partial \theta}\left(\frac{\partial f}{\partial x}\right) &= -r\sin\theta\frac{\partial}{\partial x}\left(\frac{\partial f}{\partial x}\right) + r\cos\theta\frac{\partial}{\partial y}\left(\frac{\partial f}{\partial x}\right) \\
&= -r\sin\theta\frac{\partial^2 f}{\partial x^2} + r\cos\theta\frac{\partial^2 f}{\partial y\,\partial x} \\
\frac{\partial}{\partial \theta}\left(\frac{\partial f}{\partial y}\right) &= -r\sin\theta\frac{\partial}{\partial x}\left(\frac{\partial f}{\partial y}\right) + r\cos\theta\frac{\partial}{\partial y}\left(\frac{\partial f}{\partial y}\right) \\
&= -r\sin\theta\frac{\partial^2 f}{\partial x\,\partial y} + r\cos\theta\frac{\partial^2 f}{\partial y^2}
\end{aligned}$$

Finally, we can substitute these into the second derivative:

$$\begin{aligned}
\frac{\partial^2 f}{\partial \theta^2} &= \frac{\partial}{\partial \theta}\left(-r\sin\theta\frac{\partial f}{\partial x} + r\cos\theta\frac{\partial f}{\partial y}\right) \\
&= -r\cos\theta\frac{\partial f}{\partial x} - r\sin\theta\left(-r\sin\theta\frac{\partial^2 f}{\partial x^2} + r\cos\theta\frac{\partial^2 f}{\partial y\,\partial x}\right) \\
&\quad - r\sin\theta\frac{\partial f}{\partial y} + r\cos\theta\left(-r\sin\theta\frac{\partial^2 f}{\partial x\,\partial y} + r\cos\theta\frac{\partial^2 f}{\partial y^2}\right) \\
&= -r\cos\theta\frac{\partial f}{\partial x} + r^2\sin^2\theta\frac{\partial^2 f}{\partial x^2} - r^2\sin\theta\cos\theta\frac{\partial^2 f}{\partial y\,\partial x} \\
&\quad - r\sin\theta\frac{\partial f}{\partial y} - r^2\sin\theta\cos\theta\frac{\partial^2 f}{\partial x\,\partial y} + r^2\cos^2\theta\frac{\partial^2 f}{\partial y^2} \\
&= -r\left(\cos\theta\frac{\partial f}{\partial x} + \sin\theta\frac{\partial f}{\partial y}\right) \\
&\quad + r^2\left(\sin^2\theta\frac{\partial^2 f}{\partial x^2} - 2\sin\theta\cos\theta\frac{\partial^2 f}{\partial x\,\partial y} + \cos^2\theta\frac{\partial^2 f}{\partial y^2}\right)
\end{aligned}$$

2.5.1 Implicit Differentiation

Implicit Differentiation

If $F(x, y) = 0$ is a function where $y = y(x)$, then we can use implicit differentiation to find the derivative of y with respect to x. The chain rule gives us:

$$F_x + F_y \frac{dy}{dx} = 0 \quad \Longrightarrow \quad \frac{dy}{dx} = -\frac{F_x}{F_y}$$

This can be extended to functions of more than two variables. We can start by assuming that $z = f(x, y)$ and we want to find $\frac{\partial z}{\partial x}$ and $\frac{\partial z}{\partial y}$.
To find $\frac{\partial z}{\partial x}$, we differentiate both sides wrt x:

$$\frac{\partial F}{\partial x}\frac{\partial x}{\partial x} + \frac{\partial F}{\partial y}\frac{\partial y}{\partial x} + \frac{\partial F}{\partial z}\frac{\partial z}{\partial x} = 0$$

Since $\frac{\partial x}{\partial x} = 1$ and $\frac{\partial y}{\partial x} = 0$, we get:

$$\frac{\partial z}{\partial x} = -\frac{F_x}{F_z} \quad \text{and} \quad \frac{\partial z}{\partial y} = -\frac{F_y}{F_z}$$

Example 2.12: Find $\frac{\mathrm{d}y}{\mathrm{d}x}$ for $x\cos(3y) + x^3y^5 = 3x - e^{xy}$

First, we rearrange the equation in the form $F(x, y) = 0$:

$$x\cos(3y) + x^3y^5 - 3x + e^{xy} = 0$$

Now, the derivative is:

$$\frac{\mathrm{d}y}{\mathrm{d}x} = -\frac{F_x}{F_y} = -\frac{\cos(3y) + 3x^2y^5 - 3 + ye^{xy}}{-3x\sin(3y) + 5x^3y^4 + xe^{xy}}$$

Example 2.13: Find $\frac{\partial z}{\partial x}$ and $\frac{\partial z}{\partial y}$ for $x^2\sin(2y - 5z) = 1 + y\cos(6zx)$

First, let's rearrange the equation in the form $F(x, y, z) = 0$:

$$x^2\sin(2y - 5z) - 1 - y\cos(6zx) = 0$$

Now, the derivatives are:

$$\frac{\partial z}{\partial x} = -\frac{F_x}{F_z} = -\frac{2x\sin(2y - 5z) + 6yz\sin(6zx)}{-5x^2\cos(2y - 5z) + 6xy\sin(6zx)}$$

$$\frac{\partial z}{\partial y} = -\frac{F_y}{F_z} = -\frac{2x^2\cos(2y - 5z) - \cos(6zx)}{-5x^2\cos(2y - 5z) + 6xy\sin(6zx)}$$

2.6 Directional Derivatives

Definition 2.6.1: Directional Derivative

The rate of change of $f(x, y)$ in the direction of the unit vector $\vec{u} = \langle a, b\rangle$ is called the **directional derivative** of f and is denoted by $D_{\vec{u}}f(x, y)$. The definition of the directional derivative is:

$$D_{\vec{u}}f(x, y) = \lim_{h\to 0} \frac{f(x + ah, y + bh) - f(x, y)}{h}$$

Now, in practice, finding this limit can be difficult. We can derive an equivalent formula for taking directional derivatives.
Let's define a new function of one variable:

$$g(z) = f(x_0 + az, y_0 + bz)$$

where x_0, y_0, a, b are constants. Then, by the definition of the derivative, we have

$$g'(z) = \lim_{h\to 0} \frac{g(z + h) - g(z)}{h}$$

For $z = 0$, we have:

$$g'(0) = \lim_{h\to 0} \frac{g(h) - g(0)}{h} = \lim_{h\to 0} \frac{f(x_0 + ah, y_0 + bh) - f(x_0, y_0)}{h} = D_{\vec{u}}f(x_0, y_0)$$

Thus, we have the following relationship:

$$g'(0) = D_{\vec{u}f(x_0,y_0)}$$

Now, let's rewrite $g(z)$ as follows:

$$g(z) = f(x, y) \quad \text{where } x = x_0 + az \text{ and } y = y_0 + bz$$

We can now apply the chain rule to find $g'(z)$:

$$g'(z) = \frac{\mathrm{d}g}{\mathrm{d}z} = \frac{\partial f}{\partial x}\frac{\mathrm{d}x}{\mathrm{d}z} + \frac{\partial f}{\partial y}\frac{\mathrm{d}y}{\mathrm{d}z} = f_x(x, y)a + f_y(x, y)b$$

If we take $z = 0$, we get $x = x_0$ and $y = y_0$, and finally we have:

$$D_{\vec{u}}f(x_0, y_0) = g'(0) = f_x(x_0, y_0)a + f_y(x_0, y_0)b$$

Note:-

$$D_{\vec{u}}f(x, y) = f_x(x, y)a + f_y(x, y)b$$

$$D_{\vec{u}}f(x, y, z) = f_x(x, y, z)a + f_y(x, y, z)b + f_z(x, y, z)c$$

Example 2.14: Find each of the directional derivatives:
(a) $D_{\vec{u}}f(8, 1, 2)$ **where** $f(x, y, z) = \ln\frac{x}{z} + \ln\frac{z}{y} + xy^2$ **in the direction of** $\vec{v} = \langle 1, 5, 2\rangle$
(b) $D_{\vec{u}}f(x, y)$ **where** $f(x, y) = xe^{xy} + y$ **and** $\vec{u}$ **is the unit vector in the direction of** $\theta = \frac{2\pi}{3}$

(a) First, the unit vector in the direction of $\vec{v}$ is:

$$\vec{u} = \frac{\vec{v}}{||\vec{v}||} = \frac{\langle 1,5,2\rangle}{\sqrt{1^2+5^2+2^2}} = \frac{\langle 1,5,2\rangle}{\sqrt{30}}$$

Simplifying the function, we have:

$$f(x,y,z) = \ln(x) - \ln(z) + \ln(z) - \ln(y) + xy^2 = \ln(x) - \ln(y) + xy^2$$

The directional derivative is then,

$$\begin{aligned}
D_{\vec{u}}f(x,y,z) &= \frac{1}{\sqrt{30}}\left[f_x(x,y,z) + 5f_y(x,y,z) + 2f_z(x,y,z)\right] \\
D_{\vec{u}}f(8,1,2) &= \frac{1}{\sqrt{30}}\left[1\left(\frac{1}{x}+y^2\right) + 5\left(-\frac{1}{y}+2xy\right) + 2\cdot 0\right]_{(8,1,2)} \\
&= \frac{1}{\sqrt{30}}\left(\frac{1}{8}+1-5+80\right) \\
&= \frac{609}{8\sqrt{30}}
\end{aligned}$$

(b) The unit vector in the direction of $\theta = \frac{2\pi}{3}$ is:

$$\vec{u} = \left\langle \cos\left(\frac{2\pi}{3}\right), \sin\left(\frac{2\pi}{3}\right)\right\rangle = \left\langle -\frac{1}{2}, \frac{\sqrt{3}}{2}\right\rangle$$

So, the directional derivative is:

$$\begin{aligned}
D_{\vec{u}}f(x,y) &= \left(-\frac{1}{2}\right)(e^{xy} + xye^{xy}) + \left(\frac{\sqrt{3}}{2}\right)\left(x^2e^{xy}+1\right) \\
D_{\vec{u}}f(2,0) &= \left(-\frac{1}{2}\right)(1) + \left(\frac{\sqrt{3}}{2}\right)(5) \\
&= \frac{5\sqrt{3}-1}{2}
\end{aligned}$$

Notice, the directional derivative can also be written in the following way:

$$\begin{aligned}
D_{\vec{u}}f(x,y,z) &= f_x(x,y,z)a + f_y(x,y,z)b + f_z(x,y,z)c \\
&= \langle f_x, f_y, f_z\rangle \cdot \langle a,b,c\rangle
\end{aligned}$$

In other words, the directional derivative is the dot product of the gradient vector and the unit vector in the direction of interest.

Gradient Vector

The **gradient vector** of a function $f(x,y)$ is denoted by ∇f and is defined as:

$$\nabla f = \langle f_x, f_y\rangle = \left\langle \frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}\right\rangle$$

For a function $f(x, y, z)$, the gradient vector is:

$$\nabla f = \langle f_x, f_y, f_z \rangle = \left\langle \frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z} \right\rangle$$

Directional Derivative

The directional derivative can also be expressed in terms of the gradient vector:

$$D_{\vec{u}} f(\vec{x}) = \nabla f \cdot \vec{u}$$

where $\vec{x} = \langle x, y, z \rangle$ or $\vec{x} = \langle x, y \rangle$ depending on the function and $\vec{u}$ is the unit vector in the direction of interest.

Example 2.15: Find the directional derivative

$$D_{\vec{u}} f(\vec{x}) \textbf{ for } f(x, y, z) = \sin(yz) + \ln(x^2) \textbf{ at } (1, 1, \pi)$$

in the direction of $\vec{v} = \langle 1, 1, -1 \rangle$

The gradient vector is:

$$\begin{aligned}
\nabla f(x, y, z) &= \left\langle \frac{2}{x}, z\cos(yz), y\cos(yz) \right\rangle \\
\nabla f(1, 1, \pi) &= \left\langle \frac{2}{1}, \pi\cos(\pi), \cos(\pi) \right\rangle = \langle 2, -\pi, -1 \rangle
\end{aligned}$$

The unit vector in the direction of $\vec{v}$ is:

$$\vec{u} = \frac{\vec{v}}{||\vec{v}||} = \frac{\langle 1, 1, -1 \rangle}{\sqrt{3}} = \left\langle \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, -\frac{1}{\sqrt{3}} \right\rangle$$

Hence, the directional derivative is:

$$\begin{aligned}
D_{\vec{u}} f(1, 1, \pi) &= \nabla f \cdot \vec{u} \\
&= \left\langle 2, -\pi, -1 \right\rangle \cdot \left\langle \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, -\frac{1}{\sqrt{3}} \right\rangle \\
&= 2 \cdot \frac{1}{\sqrt{3}} - \pi \cdot \frac{1}{\sqrt{3}} + 1 \cdot \frac{1}{\sqrt{3}} = \frac{3 - \pi}{\sqrt{3}}
\end{aligned}$$

Theorem 2.6.2: *The maximum value of $D_{\vec{u}} f(\vec{x})$ (and hence then the maximum rate of change of the function $f(\vec{x})$) is given by $||\nabla f(\vec{x})||$ and will occur in the direction given by $\nabla f(\vec{x})$.*

Proof:

We can use a nice fact about dot products as well as the fact that $\vec{u}$ is a unit vector to proof this theorem:

$$D_{\vec{u}} f = \nabla f \cdot \vec{u} = ||\nabla f||\, ||\vec{u}|| \cos\theta = ||\nabla f|| \cos\theta$$

where θ is the angle between the gradient and $\vec{u}$.
Now, the largest possible value of $\cos\theta$ is 1, which occurs at $\theta = 0$. Therefore, the maximum

value of $D_{\vec{u}}f(\vec{x})$ is $||\nabla f(\vec{x})||$. Also, the maximum value occurs when the angle between the gradient and $\vec{u}$ is zero, or in other words, when $\vec{u}$ is pointing in the same direction as the gradient.

Note:-

The gradient vector $\nabla f(x_0, y_0)$ is orthogonal (or perpendicular) to the level curve/contour curve $f(x, y) = k$ at the point (x_0, y_0). Likewise, the gradient vector $\nabla f(x_0, y_0, z_0)$ is orthogonal to the level surface $f(x, y, z) = k$ at the point (x_0, y_0, z_0).

Proof:

Let S be the level surface given by $f(x, y, z) = k$ and let $P(x_0, y_0, z_0)$ be a point on the surface S.

Now, let C be any curve on the surface S that contains the point P. Let $\vec{r}(t) = \langle x(t), y(t), z(t)\rangle$ be the vector equation for C and suppose that t_0 is the value of t such that $\vec{r}(t_0) = \langle x_0, y_0, z_0\rangle$.

In other words, t_0 is the value of t that gives P.

Since C lies on S, we know that the points on C must satisfy the equation for S. That is

$$f(x(t), y(t), z(t)) = k$$

Using the chain rule, we get:

$$\frac{\partial f}{\partial x}\frac{\mathrm{d}x}{\mathrm{d}t} + \frac{\partial f}{\partial y}\frac{\mathrm{d}y}{\mathrm{d}t} + \frac{\partial f}{\partial z}\frac{\mathrm{d}z}{\mathrm{d}t} = 0$$

Notice that $\nabla f = \langle f_x, f_y, f_z\rangle$ and $\vec{r}'(t) = \langle x'(t), y'(t), z'(t)\rangle$ so this becomes:

$$\nabla f \cdot \vec{r}'(t) = 0$$

At $t = t_0$,

$$\nabla f(x_0, y_0, z_0) \cdot \vec{r}'(t_0) = 0$$

This then tells us that the gradient vector at P (i.e. $\nabla f(x_0, y_0, z_0)$) is orthogonal to the tangent vector $\vec{r}'(t_0)$ to any curve C that passes through P and on the surface S and su must also be orthogonal to the surface S.

3 Applications of Partial Derivatives

3.1 Tangent Planes and Linear Approximations

Let there be a point (x_0, y_0) on function $z = f(x, y)$ in $\mathbb{R}^3$. Let C_1 be the trace to $f(x, y)$ for plane $y = y_0$ and C_2 be the trace to $f(x, y)$ for plane $x = x_0$. Now, let L_1 and L_2 be the tangent lines to C_1 and C_2 at point (x_0, y_0), respectively. The tangent plane to $f(x, y)$ at point (x_0, y_0) is defined as the plane that contains both lines L_1 and L_2.

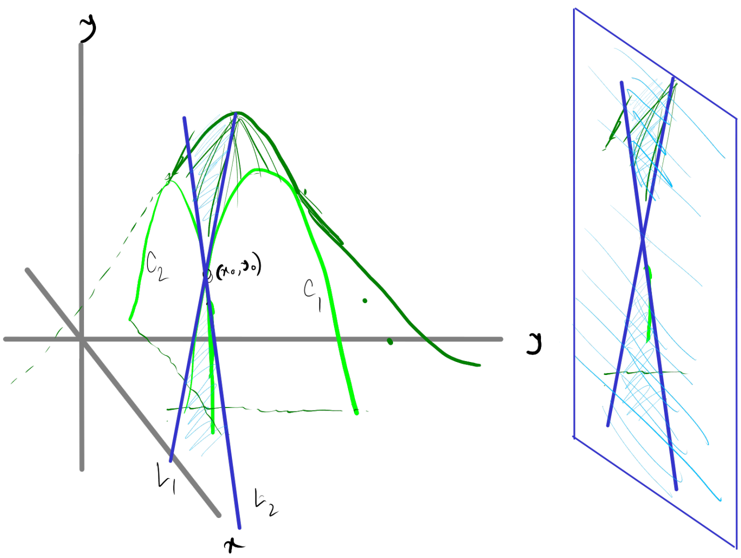


Figure 3.1.1: Tangent plane

Now, we need to find the equation of the tangent plane. The general equation of a plane is given by

$$a(x - x_0) + b(y - y_0) + c(z - z_0) = 0$$

where (x_0, y_0, z_0) is a point on the plane and a, b, and c are the coefficients of the plane. The equation can be rewritten as

$$z - z_0 = -\frac{a}{c}(x - x_0) - \frac{b}{c}(y - y_0)$$

Let's rename the constants as follows:

$$A = -\frac{a}{c}, \qquad B = -\frac{b}{c}$$

Thus we get

$$z - z_0 = A(x - x_0) + B(y - y_0)$$

Now, assuming $y = y_0$ (i.e., y is fixed) and $x = x_0$ (i.e., x is fixed) we get respectively

$$z - z_0 = A(x - x_0) \qquad \text{and} \qquad z - z_0 = B(y - y_0)$$

Note, these are the equations for the tangent lines L_1 and L_2 respectively, where the slopes are respectively $A = f_x(x_0, y_0)$ and $B = f_y(x_0, y_0)$. We also know $z_0 = f(x_0, y_0)$.
Hence, the equation of the tangent plane is given by

$$\boxed{z = f(x_0, y_0) + f_x(x_0, y_0)(x - x_0) + f_y(x_0, y_0)(y - y_0)}$$

The linear approximation for a surface "near" the point (x_0, y_0) is given then

$$L(x, y) = f(x_0, y_0) + f_x(x_0, y_0)(x - x_0) + f_y(x_0, y_0)(y - y_0)$$

Linear Approximation

The linear approximation to the function $z = f(x, y)$ at the point (x_0, y_0) is given by

$$L(x, y) = f(x_0, y_0) + f_x(x_0, y_0)(x - x_0) + f_y(x_0, y_0)(y - y_0)$$

Example 3.1: Find the linear approximation to $z = \frac{10x^2}{x-y}$ at $(4, -1)$

The linear approximation is given by

$$L(x, y) = z(4, -1) + z_x(4, -1)(x - 4) + z_y(4, -1)(y + 1)$$

Here,

$$\begin{aligned}
z(4, -1) &= \frac{10 \times 4^2}{4 - (-1)} \\
&= \frac{160}{5} = 32 \\
z_x(4, -1) &= \frac{\partial}{\partial x}\left(\frac{10x^2}{x - y}\right)\bigg|_{(4,-1)} \\
&= \frac{20x(x - y) - 10x^2}{(x - y)^2}\bigg|_{(4,-1)} \\
&= \frac{20 \times 4(4 - (-1)) - 10 \times 4^2}{(4 - (-1))^2} \\
&= \frac{80 \times 5 - 160}{25} = \frac{400 - 160}{25} = \frac{240}{25} = 9.6 \\
z_y(4, -1) &= \frac{\partial}{\partial y}\left(\frac{10x^2}{x - y}\right)\bigg|_{(4,-1)} \\
&= \frac{10x^2}{(x - y)^2}\bigg|_{(4,-1)} \\
&= \frac{10 \times 4^2}{(4 - (-1))^2} = \frac{160}{25} = 6.4
\end{aligned}$$

Hence, the linear approximation is given by

$$\boxed{L(x, y) = 32 + 9.6(x - 4) + 6.4(y + 1)}$$

3.2 Gradient Vector, Tangent Planes, and Normal Lines

Note:-

The gradient vector $\nabla f(x_0, y_0)$ is orthogonal to the level curve $f(x, y) = k$ at the point (x_0, y_0). Likewise, the gradient vector $\nabla f(x_0, y_0, z_0)$ is orthogonal to the level surface $f(x, y, z) = k$ at the point (x_0, y_0, z_0).

We know, the gradient vector is

$$\nabla f = \langle f_x, f_y, f_z \rangle$$

So, the tangent plane to the surface $f(x, y, z) = k$ at (x_0, y_0, z_0) is given by the equation

$$f_x(x_0, y_0, z_0)(x - x_0) + f_y(x_0, y_0, z_0)(y - y_0) + f_z(x_0, y_0, z_0)(z - z_0) = 0$$

This is a general form of the equation of a tangent plane than that of the previous section. However, we can also write the equation in the previous form. For that, we need to find the tangent plane to the surface given by $z = f(x, y)$ at the point (x_0, y_0, z_0), where $z_0 = f(x_0, y_0)$. We can rewrite as

$$f(x, y) - z = 0$$

Now, defining a new function as

$$F(x, y, z) = f(x, y) - z$$

The surface given by $z = f(x, y)$ is identical to the surface given by $F(x, y, z) = 0$. Now, the gradient vector for F is

$$\nabla F = \langle F_x, F_y, F_z \rangle = \langle f_x, f_y, -1 \rangle$$

Note:

$$F_x = \frac{\partial}{\partial x}\left[f(x, y) - z\right] = f_x$$
$$F_y = \frac{\partial}{\partial y}\left[f(x, y) - z\right] = f_y$$
$$F_z = \frac{\partial}{\partial z}\left[f(x, y) - z\right] = -1$$

Hence, the tangent plane is then

$$F_x(x - x_0) + F_y(y - y_0) + F_z(z - z_0) = 0$$

$$f_x(x_0, y_0)(x - x_0) + f_y(x_0, y_0)(y - y_0) - (z - z_0) = 0$$
$$\boxed{z = f(x_0, y_0) + f_x(x_0, y_0)(x - x_0) + f_y(x_0, y_0)(y - y_0)}$$

We can also find the normal line to the surface $f(x, y, z) = k$ at the point (x_0, y_0, z_0). The equation of line requires a point and a parallel vector, which is given by the gradient vector. Hence, the equation of the normal line is given by

$$\boxed{\vec{r}(t) = \langle x_0, y_0, z_0 \rangle + t\nabla f(x_0, y_0, z_0)}$$

Tangent Plane

The tangent plane to the surface given by $F(x, y, z) = f(x, y) - z$ at the point (x_0, y_0, z_0) is given by

$$F_x(x_0, y_0, z_0)(x - x_0) + F_y(x_0, y_0, z_0)(y - y_0) + F_z(x_0, y_0, z_0)(z - z_0) = 0$$

Normal Line

The normal line to the surface given by $F(x, y, z) = f(x, y) - z$ at the point (x_0, y_0, z_0) is given by

$$\vec{r}(t) = \langle x_0, y_0, z_0 \rangle + t\nabla F(x_0, y_0, z_0)$$

Example 3.2: Find the tangent plane and normal line to $9yz - \sqrt{x^2 - 8z} = xy^2 - 26$ at $(3, 1, -2)$

Rearranging the equation, we get

$$xy^2 + \sqrt{x^2 - 8z} - 9yz - 26 = 0$$

The gradient vector is given by

$$\begin{aligned}
\nabla F &= \langle y^2 + \frac{x}{\sqrt{x^2-8z}}, 2xy - 9z, -\frac{4}{\sqrt{x^2-8z}} - 9y\rangle \\
\nabla F(3,1,-2) &= \langle 1 + \frac{3}{5}, 6 + 18, -\frac{4}{5} - 9\rangle \\
&= \langle \frac{8}{5}, 24, \frac{-49}{5}\rangle
\end{aligned}$$

The tangent plane, hence, would be

$$\begin{aligned}
\frac{8}{5}(x-3) + 24(y-1) - \frac{49}{5}(z+2) = 0 \\
8x - 24 + 120y - 120 - 49z - 98 = 0 \\
8x + 120y - 49z = 242
\end{aligned}$$

And the normal line is given by the equation

$$\frac{x-3}{\frac{8}{5}} = \frac{y-1}{24} = \frac{z+2}{-\frac{49}{5}}$$

$$\frac{x-3}{8} = \frac{y-1}{120} = -\frac{z+2}{49}$$

Example 3.3: Find the point(s) on $6x^2 + y^2 - 3z^2 = 4$ where the tangent plane to the surface is parallel to the plane given by $2x + 7y - z = 6$.
The gradient vector of the surface is given by

$$\nabla F = \langle 12x, 2y, -6z\rangle$$

The normal vector to the parallel plane is

$$\vec{n} = \langle 2, 7, -1\rangle$$

Since the planes are parallel, the gradient vector must be a scalar multiple of the normal vector, i.e.,

$$\langle 12x, 2y, -6z\rangle = k\langle 2, 7, -1\rangle$$

This gives us the following equations:

$$12x = 2k \implies k = 6x$$

$$2y = 7k = 42x \implies y = 21x$$

$$-6z = -k \implies z = x$$

Now, substituting these values in the equation of the surface, we get

$$\begin{aligned}
6x^2 + (21x)^2 - 3x^2 = 4 \\
x^2 = \frac{4}{444}
\end{aligned}$$

$$\therefore x = \pm\frac{1}{\sqrt{111}}, \qquad y = \pm\frac{21}{\sqrt{111}}, \qquad z = \pm\frac{1}{\sqrt{111}}$$

Hence, the points are

$$\left(\frac{1}{\sqrt{111}}, \frac{21}{\sqrt{111}}, \frac{1}{\sqrt{111}}\right) \text{ and } \left(-\frac{1}{\sqrt{111}}, -\frac{21}{\sqrt{111}}, -\frac{1}{\sqrt{111}}\right)$$

3.3 Relative Minimums and Maximums

Definition 3.3.1: Relative Minimum and Maximum

A function $f(x, y)$ has a **relative minimum** at the point (a, b) if $f(x, y) \geq f(a, b)$ for all points (x, y) in some region around (a, b).

A function $f(x, y)$ has a **relative maximum** at the point (a, b) if $f(x, y) \leq f(a, b)$ for all points (x, y) in some region around (a, b).

Definition 3.3.2: Critical Point

The point (a, b) is a **critical point** (or a **stationary point**) of the function $f(x, y)$ provided one of the following is true:

1. $\nabla f(a, b) = \vec{0}$ (i.e., $f_x(a, b) = 0$ and $f_y(a, b) = 0$)
2. $f_x(a, b)$ and/or $f_y(a, b)$ do not exist

Note:-

If the point (a, b) is a relative extrema of the function $f(x, y)$ and the first order derivative of $f(x, y)$ exists at (a, b), then (a, b) is a critical point of $f(x, y)$. However, the converse is not true. That is, a critical point need not be a relative extrema.

Proof:

Let $g(x) = f(x, y)$ and suppose that $f(x, y)$ has a relative extrema at (a, b). However, this also means that $g(x)$ also has a relative extrema at $x = a$. By Fermat's theorem, we know that $g'(a) = 0$. But we also know that $g'(a) = f_x(a, b)$ and so we have that $f_x(a, b) = 0$.
If we now define $h(y) = f(a, b)$, then we can similarly show that $f_y(a, b) = 0$.
So, putting all these together means that $\nabla f(a, b) = \vec{0}$ and so $f(x, y)$ has a critical point at (a, b).

Example of a point that is a critical point but not a relative extrema is the point $(0, 0)$ for the function $f(x, y) = x^2 - y^2$. In the x-axis, the function becomes $f(x, 0) = x^2$, which is a parabola opening upwards. So, the point $(0, 0)$ looks like a minimum along this path. In the y-axis, the function becomes $f(0, y) = -y^2$, which is a parabola opening downwards. So, the point $(0, 0)$ looks like a maximum along this path. Since the point $(0, 0)$ is a minimum in one direction and a maximum in another, it's neither a relative minimum nor a relative maximum. This type of critical point is also called a **saddle point**.

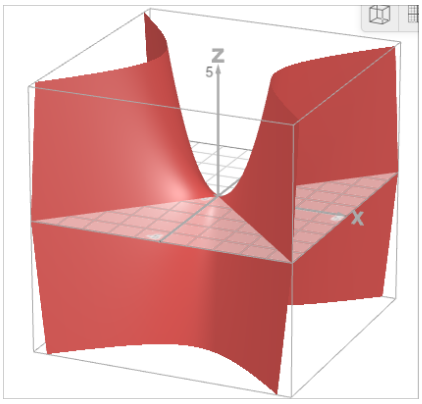


Figure 3.3.1: Saddle point

Definition 3.3.3: Saddle Point

A point (a, b) is a **saddle point** of the function $f(x, y)$ if it is a critical point but neither a relative minimum nor a relative maximum.

Definition 3.3.4: Hessian Matrix

The Hessian matrix of a function $f(x, y)$ is given by

$$H = \begin{bmatrix} f_{xx} & f_{xy} \\ f_{xy} & f_{yy} \end{bmatrix}$$

where f_{xx}, f_{yy}, and f_{xy} are the second order partial derivatives of $f(x, y)$.
The Hessian matrix is used in the second derivative test to classify critical points of the function $f(x, y)$.

Relative Extrema

Suppose that (a, b) is a critical point of $f(x, y)$ and that the second order partial derivatives are continuous in some region that contains (a, b). Next, let's use the determinant of the Hessian matrix, denoted by D, to classify the critical point (a, b).

$$D = D(a, b) = f_{xx}(a, b)f_{yy}(a, b) - \left[f_{xy}(a, b)\right]^2$$

We then have the following classifications of the critical point:

1. If $D > 0$ and $f_{xx}(a, b) > 0$ then there is a relative minimum at (a, b).
2. If $D > 0$ and $f_{xx}(a, b) < 0$ then there is a relative maximum at (a, b).
3. If $D < 0$ then there is a saddle point at (a, b).
4. If $D = 0$ then the test is inconclusive. In this case, we need to use other methods to determine the nature of the critical point.

Note that if $D > 0$ then both $f_{xx}(a, b)$ and $f_{yy}(a, b)$ will have the same sign and so in the first two cases, we can conclude that both $f_{xx}(a, b)$ and $f_{yy}(a, b)$ are either both positive or both negative.

Example 3.4: Find and classify all the critical points for $f(x, y) = 3x^2y + y^3 - 3x^2 - 3y^2 + 2$

First, we need to find the first and second order partial derivatives of the function:

$$f_x = 6xy - 6x, \qquad f_y = 3x^2 + 3y^2 - 6y$$

$$f_{xx} = 6y - 6, \qquad f_{yy} = 6y - 6, \qquad f_{xy} = 6x$$

Now, we need to find the critical points by solving the equations $f_x = 0$ and $f_y = 0$ simultaneously:

$$\begin{aligned} 6xy - 6x &= 0 \\ 3x^2 + 3y^2 - 6y &= 0 \end{aligned}$$

From the first equation,

$$6x(y - 1) = 0 \implies x = 0, y = 1$$

For $x = 0$:

$$3y^2 - 6y = 3y(y - 2) = 0 \implies y = 0, 2$$

For $y = 1$:
$$3x^2 + 3 - 6 = 0 \implies x^2 = 1 \implies x = 1, -1$$
Hence, the critical points are $(0,0)$, $(0,2)$, $(1,1)$, and $(-1,1)$. Now, we need to classify these critical points using the second derivative test:
$$D(x,y) = f_{xx}(x,y)f_{yy}(x,y) - \left[f_{xy}(x,y)\right]^2 = (6y-6)^2 - 36x^2$$
For $(0,0)$:
$$D(0,0) = 36 > 0 \qquad f_{xx}(0,0) = -6 < 0$$
For $(0,2)$:
$$D(0,2) = 36 > 0 \qquad f_{xx}(0,2) = 6 > 0$$
For $(1,1)$:
$$D(1,1) = -36 < 0$$
For $(-1,1)$:
$$D(-1,1) = -36 < 0$$
So, we have the following classifications:

- $(0,0)$ is a relative maximum.
- $(0,2)$ is a relative minimum.
- $(1,1)$ and $(-1,1)$ are saddle points.

Example 3.5: Determine the point on the plane $4x - 2y + z = 1$ that is closest to the point $(-2,-1,5)$
We can rewrite the equation of the plane as
$$z = 1 - 4x + 2y$$
The distance between a point (x,y,z) on the plane and the point $(-2,-1,5)$ is given by
$$d = \sqrt{(x+2)^2 + (y+1)^2 + (2y-4x-4)^2}$$
Now, since the finding the mininmum value of d is equivalent to finding the minimum value of d^2, let
$$\begin{aligned} f(x,y) = d^2 &= (x+2)^2 + (y+1)^2 + (2y-4x-4)^2 \\ &= (x^2+4x+4) + (y^2+2y+1) + (16x^2+4y^2-16xy+32x-16y+16) \\ f(x,y) &= 17x^2 + 5y^2 - 16xy + 36x - 14y + 21 \end{aligned}$$
We can now find the derivatives:
$$f_x(x,y) = 34x - 16y + 36, \qquad f_y(x,y) = 10y - 16x - 14$$
$$f_{xx}(x,y) = 34, \qquad f_{yy}(x,y) = 10, \qquad f_{xy} = -16$$
Before finding the critical points, notice that
$$\begin{aligned} D(x,y) &= f_{xx}f_{yy} - f_{xy}^2 \\ &= 34 \times 10 - (-16)^2 \\ &= 340 - 256 = 84 > 0 \end{aligned}$$

Since $D > 0$ and $f_{xx} > 0$, all the critical points will be relative minimums. Now, to find the critical points, solve the equations

$$\begin{aligned} f_x &= 34x - 16y + 36 = 0 \\ f_y &= 10y - 16x - 14 = 0 \end{aligned}$$

From the first equation, we get

$$x = \frac{1}{34}(16y - 36) = \frac{1}{17}(8y - 18)$$

Substituting this in the second equation, we get

$$\begin{aligned} 10y - \frac{16}{17}(8y - 18) - 14 &= 0 \\ 170y - 16(8y - 18) - 238 &= 0 \\ 42y + 50 &= 0 \\ \therefore y &= -\frac{25}{21} \end{aligned}$$

And substituting this value in the equation for x, we get

$$\begin{aligned} x &= \frac{1}{17}\left(\frac{-200}{21} - 18\right) \\ &= \frac{-200 - 378}{17 \times 21} \\ \therefore x &= -\frac{34}{21} \end{aligned}$$

Finally, substituting these values in the equation of the plane, we get

$$z = 1 + 4 \times \frac{34}{21} - 2 \times \frac{25}{21} = \frac{107}{21}$$

So, the point on the plane that is closest to the point $(-2, -1, 5)$ is $\left(-\frac{34}{21}, -\frac{25}{21}, \frac{107}{21}\right)$

3.4 Absolute Extrema

Definition 3.4.1: Closed, Open, and Bounded Region

1. A region $\mathbb{R}^2$ is called **closed** if it contains all its boundary points.
2. A region $\mathbb{R}^2$ is called **open** if it does not contain any of its boundary points.
3. A region $\mathbb{R}^2$ is called **bounded** if it can be contained in a circle of finite radius.

Theorem 3.4.2* (Extreme Value Theorem)*: *If $f(x, y)$ is continuous on a closed and bounded region D in $\mathbb{R}^2$, then there are two points (x_1, y_1) and (x_2, y_2) in region D such that $f(x_1, y_1)$ is the absolute maximum and $f(x_2, y_2)$ is the absolute minimum of $f(x, y)$ in D.*

Finding Absolute Extrema

1. Find all the critical points of the function $f(x, y)$ that lie in the region D and determine the function value at each of these points.
2. Find all extrema of the function $f(x, y)$ on the boundary of the region D and determine the function value at each of these points.
3. The largest and the smallest values found in the first two steps are the absolute minimum and the absolute maximum of the function $f(x, y)$ in the region D.

Example 3.6: Find the absolute minimum and absolute maximum of $f(x, y) = x^2 + 4y^2 - 2x^2y + 4$ on the rectangle given by $-1 \le x \le 1$ and $-1 \le y \le 1$.

First, we need to find the critical points of the function $f(x, y)$ in the region D. The first order partial derivatives are given by

$$\begin{aligned} f_x &= 2x - 4xy \\ f_y &= 8y - 2x^2 \end{aligned}$$

Setting these equal to zero, we get the following equations:

$$\begin{aligned} 2x - 4xy &= 0 \\ 8y - 2x^2 &= 0 \end{aligned}$$

Solving the equations, we get

$$(x, y) = (0, 0), \left(\sqrt{2}, \frac{1}{2}\right), \left(-\sqrt{2}, \frac{1}{2}\right)$$

Since $-1 \le x \le 1$, the only critical point in the region D is $(0, 0)$. Value of the function at this point is

$$f(0, 0) = 0^2 + 4(0)^2 - 2(0)^2(0) + 4 = 4$$

Now, we need to find the extrema on the boundary of the region. The boundary consists of four line segments:

1. $(-1, -1)$ to $(1, -1)$ where $y = -1$ and $-1 \le x \le 1$
2. $(1, -1)$ to $(1, 1)$ where $x = 1$ and $-1 \le y \le 1$
3. $(1, 1)$ to $(-1, 1)$ where $y = 1$ and $-1 \le x \le 1$
4. $(-1, 1)$ to $(-1, -1)$ where $x = -1$ and $-1 \le y \le 1$

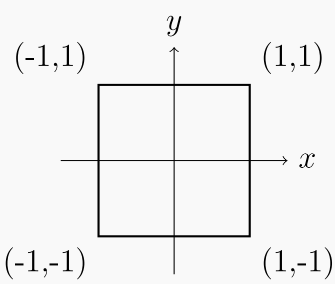


Figure 3.4.1: Boundary of the region

For segment 1:
Let $g(x) = f(x, -1) = 3x^2 + 8$
For critical point on the edge, $g'(x) = 0$ or $6x = 0 \implies x = 0$.

$$\begin{aligned} g(0) &= f(0, -1) = 8 \\ g(-1) &= f(-1, -1) = 11 \\ g(1) &= f(1, -1) = 11 \end{aligned}$$

For segment 2:

Let $h(y) = f(1, y) = 1 + 4y^2 - 2(1)y + 4 = 5 + 4y^2 - 2y$
For critical point on the edge, $h'(y) = 0$ or $8y - 2 = 0 \implies y = \frac{1}{4}$.

$$\begin{aligned} h\left(\frac{1}{4}\right) &= f(1, \frac{1}{4}) = \frac{19}{4} = 4.75 \\ h(-1) &= f(1, -1) = 11 \\ h(1) &= f(1, 1) = 7 \end{aligned}$$

For segment 3:
Let $k(x) = f(x, 1) = x^2 + 4(1)^2 - 2x^2(1) + 4 = -x^2 + 8$
For critical point on the edge, $k'(x) = 0$ or $-2x = 0 \implies x = 0$.

$$\begin{aligned} k(0) &= f(0, 1) = 8 \\ k(-1) &= f(-1, 1) = 7 \\ k(1) &= f(1, 1) = 7 \end{aligned}$$

For segment 4:
Let $l(y) = f(-1, y) = (-1)^2 + 4y^2 - 2(-1)y + 4 = 5 + 4y^2 + 2y$
For critical point on the edge, $l'(y) = 0$ or $8y + 2 = 0 \implies y = -\frac{1}{4}$.

$$\begin{aligned} l\left(-\frac{1}{4}\right) &= f(-1, -\frac{1}{4}) = \frac{19}{4} = 4.75 \\ l(-1) &= f(-1, -1) = 11 \\ l(1) &= f(-1, 1) = 7 \end{aligned}$$

Now, we can summarize the function values at the critical point and the boundary points:

$$\begin{aligned} &f(0,0) = 4, \\ &f(0,-1) = 8, && f(-1,-1) = 11, && f(1,-1) = 11, \\ &f\left(1, \frac{1}{4}\right) = 4.75, && f(1,-1) = 11, && f(1,1) = 7, \\ &f(0,1) = 8, && f(-1,1) = 7, && f(1,1) = 7, \\ &f\left(-1, -\frac{1}{4}\right) = 4.75, && f(-1,-1) = 11, && f(-1,1) = 7 \end{aligned}$$

Hence, the absolute extrema are given by

$$\begin{aligned} &\text{Absolute Maximum: } (-1,-1) \text{ and } (1,-1) \text{ with value } 11, \\ &\text{Absolute Minimum: } (0,0) \text{ with value } 4 \end{aligned}$$

Example 3.7: Find the absolute minimum and absolute maximum of $f(x, y) = 2x^2 - y^2 + 6y$ **on the disk of radius** 4, $x^2 + y^2 \leq 16$

First, we need to find the critical points of the function $f(x, y)$ in the region D. The first order partial derivatives are given by

$$\begin{aligned} f_x &= 4x \\ f_y &= -2y + 6 \end{aligned}$$

Setting these equal to zero, we get the following equations:

$$4x = 0 \implies x = 0$$
$$-2y + 6 = 0 \implies y = 3$$

So, the only critical point in the region D is $(0, 3)$. The value of the function at this point is

$$f(0, 3) = 2(0)^2 - (3)^2 + 6(3) = 9$$

Now, we need to find the extrema on the boundary of the region. The boundary consists of the circle $x^2 + y^2 = 16$. We can rewrite it as

$$x^2 = 16 - y^2$$

Let

$$\begin{aligned} g(y) &= f\left(\sqrt{16 - y^2}, y\right) \\ &= 2(16 - y^2) - y^2 + 6y \\ &= 32 - 3y^2 + 6y \end{aligned}$$

The critical points on the boundary are given by

$$g'(y) = -6y + 6 = 0 \implies y = 1$$

$$x = \pm\sqrt{16 - y^2} = \pm\sqrt{15}$$

Now, we can find the value of the function at this point and at the endpoints of the boundary:

$$g(1) = f(\pm\sqrt{15}, 1) = 35$$
$$g(-4) = f(0, -4) = -40$$
$$g(4) = f(0, 4) = 8$$

Now, we can summarize the function values at the critical point and the boundary points:

$$\begin{aligned} &f(0, 3) = 9, && \\ &f(\sqrt{15}, 1) = 35, && f(-\sqrt{15}, 1) = 35, \\ &f(0, -4) = -40, && f(0, 4) = 8 \end{aligned}$$

Hence, the absolute extrema are given by

$$\begin{aligned} &\text{Absolute Maximum: } (\sqrt{15}, 1) \text{ and } (-\sqrt{15}, 1) \text{ with value } 35, \\ &\text{Absolute Minimum: } (0, -4) \text{ with value } -40 \end{aligned}$$

Example 3.8: Find the absolute minimum and absolute maximum of $f(x, y) = 18x^2 + 4y^2 - y^2x - 2$ on the triangle with vertices $(-1, -1)$, $(5, -1)$, and $(5, 17)$.

First, we need to find the critical points of the function $f(x, y)$ in the region D. The first order partial derivatives are given by

$$f_x = 36x - y^2$$
$$f_y = 8y - 2xy$$

Setting these equal to zero, we get the following equations:

$$36x - y^2 = 0$$
$$8y - 2xy = 0$$

Solving the equations, we get

$$(x, y) = (0, 0), (4, -12), (4, 12)$$

Notice that the triangle is bounded by

1. $y = -1$
2. $x = 5$
3. $y + 1 = 3(x + 1) \implies y = 3x + 2$

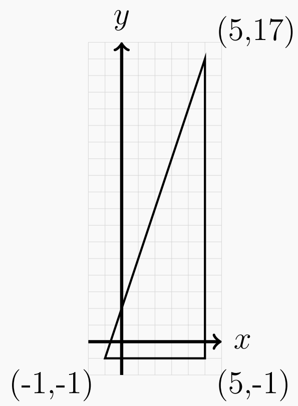


Figure 3.4.2: Triangle region

Among the critical points,

1. $(0, 0)$ lies inside the triangle.
2. $(4, 12)$ lies inside the triangle. $[3 \times 4 + 2 = 14 > 12]$
3. $(4, -12)$ lies outside the triangle. $[-1 \leq y \leq 17]$

The values of the function at the critical points are:

$$f(0, 0) = -2 \qquad \text{and} \qquad f(4, 12) = 286$$

Now, we need to find the extrema on the boundary of the region. The boundary consists of three line segments:

1. $(-1, -1)$ to $(5, -1)$ where $y = -1$ and $-1 \leq x \leq 5$
2. $(5, -1)$ to $(5, 17)$ where $x = 5$ and $-1 \leq y \leq 17$
3. $(-1, -1)$ to $(5, 17)$ where $y = 3x + 2$ and $-1 \leq x \leq 5$

For segment 1:
Let $g(x) = f(x, -1) = 18x^2 + 4 - x - 2 = 18x^2 - x + 2$
For critical point on the edge, $g'(x) = 0$ or $36x - 1 = 0 \implies x = \frac{1}{36}$.

$$g\left(\frac{1}{36}\right) = f\left(\frac{1}{36}, -1\right) \approx 1.986$$
$$g(-1) = f(-1, -1) = 21$$
$$g(5) = f(5, -1) = 447$$

For segment 2:
Let $h(y) = f(5, y) = 450 + 4y^2 - 5y^2 - 2 = 448 - y^2$

For critical point on the edge, $h'(y) = -2y = 0 \implies y = 0$

$$\begin{aligned} h(0) &= f(5,0) = 448, \\ h(-1) &= f(5,-1) = 447 \\ h(17) &= f(5,17) = 159 \end{aligned}$$

For segment 3:
Let $k(x) = f(x, 3x+2) = 18x^2 + 4(3x+2)^2 - (3x+2)^2x - 2$

$$\begin{aligned} k(x) &= 18x^2 + 4(3x+2)^2 - (3x+2)^2x - 2 \\ &= 18x^2 + 4(9x^2 + 12x + 4) - (9x^3 + 12x^2 + 4x) - 2 \\ &= 18x^2 + 36x^2 + 48x + 16 - 9x^3 - 12x^2 - 4x - 2 \\ &= -9x^3 + 42x^2 + 44x + 14 \end{aligned}$$

For critical point on the edge, $k'(x) = 0$ or $-27x^2 + 84x + 44 = 0$. This gives us the points $(x, y) \approx (3.568, 12.704), (-0.457, 0.629)$

$$\begin{aligned} k(3.568) &= f(3.568, 12.704) \approx 296.872 \\ k(-0.457) &= f(-0.457, 0.629) \approx 3.523 \end{aligned}$$

Hence, the absolute extrema are given by

$$\begin{aligned} &\text{Absolute Maximum: } (5,-1) \text{ with value } 447, \\ &\text{Absolute Minimum: } (0,0) \text{ with value } -2 \end{aligned}$$

3.5 Lagrange Multipliers

Method of Lagrange Multipliers

Given a function $f(x, y, z)$ and a constraint $g(x, y, z) = k$. To find the absolute extrema:

1. Solve the following system of equations:

$$\begin{aligned} \nabla f(x,y,z) &= \lambda \nabla g(x,y,z) \\ g(x,y,z) &= k \end{aligned}$$

2. Plug in all solutions, (x, y, z), from the first step into the function $f(x, y, z)$ and identify the minimum and maximum values, provided they exist and $\nabla g \neq \vec{0}$ at the point.

The constant λ is called the **Lagrange Multiplier**.

Lagrange Multipliers for Multiple Constraints

If there are multiple constraints, say $g_1(x, y, z) = k_1$ and $g_2(x, y, z) = k_2$, then the system of equations becomes

$$\begin{aligned} \nabla f(x,y,z) &= \lambda_1 \nabla g_1(x,y,z) + \lambda_2 \nabla g_2(x,y,z) \\ g_1(x,y,z) &= k_1 \\ g_2(x,y,z) &= k_2 \end{aligned}$$

where λ_1 and λ_2 are the Lagrange Multipliers for the constraints g_1 and g_2, respectively.

The method can be extended to any number of constraints by adding more terms to the right-hand side of the first equation.

Example 3.9: Find the maximum and minimum values of $f(x,y)=4x^2+10y^2$ on the disk $x^2+y^2\leq 4$.

Using Lagrange multipliers, we need to solve the following system of equations:

$$\begin{aligned}\nabla f(x,y) &= \lambda\nabla g(x,y)\\ g(x,y) &= x^2+y^2=4\end{aligned}$$

That is, we need to solve the following equations:

$$\begin{aligned}8x &= 2\lambda x\\ 20y &= 2\lambda y\\ x^2+y^2 &= 4\end{aligned}$$

From the first equation, we get:

$$2x(4-\lambda)=0 \implies x=0 \text{ or } \lambda=4$$

If we have $x=0$ then the constraint gives us

$$y^2=4-x^2 \implies y=\pm\sqrt{4-x^2}=\pm 2$$

If we have $\lambda=4$ the second equation gives us

$$20y=8y \implies y=0$$

Then the constraint gives us

$$x^2+0^2=4 \implies x=\pm 2$$

Now, we can find the values of the function at these points:

$$\begin{aligned}&f(0,0)=0 &&\\ &f(0,2)=40 && f(0,-2)=40\\ &f(2,0)=16 && f(-2,0)=16\end{aligned}$$

Hence, the maximum value is 40 at the points $(0,2)$ and $(0,-2)$, and the minimum value is 0 at the point $(0,0)$.

Example 3.10: Find the maximum and minimum values of $f(x,y,z)=4y-2z$ subject to the constraints $2x-y-z=2$ and $x^2+y^2=1$.

Using Lagrange multipliers method, we need to solve the following system of equations:

$$\begin{aligned}\nabla f(x,y,z) &= \lambda\nabla g(x,y,z)+\mu\nabla h(x,y,z)\\ g(x,y,z) &= 2x-y-z=2\\ h(x,y,z) &= x^2+y^2=1\end{aligned}$$

That is, we need to solve the following equations:

$$\begin{aligned}
0 &= 2\lambda + 2\mu x \\
4 &= -\lambda + 2\mu y \\
-2 &= -\lambda \\
2x - y - z &= 2 \\
x^2 + y^2 &= 1
\end{aligned}$$

From the third equation, we get $\lambda = 2$. Plugging this into the first and second equations, we get respectively

$$\begin{aligned}
-4 = 2\mu x &\implies x = -\frac{2}{\mu} \\
6 = 2\mu y &\implies y = \frac{3}{\mu}
\end{aligned}$$

Now, substituting these values in the fifth equation, we get

$$\frac{4}{\mu^2} + \frac{9}{\mu^2} = 1 \implies \mu = \pm\sqrt{13}$$

Now, for $\mu = \sqrt{13}$, we have

$$\begin{aligned}
x &= -\frac{2}{\sqrt{13}} \\
y &= \frac{3}{\sqrt{13}} \\
z &= 2x - y - 2 = -\frac{4}{\sqrt{13}} - \frac{3}{\sqrt{13}} - 2 = -2 - \frac{7}{\sqrt{13}}
\end{aligned}$$

And for $\mu = -\sqrt{13}$, we have

$$\begin{aligned}
x &= \frac{2}{\sqrt{13}} \\
y &= -\frac{3}{\sqrt{13}} \\
z &= 2x - y - 2 = \frac{4}{\sqrt{13}} + \frac{3}{\sqrt{13}} - 2 = -2 + \frac{7}{\sqrt{13}}
\end{aligned}$$

Now, we can find the values of the function at these points:

$$\begin{aligned}
f\left(-\frac{2}{\sqrt{13}}, \frac{3}{\sqrt{13}}, -2 - \frac{7}{\sqrt{13}}\right) &\approx 11.211 \\
f\left(\frac{2}{\sqrt{13}}, -\frac{3}{\sqrt{13}}, -2 + \frac{7}{\sqrt{13}}\right) &\approx -3.211
\end{aligned}$$

Hence, the maximum value is approximately 11.211 at the point $\left(-\frac{2}{\sqrt{13}}, \frac{3}{\sqrt{13}}, -2 - \frac{7}{\sqrt{13}}\right)$, and the minimum value is approximately -3.211 at the point $\left(\frac{2}{\sqrt{13}}, -\frac{3}{\sqrt{13}}, -2 + \frac{7}{\sqrt{13}}\right)$.

4 Multiple Integrals

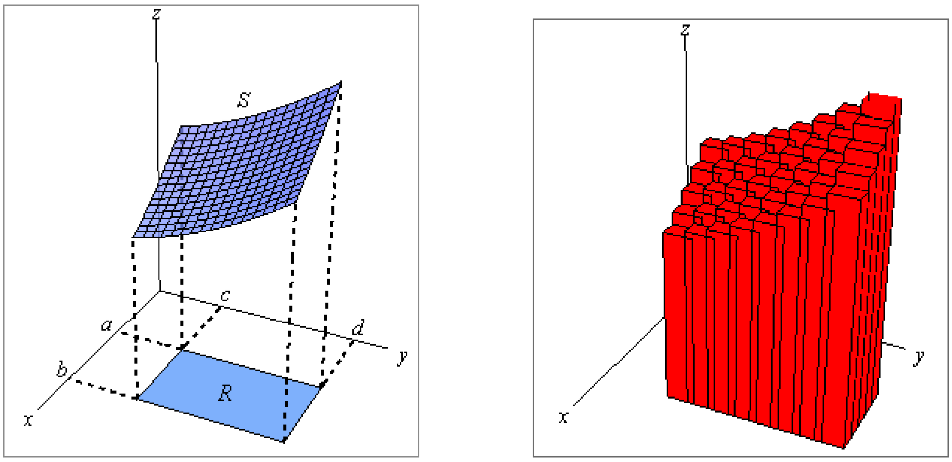
4.1 Double Integrals

If $f(x,y)$ is a function of two variables and $R = [a,b] \times [c,d]$ (that means the ranges for x and y are $a \leq x \leq b$ and $c \leq y \leq d$) is the projection of the region in the xy-plane, then we can approximate the volume under the surface $S = f(x,y)$ over the region R as:

$$V \approx \sum_{i=1}^{n} \sum_{j=1}^{m} f(x_i^*, y_j^*) \Delta A_{ij}$$

where (x_i^*, y_j^*) is a point on the Region R and ΔA_{ij} is the area of the rectangle in the xy-plane with vertices (x_i, y_j), (x_i, y_{j+1}), (x_{i+1}, y_j), and (x_{i+1}, y_{j+1}). We can then take the limit as $n \to \infty$ and $m \to \infty$ to get the volume as

$$V = \lim_{n,m \to \infty} \sum_{i=1}^{n} \sum_{j=1}^{m} f(x_i^*, y_j^*) \Delta A_{ij}$$



(a) (b)

Figure 4.1.1

Double Integral

$$\text{Volume} = \iint_R f(x,y)\,\mathrm{d}A$$

We can use this double sum in the definition to estimate the value of the double indegral. We can do so by choosing (x_i^*, y_j^*) to be the midpoint of the rectangle $(\bar{x}_i, \bar{y}_j)$. This leads us to the **Midpoint Rule** for double integrals:

$$\iint_R f(x,y)\,\mathrm{d}A \approx \sum_{i=1}^{n} \sum_{j=1}^{m} f(\bar{x}_i, \bar{y}_j) \Delta A_{ij}$$

4.2 Iterated Integrals

***Theorem 4.2.1* (Fubini's Theorem):** *If $f(x, y)$ is continuous on the rectangle $R = [a, b] \times [c, d]$, then*

$$\iint\limits_R f(x,y)\,\mathrm{d}A = \int_c^d \int_a^b f(x,y)\,\mathrm{d}x\,\mathrm{d}y = \int_a^b \int_c^d f(x,y)\,\mathrm{d}y\,\mathrm{d}x$$

*These integrals are called **iterated integrals**.*

Note:-

If $f(x, y) = g(x)h(y)$ and we're integrating over the rectangle $R = [a, b] \times [c, d]$, then

$$\iint\limits_R f(x,y)\,\mathrm{d}A = \iint\limits_R g(x)h(y)\,\mathrm{d}A = \int_a^b g(x)\,\mathrm{d}x \int_c^d h(y)\,\mathrm{d}y$$

Example 4.1: Evaluate $\displaystyle\iint\limits_R x\cos^2(y)\,\mathrm{d}A\,,\, R = [-2, 3] \times [0, \frac{\pi}{2}]$

Since the integrand is a function of x times a function of y, we can use the fact.

$$\begin{aligned}
\iint\limits_R x\cos^2(y)\,\mathrm{d}x &= \left(\int_{-2}^{3} x\,\mathrm{d}x\right)\left(\int_0^{\pi/2} \cos^2(y)\,\mathrm{d}y\right) \\
&= \left(\frac{1}{2}x^2\right)\Bigg|_{-2}^{3} \left(\frac{1}{2}\int_0^{\pi/2} (1+\cos(2y))\,\mathrm{d}y\right) \\
&= \frac{5}{4}\left(y + \frac{1}{2}\sin(2y)\right)\Bigg|_0^{\pi/2} \\
&= \frac{5\pi}{8}
\end{aligned}$$

Example 4.2: Determine the volume that lies under $f(x, y) = 9x^2+4xy+4$ and above the rectangle given by $[-1, 1] \times [0, 2]$ in the xy-plane

$$\begin{aligned}
V &= \iint\limits_R (9x^2 + 4xy + 4)\,\mathrm{d}A \\
&= \int_0^2 \int_{-1}^{1} (9x^2 + 4xy + 4)\,\mathrm{d}x\,\mathrm{d}y \\
&= \int_0^2 \left(3x^3 + 2x^2y + 4x\right)\Big|_{-1}^{1}\,\mathrm{d}y \\
&= \int_0^2 \big((3 + 2y + 4) - (-3 + 2y - 4)\big)\,\mathrm{d}y \\
&= \int_0^2 (6 + 8)\,\mathrm{d}y = 14y\Big|_0^2 = 28
\end{aligned}$$

4.3 Single Indefinite Integrals of a Multivariable Function

Single Indefinite Integral of a Multivariable Function

Let $f(x, y)$ be a function of two variables. Then the **single indefinite integral** of f with respect to x is given by

$$\int f(x,y)\,\mathrm{d}x = F(x,y) + C(y)$$

where $F(x, y)$ is any function such that $\frac{\partial F}{\partial x} = f(x, y)$ and $C(y)$ is any function of y. Similarly, the single indefinite integral of f with respect to y is given by

$$\int f(x,y)\,\mathrm{d}y = G(x,y) + D(x)$$

where $G(x, y)$ is any function such that $\frac{\partial G}{\partial y} = f(x, y)$ and $D(x)$ is any function of x.

Example 4.3: Evaluate $\int (x^2y + 3xy^2)\,\mathrm{d}x$ **and** $\int (x^2y + 3xy^2)\,\mathrm{d}y$

$$\begin{aligned}
\int (x^2y + 3xy^2)\,\mathrm{d}x &= \int x^2y\,\mathrm{d}x + \int 3xy^2\,\mathrm{d}x \\
&= y\int x^2\,\mathrm{d}x + 3y^2\int x\,\mathrm{d}x \\
&= y\left(\frac{1}{3}x^3\right) + 3y^2\left(\frac{1}{2}x^2\right) + C(y) \\
&= \frac{1}{3}x^3y + \frac{3}{2}x^2y^2 + C(y)
\end{aligned}$$

$$\begin{aligned}
\int (x^2y + 3xy^2)\,\mathrm{d}y &= \int x^2y\,\mathrm{d}y + \int 3xy^2\,\mathrm{d}y \\
&= x^2\int y\,\mathrm{d}y + 3x\int y^2\,\mathrm{d}y \\
&= x^2\left(\frac{1}{2}y^2\right) + 3x\left(\frac{1}{3}y^3\right) + D(x) \\
&= \frac{1}{2}x^2y^2 + xy^3 + D(x)
\end{aligned}$$

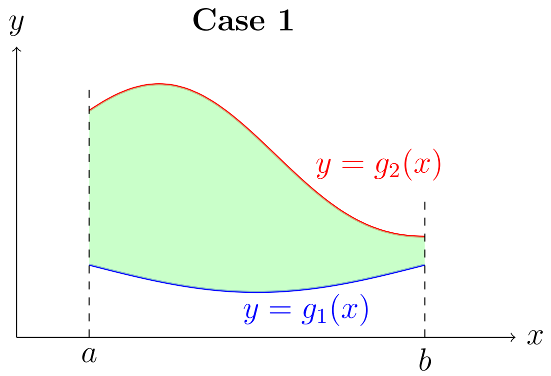
4.4 Double Integrals Over General Regions

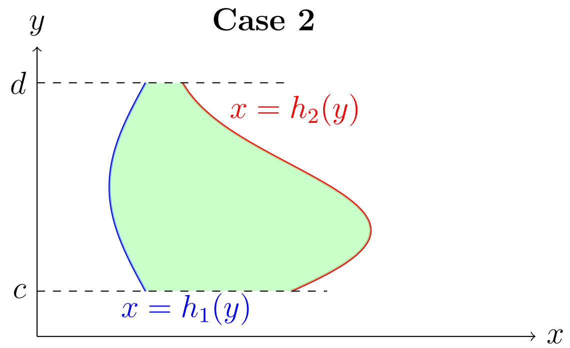
In previous section, double integrals over rectangular regions were discussed. But most regions are not rectangular, so we need to look at the following double integral now:

$$\iint\limits_D f(x,y)\,\mathrm{d}A$$

There are two types of regions that we need to look at. Here are diagrams for both cases: We can use set builder method to describe these regions as follows:

- Case 1: $D = \{(x, y)|a \le x \le b, g_1(x) \le y \le g_2(x)\}$
- Case 2: $D = \{(x, y)|c \le y \le d, h_1(y) \le x \le h_2(y)\}$





Note:-

The double integral for both of these cases are defined in terms of iterated integrals as follows:

- Case 1: $D = \{(x, y)|a \leq x \leq b, g_1(x) \leq y \leq g_2(x)\}$

$$\iint\limits_D f(x, y)\,\mathrm{d}A = \int_a^b \int_{g_1(x)}^{g_2(x)} f(x, y)\,\mathrm{d}y\,\mathrm{d}x$$

- Case 2: $D = \{(x, y)|c \leq y \leq d, h_1(y) \leq x \leq h_2(y)\}$

$$\iint\limits_D f(x, y)\,\mathrm{d}A = \int_c^d \int_{h_1(y)}^{h_2(y)} f(x, y)\,\mathrm{d}x\,\mathrm{d}y$$

Properties

1. $\displaystyle\iint\limits_D \big(f(x, y) + g(x, y)\big)\,\mathrm{d}A = \iint\limits_D f(x, y)\,\mathrm{d}A + \iint\limits_D g(x, y)\,\mathrm{d}A$

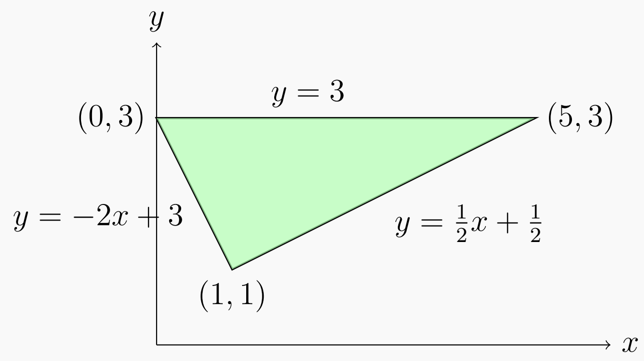
2. $\displaystyle\iint\limits_D kf(x, y)\,\mathrm{d}A = k\iint\limits_D f(x, y)\,\mathrm{d}A$

3. If the region D can be split into two separate regions D_1 and D_2 then the integral can be written as
$$\iint\limits_D f(x, y)\,\mathrm{d}A = \iint\limits_{D_1} f(x, y)\,\mathrm{d}A + \iint\limits_{D_2} f(x, y)\,\mathrm{d}A$$

Example 4.4: Evaluate the integral $\displaystyle\iint\limits_D (6x^2 - 40y)\,\mathrm{d}A$, where D is the triangle with vertices $(0, 3)$, $(1, 1)$, and $(5, 3)$

We can break the region into two parts and evaluate the integral as follows:

$$D_1 = \{(x, y)|0 \leq x \leq 1, -2x + 3 \leq y \leq 3\}$$
$$D_2 = \{(x, y)|1 \leq x \leq 5, \frac{1}{2}x + \frac{1}{2} \leq y \leq 3\}$$



Solution 1:

$$\begin{aligned}
\iint\limits_D (6x^2 - 40y)\,\mathrm{d}A &= \iint\limits_{D_1} (6x^2 - 40y)\,\mathrm{d}A + \iint\limits_{D_2} (6x^2 - 40y)\,\mathrm{d}A \\
&= \int_0^1 \int_{-2x+3}^3 (6x^2 - 40y)\,\mathrm{d}y\,\mathrm{d}x + \int_1^5 \int_{\frac{1}{2}x+\frac{1}{2}}^3 (6x^2 - 40y)\,\mathrm{d}y\,\mathrm{d}x \\
&= \int_0^1 (6x^2y - 20y^2)\Big|_{-2x+3}^3\,\mathrm{d}x + \int_1^5 (6x^2y - 20y^2)\Big|_{\frac{1}{2}x+\frac{1}{2}}^3\,\mathrm{d}x \\
&= \int_0^1 (12x^3 - 180 + 20(3-2x)^2)\,\mathrm{d}x + \int_1^5 \left(-3x^3 + 25x^2 - 180 + 20\left(\frac{1}{2}x + \frac{1}{2}\right)^2\right)\mathrm{d}x \\
&= \left(3x^4 - 180x - \frac{10}{3}(3-2x)^3\right)\Bigg|_0^1 + \left(\frac{-3}{4}x^4 + 5x^3 - 180x + \frac{40}{3}\left(\frac{1}{2}x + \frac{1}{2}\right)^3\right)\Bigg|_1^5 \\
&= -\frac{935}{3}
\end{aligned}$$

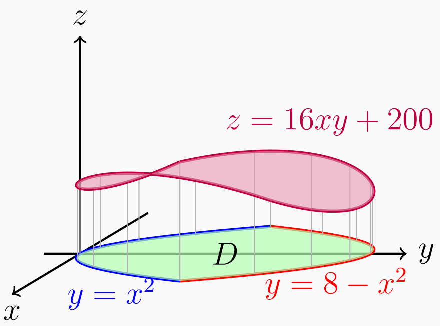
However, we can also evaluate the integral using a different approach.
Solution 2:

$$\begin{aligned}
\iint\limits_D (6x^2 - 40y)\,\mathrm{d}A &= \int_1^3 \int_{-\frac{1}{2}y+\frac{3}{2}}^{2y-1} (6x^2 - 40y)\,\mathrm{d}x\,\mathrm{d}y \\
&= \int_1^3 (2x^3 - 40xy)\Big|_{-\frac{1}{2}y+\frac{3}{2}}^{2y-1}\,\mathrm{d}y \\
&= \int_1^3 \left[100y - 100y^2 + 2(2y-1)^3 - 2\left(-\frac{1}{2}y + \frac{3}{2}\right)^3\right]\mathrm{d}y \\
&= \left[50y^2 - \frac{100}{3}y^3 + \frac{1}{4}(2y-1)^4 + \left(-\frac{1}{2}y + \frac{3}{2}\right)^4\right]\Bigg|_1^3 \\
&= -\frac{935}{3}
\end{aligned}$$

Example 4.5: Find the volume of the solid that lies below the surface given by $z = 16xy + 200$ and lies above the region in the xy-plane bounded by $y = x^2$ and $y = 8 - x^2$.

The volume is given by

$$
\begin{aligned}
V &= \iint\limits_D (16xy + 200)\,\mathrm{d}A \\
&= \int_{-2}^{2}\int_{x^2}^{8-x^2} (16xy + 200)\,\mathrm{d}y\,\mathrm{d}x \\
&= \int_{-2}^{2} (8xy^2 + 200y)\Big|_{x^2}^{8-x^2}\,\mathrm{d}x \\
&= \int_{-2}^{2} (-128x^3 - 400x^2 + 512x + 1600)\,\mathrm{d}x \\
&= \left(-32x^4 - \frac{400}{3}x^3 + 256x^2 + 1600x\right)\Bigg|_{-2}^{2} \\
&= \frac{12800}{3}
\end{aligned}
$$



Note:-

In general,

$$V = \iint\limits_D f(x,y)\,\mathrm{d}A$$

gives the net volume between the graph of $z = f(x,y)$ and the region D in the xy-plane. Regions that are below the xy-plane, have a negative volume, and the regions that are above the xy-plane, have a positive volume.

Note:-

$$\text{Area of region } D = \iint\limits_D \mathrm{d}A$$

4.5 Double Integrals in Polar Coordinates

Some regions are much easier to describe in terms of polar coordinates rather than Cartesian coordinates. For instance, we might have a region that is a disk, ring, or a portion of a disk or ring.

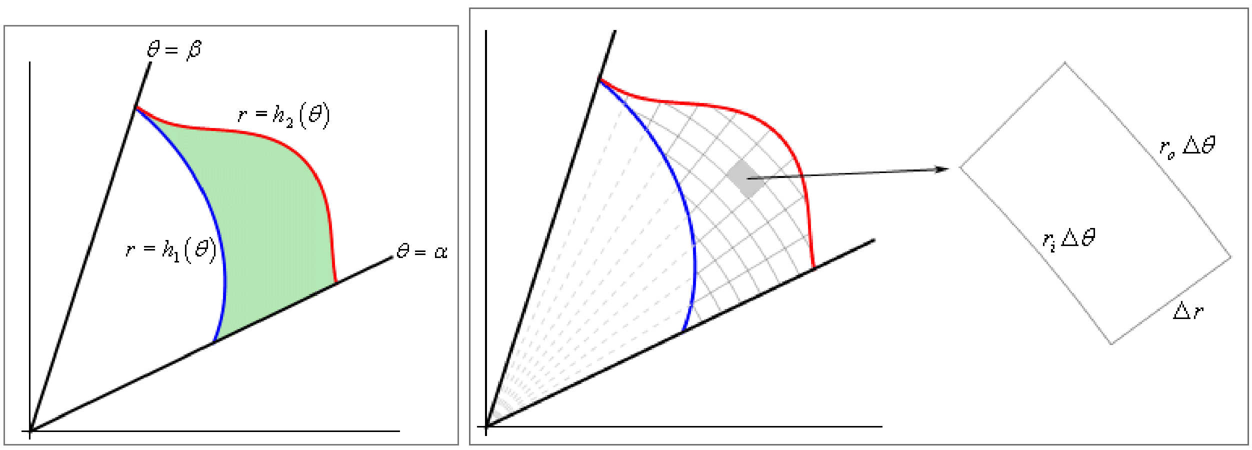


Figure 4.5.1: A region in polar coordinates and its mesh approximation

In Figure 4.5.1, the region is broke up into a mesh of radial lines and arcs. The area of each piece is ΔA. Two sides of the piece both have length $\Delta r = r_o - r_i$. And the other two sides are arcs $r_o\Delta\theta$ and $r_i\Delta\theta$. We can assume that the mesh is so small that $r_i \approx r_o \approx r$. Hence, the area of each piece is approximately

$$\Delta A \approx r\Delta r\Delta\theta \implies \mathrm{d}A = r\,\mathrm{d}r\,\mathrm{d}\theta$$

Note:-

$$\mathrm{d}A = r\,\mathrm{d}r\,\mathrm{d}\theta$$

Polar Rectangle and Area Element

A **polar rectangle** R is a region in the plane defined by

$$R = \{(r,\theta) \mid a \le r \le b, \alpha \le \theta \le \beta\}$$

where $0 \le \beta - \alpha \le 2\pi$.

Double Integral for General Polar Regions

For general regions in polar coordinates, if $D = \{(r,\theta) \mid \alpha \le \theta \le \beta, h_1(\theta) \le r \le h_2(\theta)\}$, then

$$\iint\limits_D f(x,y)\,\mathrm{d}A = \int_\alpha^\beta \int_{h_1(\theta)}^{h_2(\theta)} f(r\cos\theta, r\sin\theta)\,r\,\mathrm{d}r\,\mathrm{d}\theta$$

Example 4.6: Evaluate $\iint\limits_R (3x+4y^2)\,\mathrm{d}A$, where R is the region in the upper half-plane bounded by the circles $x^2+y^2=1$ and $x^2+y^2=4$.

The region R can be described in polar coordinates as

$$R = \big\{(r,\theta) \mid 1 \le r \le 2, 0 \le \theta \le \pi\big\}$$

Converting to polar coordinates: $x = r\cos\theta$ and $y = r\sin\theta$, and $\mathrm{d}A = r\,\mathrm{d}r\,\mathrm{d}\theta$.

$$\begin{aligned}
\iint\limits_R (3x+4y^2)\,\mathrm{d}A &= \int_0^\pi \int_1^2 (3r\cos\theta + 4r^2\sin^2\theta)r\,\mathrm{d}r\,\mathrm{d}\theta \\
&= \int_0^\pi \int_1^2 (3r^2\cos\theta + 4r^3\sin^2\theta)\,\mathrm{d}r\,\mathrm{d}\theta \\
&= \int_0^\pi \left(r^3\cos\theta + r^4\sin^2\theta\right)\Big|_1^2\,\mathrm{d}\theta \\
&= \int_0^\pi \big((8\cos\theta + 16\sin^2\theta) - (\cos\theta + \sin^2\theta)\big)\,\mathrm{d}\theta \\
&= \int_0^\pi (7\cos\theta + 15\sin^2\theta)\,\mathrm{d}\theta \\
&= \int_0^\pi \left(7\cos\theta + 15\cdot\frac{1-\cos(2\theta)}{2}\right)\mathrm{d}\theta \\
&= \left(7\sin\theta + \frac{15}{2}\theta - \frac{15}{4}\sin(2\theta)\right)\Bigg|_0^\pi \\
&= \left(0 + \frac{15\pi}{2} - 0\right) - (0+0-0) = \frac{15\pi}{2}
\end{aligned}$$

Example 4.7: Find the volume of the solid that lies under the paraboloid $z = x^2 + y^2$ and above the disk $x^2 + y^2 \leq 9$.

In polar coordinates, the disk is $D = \{(r, \theta) \mid 0 \leq r \leq 3, 0 \leq \theta \leq 2\pi\}$ and $z = x^2 + y^2 = r^2$.

$$
\begin{aligned}
V &= \iint_D (x^2 + y^2)\,\mathrm{d}A = \int_0^{2\pi} \int_0^3 r^2 \cdot r\,\mathrm{d}r\,\mathrm{d}\theta \\
&= \int_0^{2\pi} \int_0^3 r^3\,\mathrm{d}r\,\mathrm{d}\theta = \int_0^{2\pi} \left(\frac{r^4}{4}\right)\bigg|_0^3 \mathrm{d}\theta \\
&= \int_0^{2\pi} \frac{81}{4}\,\mathrm{d}\theta = \frac{81}{4} \cdot 2\pi = \frac{81\pi}{2}
\end{aligned}
$$

Example 4.8: Evaluate $\iint_D e^{x^2+y^2}\,\mathrm{d}A$, where D is the region bounded by the semicircle $x = \sqrt{4 - y^2}$ and the y-axis in the first quadrant.

The region D is a quarter circle of radius 2 in the first quadrant. In polar coordinates:

$$
D = \left\{(r, \theta) \mid 0 \leq r \leq 2, 0 \leq \theta \leq \frac{\pi}{2}\right\}
$$

Since $x^2 + y^2 = r^2$, we have $e^{x^2+y^2} = e^{r^2}$.

$$
\iint_D e^{x^2+y^2}\,\mathrm{d}A = \int_0^{\pi/2} \int_0^2 e^{r^2} \cdot r\,\mathrm{d}r\,\mathrm{d}\theta
$$

Using the substitution $u = r^2$, so $\mathrm{d}u = 2r\,\mathrm{d}r$:

$$
\begin{aligned}
&= \int_0^{\pi/2} \left(\frac{1}{2}\int_0^4 e^u\,\mathrm{d}u\right) \mathrm{d}\theta \\
&= \int_0^{\pi/2} \frac{1}{2}\,(e^u)\Big|_0^4 \mathrm{d}\theta \\
&= \int_0^{\pi/2} \frac{1}{2}(e^4 - 1)\,\mathrm{d}\theta \\
&= \frac{e^4 - 1}{2} \cdot \frac{\pi}{2} = \frac{\pi(e^4 - 1)}{4}
\end{aligned}
$$

When to Use Polar Coordinates

Polar coordinates are particularly useful when:

1. The region D is a circle, disk, annulus, or sector of a circle
2. The function $f(x, y)$ can be simplified using $x^2 + y^2 = r^2$ (e.g., $f(x, y) = \sqrt{x^2 + y^2}$ becomes $f(r, \theta) = r$)
3. The integrand contains expressions like $x^2 + y^2$, $\sqrt{x^2 + y^2}$, or $\arctan(y/x)$

4.6 Triple Integrals

Triple integrals extend the concept of double integrals to three dimensions. Just as double integrals compute volumes under surfaces in 2D regions, triple integrals can compute various quantities over three-dimensional regions, including volumes and masses.

Triple Integral

For a function $f(x, y, z)$ defined over a three-dimensional region E, the triple integral is denoted:

$$\iiint\limits_E f(x, y, z)\,\mathrm{d}V$$

where $\mathrm{d}V = \mathrm{d}x\,\mathrm{d}y\,\mathrm{d}z$ represents the differential volume element.

Volume of 3D Region

The volume of a three-dimensional region E is given by the triple integral

$$V = \iiint\limits_E \mathrm{d}V$$

Triple Integrals over Boxes

For a rectangular box $B = [a, b] \times [c, d] \times [r, s]$, the triple integral can be evaluated as an iterated integral

$$\iiint\limits_B f(x, y, z)\,\mathrm{d}V = \int_r^s \int_c^d \int_a^b f(x, y, z)\,\mathrm{d}x\,\mathrm{d}y\,\mathrm{d}z$$

Example 4.9: Evaluate $\iiint\limits_B 8xyz^2\,\mathrm{d}V$ where $B = [1, 2] \times [0, 1] \times [-1, 1]$.

We can use any order of integration. Let's use $\mathrm{d}x\,\mathrm{d}y\,\mathrm{d}z$:

$$\begin{aligned}
\iiint\limits_B 8xyz^2\,\mathrm{d}V &= \int_{-1}^1 \int_0^1 \int_1^2 8xyz^2\,\mathrm{d}x\,\mathrm{d}y\,\mathrm{d}z \\
&= \int_{-1}^1 \int_0^1 4x^2yz^2\Big|_{x=1}^{x=2}\,\mathrm{d}y\,\mathrm{d}z \\
&= \int_{-1}^1 \int_0^1 (16yz^2 - 4yz^2)\,\mathrm{d}y\,\mathrm{d}z \\
&= \int_{-1}^1 \int_0^1 12yz^2\,\mathrm{d}y\,\mathrm{d}z \\
&= \int_{-1}^1 6y^2z^2\Big|_{y=0}^{y=1}\,\mathrm{d}z \\
&= \int_{-1}^1 6z^2\,\mathrm{d}z \\
&= 2z^3\Big|_{-1}^{1} \\
&= 2(1) - 2(-1) = 4
\end{aligned}$$

4.6.1 Triple Integrals over General Regions

For more general regions, we define three types of regions that allow us to set up triple integrals with variable limits.

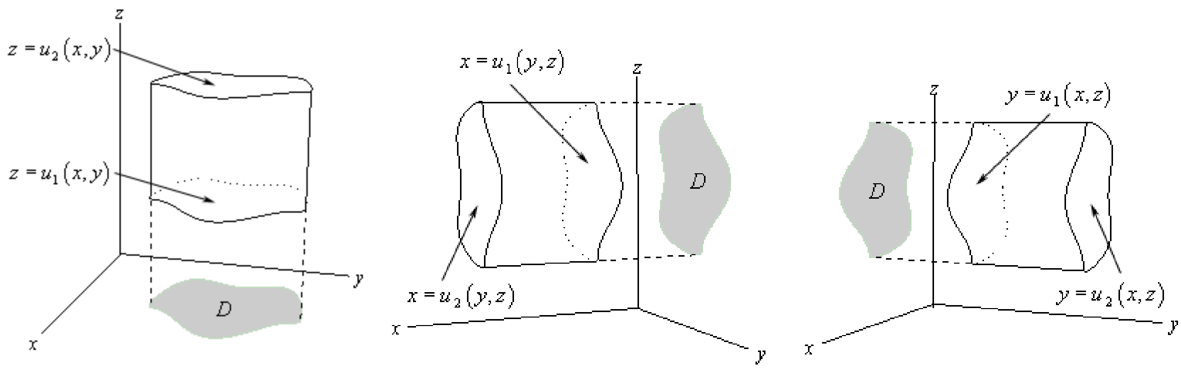


Figure 4.6.1: Types I, II, and III regions for triple integrals

Type I Region

A solid region E is of **Type I** if it lies between the graphs of two continuous functions of x and y:

$$E = \{(x, y, z) \mid (x, y) \in D,\, u_1(x, y) \leq z \leq u_2(x, y)\}$$

where D is the projection of E onto the xy-plane.
For such regions:

$$\iiint\limits_E f(x, y, z)\,\mathrm{d}V = \iint\limits_D \left[\int_{u_1(x,y)}^{u_2(x,y)} f(x, y, z)\,\mathrm{d}z\right] \mathrm{d}A$$

Type II Region

A solid region E is of **Type II** if it lies between the graphs of two continuous functions of y and z:

$$E = \{(x, y, z) \mid (y, z) \in D,\, u_1(y, z) \leq x \leq u_2(y, z)\}$$

where D is the projection of E onto the yz-plane.
For such regions:

$$\iiint\limits_E f(x, y, z)\,\mathrm{d}V = \iint\limits_D \left[\int_{u_1(y,z)}^{u_2(y,z)} f(x, y, z)\,\mathrm{d}x\right] \mathrm{d}A$$

Type III Region

A solid region E is of **Type III** if it lies between the graphs of two continuous functions of x and z:

$$E = \{(x, y, z) \mid (x, z) \in D,\, u_1(x, z) \leq y \leq u_2(x, z)\}$$

where D is the projection of E onto the xz-plane.
For such regions:

$$\iiint\limits_E f(x, y, z)\,\mathrm{d}V = \iint\limits_D \left[\int_{u_1(x,z)}^{u_2(x,z)} f(x, y, z)\,\mathrm{d}y\right] \mathrm{d}A$$

Example 4.10: Find the volume of the tetrahedron bounded by the plane $x+2y+z = 2$ and the coordinate planes.

This is a Type I region. First, we need to find the projection D onto the xy-plane. The plane intersects:

- x-axis at $(2, 0, 0)$ when $y = z = 0$
- y-axis at $(0, 1, 0)$ when $x = z = 0$
- z-axis at $(0, 0, 2)$ when $x = y = 0$

The projection D onto the xy-plane is the triangle with vertices $(0, 0)$, $(2, 0)$, and $(0, 1)$, bounded by:

- $x = 0$ (the y-axis)
- $y = 0$ (the x-axis)
- $x + 2y = 2$ (or $y = \frac{2-x}{2}$)

For a Type I region, z ranges from the lower surface $z = 0$ to the upper surface $z = 2 - x - 2y$.

The volume is:

$$
\begin{aligned}
V = \iiint\limits_E \mathrm{d}V &= \int_0^2 \int_0^{(2-x)/2} \int_0^{2-x-2y} \mathrm{d}z \, \mathrm{d}y \, \mathrm{d}x \\
&= \int_0^2 \int_0^{(2-x)/2} [z]_0^{2-x-2y} \, \mathrm{d}y \, \mathrm{d}x \\
&= \int_0^2 \int_0^{(2-x)/2} (2 - x - 2y) \, \mathrm{d}y \, \mathrm{d}x \\
&= \int_0^2 \left[(2 - x)y - y^2\right]_0^{(2-x)/2} \mathrm{d}x \\
&= \int_0^2 \left[(2 - x) \cdot \frac{2 - x}{2} - \left(\frac{2 - x}{2}\right)^2\right] \mathrm{d}x \\
&= \int_0^2 \left[\frac{(2 - x)^2}{2} - \frac{(2 - x)^2}{4}\right] \mathrm{d}x \\
&= \int_0^2 \frac{(2 - x)^2}{4} \, \mathrm{d}x \\
&= \frac{1}{4} \int_0^2 (2 - x)^2 \, \mathrm{d}x
\end{aligned}
$$

Let $u = 2 - x$, so $\mathrm{d}u = -\,\mathrm{d}x$. When $x = 0$, $u = 2$; when $x = 2$, $u = 0$:

$$
\begin{aligned}
V &= \frac{1}{4} \int_2^0 u^2 (-\,\mathrm{d}u) = \frac{1}{4} \int_0^2 u^2 \, \mathrm{d}u \\
&= \frac{1}{4} \left[\frac{u^3}{3}\right]_0^2 = \frac{1}{4} \cdot \frac{8}{3} = \frac{2}{3}
\end{aligned}
$$

Example 4.11: Evaluate $\iiint\limits_E z\,\mathrm{d}V$ where E is bounded by $y = 0$, $z = 0$, $y = x$, and $x + z = 1$ in the first octant.

This region is bounded by four planes. Let's use a Type II approach where we integrate with respect to x first.
The region can be described as:

- The projection onto the yz-plane is bounded by $z = 0$, $y = 0$, $y + z \leq 1$
- For each (y, z) in this projection, x ranges from y to $1 - z$

However, we need $y \leq 1 - z$ for the region to exist, which gives us $y \leq 1 - z$.
Setting up as Type I (integrating z first) is easier:

- Project onto the xy-plane: the region is bounded by $y = 0$, $y = x$, and $x \leq 1$
- For each (x, y), z ranges from 0 to $1 - x$

$$
\begin{aligned}
\iiint\limits_E z\,\mathrm{d}V &= \int_0^1 \int_0^x \int_0^{1-x} z\,\mathrm{d}z\,\mathrm{d}y\,\mathrm{d}x \\
&= \int_0^1 \int_0^x \left[\frac{z^2}{2}\right]_0^{1-x} \mathrm{d}y\,\mathrm{d}x \\
&= \int_0^1 \int_0^x \frac{(1-x)^2}{2}\,\mathrm{d}y\,\mathrm{d}x \\
&= \int_0^1 \frac{(1-x)^2}{2}[y]_0^x\,\mathrm{d}x \\
&= \int_0^1 \frac{x(1-x)^2}{2}\,\mathrm{d}x \\
&= \frac{1}{2}\int_0^1 x(1 - 2x + x^2)\,\mathrm{d}x \\
&= \frac{1}{2}\int_0^1 (x - 2x^2 + x^3)\,\mathrm{d}x \\
&= \frac{1}{2}\left[\frac{x^2}{2} - \frac{2x^3}{3} + \frac{x^4}{4}\right]_0^1 \\
&= \frac{1}{2}\left(\frac{1}{2} - \frac{2}{3} + \frac{1}{4}\right) \\
&= \frac{1}{2}\left(\frac{6 - 8 + 3}{12}\right) = \frac{1}{2} \cdot \frac{1}{12} = \frac{1}{24}
\end{aligned}
$$

4.7 Triple Integrals in Cylindrical Coordinates

Cylindrical coordinates are an extension of polar coordinates into three dimensions, making them particularly useful for regions with circular symmetry about the z-axis.

Recall: Cylindrical Coordinates

The cylindrical coordinate system is defined by:

$$x = r\cos\theta, \quad y = r\sin\theta, \quad z = z$$

where $r \geq 0$ is the distance from the z-axis, θ is the angle from the positive x-axis, and z is the height.
The volume element in cylindrical coordinates is:

$$\mathrm{d}V = r\,\mathrm{d}z\,\mathrm{d}r\,\mathrm{d}\theta$$

Triple Integral in Cylindrical Coordinates

For a region E where $(x, y) \in D$ in the xy-plane and $u_1(x, y) \leq z \leq u_2(x, y)$, the triple integral becomes:

$$\iiint\limits_E f(x,y,z)\,\mathrm{d}V = \int_\alpha^\beta \int_{h_1(\theta)}^{h_2(\theta)} \int_{u_1(r\cos\theta, r\sin\theta)}^{u_2(r\cos\theta, r\sin\theta)} f(r\cos\theta, r\sin\theta, z)\cdot r\,\mathrm{d}z\,\mathrm{d}r\,\mathrm{d}\theta$$

Example 4.12: Evaluate $\iiint\limits_E y\,\mathrm{d}V$ where E is the region below $z = x + 2$, above the xy-plane, and between the cylinders $x^2 + y^2 = 1$ and $x^2 + y^2 = 4$.

First, convert the bounds to cylindrical coordinates:

- The plane $z = x + 2$ becomes $z = r\cos\theta + 2$
- Above the xy-plane means $z \geq 0$
- Between cylinders: $1 \leq r \leq 2$
- Full rotation: $0 \leq \theta \leq 2\pi$

Converting $y = r\sin\theta$:

$$\begin{aligned}
\iiint\limits_E y\,\mathrm{d}V &= \int_0^{2\pi} \int_1^2 \int_0^{r\cos\theta+2} r(r\sin\theta)\,\mathrm{d}z\,\mathrm{d}r\,\mathrm{d}\theta \\
&= \int_0^{2\pi} \int_1^2 r^2\sin\theta(r\cos\theta + 2)\,\mathrm{d}r\,\mathrm{d}\theta \\
&= \int_0^{2\pi} \int_1^2 \left(r^3\sin\theta\cos\theta + 2r^2\sin\theta\right)\mathrm{d}r\,\mathrm{d}\theta \\
&= \int_0^{2\pi} \int_1^2 \left(\frac{1}{2}r^3\sin(2\theta) + 2r^2\sin\theta\right)\mathrm{d}r\,\mathrm{d}\theta \\
&= \int_0^{2\pi} \left[\frac{1}{8}r^4\sin(2\theta) + \frac{2}{3}r^3\sin\theta\right]_1^2 \mathrm{d}\theta \\
&= \int_0^{2\pi} \left(\frac{15}{8}\sin(2\theta) + \frac{14}{3}\sin\theta\right)\mathrm{d}\theta \\
&= \left[-\frac{15}{16}\cos(2\theta) - \frac{14}{3}\cos\theta\right]_0^{2\pi} \\
&= 0
\end{aligned}$$

4.8 Triple Integrals in Spherical Coordinates

Spherical coordinates are particularly useful for regions with spherical symmetry, such as spheres, cones, and combinations thereof.

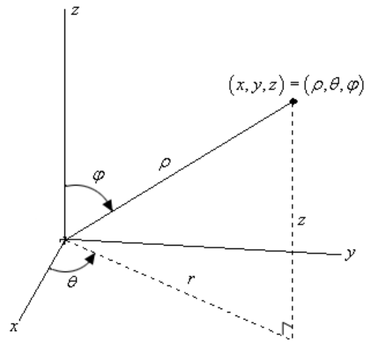


Figure 4.8.1: Spherical coordinates representation

Recall: Spherical Coordinates

The spherical coordinate system is defined by:

$$x = \rho \sin\varphi \cos\theta, \quad y = \rho \sin\varphi \sin\theta, \quad z = \rho\cos\varphi$$

$$x^2 + y^2 + z^2 = \rho^2$$

where:

- $\rho \geq 0$ is the distance from the origin
- θ is the angle in the xy-plane from the positive x-axis ($0 \leq \theta \leq 2\pi$)
- φ is the angle from the positive z-axis ($0 \leq \varphi \leq \pi$)

The volume element in spherical coordinates is:

$$\mathrm{d}V = \rho^2 \sin\varphi \,\mathrm{d}\rho\,\mathrm{d}\theta\,\mathrm{d}\varphi$$

Triple Integral in Spherical Coordinates

For a spherical wedge with $a \leq \rho \leq b$, $\alpha \leq \theta \leq \beta$, and $\delta \leq \varphi \leq \gamma$:

$$\iiint\limits_E f(x,y,z)\,\mathrm{d}V = \int_\delta^\gamma \int_\alpha^\beta \int_a^b f(\rho\sin\varphi\cos\theta, \rho\sin\varphi\sin\theta, \rho\cos\varphi)\cdot\rho^2\sin\varphi\,\mathrm{d}\rho\,\mathrm{d}\theta\,\mathrm{d}\varphi$$

Example 4.13: Evaluate $\iiint\limits_E 16z\,\mathrm{d}V$ where E is the upper half of the sphere $x^2 + y^2 + z^2 = 1$.

The upper half of the unit sphere has limits:

- $0 \leq \rho \leq 1$ (radius from 0 to 1)
- $0 \leq \theta \leq 2\pi$ (full rotation)
- $0 \leq \varphi \leq \frac{\pi}{2}$ (upper half, from z-axis to xy-plane)

Converting $z = \rho\cos\varphi$:

$$\begin{aligned}
\iiint\limits_E 16z\,\mathrm{d}V &= \int_0^{\pi/2}\int_0^{2\pi}\int_0^1 \rho^2\sin\varphi(16\rho\cos\varphi)\,\mathrm{d}\rho\,\mathrm{d}\theta\,\mathrm{d}\varphi \\
&= \int_0^{\pi/2}\int_0^{2\pi}\int_0^1 16\rho^3\sin\varphi\cos\varphi\,\mathrm{d}\rho\,\mathrm{d}\theta\,\mathrm{d}\varphi \\
&= \int_0^{\pi/2}\int_0^{2\pi}\int_0^1 8\rho^3\sin(2\varphi)\,\mathrm{d}\rho\,\mathrm{d}\theta\,\mathrm{d}\varphi \\
&= \int_0^{\pi/2}\int_0^{2\pi} 2\sin(2\varphi)\,\mathrm{d}\theta\,\mathrm{d}\varphi \\
&= \int_0^{\pi/2} 4\pi\sin(2\varphi)\,\mathrm{d}\varphi \\
&= \big[-2\pi\cos(2\varphi)\big]_0^{\pi/2} \\
&= -2\pi(-1) - (-2\pi)(1) = 4\pi
\end{aligned}$$

Example 4.14: Evaluate $\iiint\limits_E zx\,\mathrm{d}V$ where E is inside $x^2+y^2+z^2=4$ and the upward-pointing cone making angle $\frac{\pi}{3}$ with the negative z-axis, with $x \le 0$.

This region is an upside-down ice cream cone, cut in half with $x \le 0$ remaining.
Setting up the limits:

- $0 \le \rho \le 2$ (inside sphere of radius 2)
- $\frac{\pi}{2} \le \theta \le \frac{3\pi}{2}$ (left half-plane where $x \le 0$)
- The cone makes angle $\frac{\pi}{3}$ with negative z-axis, so from positive z-axis it's at angle $\pi - \frac{\pi}{3} = \frac{2\pi}{3}$, giving $\frac{2\pi}{3} \le \varphi \le \pi$

$$\begin{aligned}
\iiint\limits_E zx\,\mathrm{d}V &= \int_{2\pi/3}^{\pi}\int_{\pi/2}^{3\pi/2}\int_0^2 (\rho\cos\varphi)(\rho\sin\varphi\cos\theta)\rho^2\sin\varphi\,\mathrm{d}\rho\,\mathrm{d}\theta\,\mathrm{d}\varphi \\
&= \int_{2\pi/3}^{\pi}\int_{\pi/2}^{3\pi/2}\int_0^2 \rho^4\cos\varphi\sin^2\varphi\cos\theta\,\mathrm{d}\rho\,\mathrm{d}\theta\,\mathrm{d}\varphi \\
&= \int_{2\pi/3}^{\pi}\int_{\pi/2}^{3\pi/2} \frac{32}{5}\cos\varphi\sin^2\varphi\cos\theta\,\mathrm{d}\theta\,\mathrm{d}\varphi \\
&= \int_{2\pi/3}^{\pi} \frac{32}{5}\cos\varphi\sin^2\varphi[\sin\theta]_{\pi/2}^{3\pi/2}\,\mathrm{d}\varphi \\
&= \int_{2\pi/3}^{\pi} \frac{32}{5}\cos\varphi\sin^2\varphi(-1-1)\,\mathrm{d}\varphi \\
&= -\frac{64}{5}\int_{2\pi/3}^{\pi}\cos\varphi\sin^2\varphi\,\mathrm{d}\varphi \\
&= -\frac{64}{5}\left[\frac{\sin^3\varphi}{3}\right]_{2\pi/3}^{\pi} = -\frac{64}{15}\left(0 - \frac{3\sqrt{3}}{8}\right) = \frac{8\sqrt{3}}{5}
\end{aligned}$$

4.9 Change of Variables

Just as we had substitution in Calculus I to transform integrals into simpler forms, we can perform similar transformations for double and triple integrals.

Transformations and Regions

When performing a change of variables, we use a **transformation** to convert from one coordinate system to another. Typically, we start with a region R in xy-coordinates and transform it into a region S in uv-coordinates.

Definition 4.9.1: Transformation

A **transformation** $T : S \to R$ is given by equations:

$$x = g(u, v), \quad y = h(u, v)$$

that map points (u, v) in region S to points (x, y) in region R.

Example 4.15: An ellipse R given by $x^2 + \frac{y^2}{36} = 1$ transforms into a circle under $x = \frac{u}{2}$, $y = 3v$.

Applying the transformation:

$$\begin{aligned}
\left(\frac{u}{2}\right)^2 + \frac{(3v)^2}{36} &= 1 \\
\frac{u^2}{4} + \frac{9v^2}{36} &= 1 \\
\frac{u^2}{4} + \frac{v^2}{4} &= 1 \\
u^2 + v^2 &= 4
\end{aligned}$$

The ellipse in the xy-plane transforms into a disk of radius 2 in the uv-plane.

The Jacobian

The key to changing variables in multiple integrals is the **Jacobian**, which measures how area or volume scales under the transformation.

Definition 4.9.2: Jacobian for Double Integrals

The **Jacobian** of the transformation $x = g(u, v)$, $y = h(u, v)$ is:

$$\frac{\partial(x, y)}{\partial(u, v)} = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{vmatrix} = \frac{\partial x}{\partial u}\frac{\partial y}{\partial v} - \frac{\partial x}{\partial v}\frac{\partial y}{\partial u}$$

***Theorem 4.9.3* (Change of Variables Formula for Double Integrals)*:** *If $T : S \to R$ is a transformation given by $x = g(u, v)$ and $y = h(u, v)$, then:*

$$\iint\limits_R f(x, y)\,\mathrm{d}A = \iint\limits_S f(g(u, v), h(u, v)) \left|\frac{\partial(x, y)}{\partial(u, v)}\right| \mathrm{d}\overline{A}$$

where $\mathrm{d}\overline{A} = \mathrm{d}u\,\mathrm{d}v$ in the transformed integral.
Equivalently:

$$\mathrm{d}A = \left|\frac{\partial(x, y)}{\partial(u, v)}\right| \mathrm{d}\overline{A}$$

Example 4.16: Verify $\mathrm{d}A = r\,\mathrm{d}r\,\mathrm{d}\theta$ for polar coordinates.
The polar coordinate transformation is:

$$x = r\cos\theta, \quad y = r\sin\theta$$

Compute the Jacobian:

$$\begin{aligned}\frac{\partial(x, y)}{\partial(r, \theta)} &= \begin{vmatrix} \frac{\partial x}{\partial r} & \frac{\partial x}{\partial \theta} \\ \frac{\partial y}{\partial r} & \frac{\partial y}{\partial \theta} \end{vmatrix} = \begin{vmatrix} \cos\theta & -r\sin\theta \\ \sin\theta & r\cos\theta \end{vmatrix} \\ &= r\cos^2\theta - (-r\sin^2\theta) \\ &= r\end{aligned}$$

Therefore:

$$\mathrm{d}A = \left|\frac{\partial(x, y)}{\partial(r, \theta)}\right| \mathrm{d}r\,\mathrm{d}\theta = |r|\,\mathrm{d}r\,\mathrm{d}\theta = r\,\mathrm{d}r\,\mathrm{d}\theta$$

since $r \geq 0$ in polar coordinates.

Example 4.17: Evaluate $\iint\limits_R (x+y)\,\mathrm{d}A$ where R is the trapezoidal region with vertices $(0, 0)$, $(5, 0)$, $\left(\frac{5}{2}, \frac{5}{2}\right)$, $\left(\frac{5}{2}, -\frac{5}{2}\right)$ using transformation $x = 2u + 3v$, $y = 2u - 3v$.
First, find the equations of the sides of R:

- From $(0, 0)$ to $\left(\frac{5}{2}, \frac{5}{2}\right)$: $y = x$
- From $(0, 0)$ to $\left(\frac{5}{2}, -\frac{5}{2}\right)$: $y = -x$
- From $\left(\frac{5}{2}, \frac{5}{2}\right)$ to $(5, 0)$: $y = -x + 5$
- From $\left(\frac{5}{2}, -\frac{5}{2}\right)$ to $(5, 0)$: $y = x - 5$

Transform each edge:

- $y = x$: $2u - 3v = 2u + 3v \Rightarrow v = 0$
- $y = -x$: $2u - 3v = -(2u + 3v) \Rightarrow u = 0$
- $y = -x + 5$: $2u - 3v = -(2u + 3v) + 5 \Rightarrow u = \frac{5}{4}$

- $y = x - 5$: $2u - 3v = (2u + 3v) - 5 \Rightarrow v = \frac{5}{6}$

The region S is the rectangle: $0 \leq u \leq \frac{5}{4}$, $0 \leq v \leq \frac{5}{6}$.
Compute the Jacobian:

$$\frac{\partial(x,y)}{\partial(u,v)} = \begin{vmatrix} 2 & 3 \\ 2 & -3 \end{vmatrix} = -6 - 6 = -12$$

Evaluate the integral:

$$\begin{aligned}\iint\limits_R (x+y)\,\mathrm{d}A &= \int_0^{5/6}\int_0^{5/4} [(2u+3v)+(2u-3v)]|-12|\,\mathrm{d}u\,\mathrm{d}v \\ &= \int_0^{5/6}\int_0^{5/4} 48u\,\mathrm{d}u\,\mathrm{d}v \\ &= \int_0^{5/6} \left[24u^2\right]_0^{5/4} \mathrm{d}v \\ &= \int_0^{5/6} \frac{75}{2}\,\mathrm{d}v \\ &= \frac{75}{2}\cdot\frac{5}{6} = \frac{125}{4}\end{aligned}$$

Jacobian for Triple Integrals

For triple integrals with transformation $x = g(u,v,w)$, $y = h(u,v,w)$, $z = k(u,v,w)$:

Definition 4.9.4: Jacobian for Triple Integrals

The **Jacobian** for a transformation from (u,v,w) to (x,y,z) is:

$$\frac{\partial(x,y,z)}{\partial(u,v,w)} = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} & \frac{\partial x}{\partial w} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} & \frac{\partial y}{\partial w} \\ \frac{\partial z}{\partial u} & \frac{\partial z}{\partial v} & \frac{\partial z}{\partial w} \end{vmatrix}$$

Theorem 4.9.5* (Change of Variables Formula for Triple Integrals)*: *If $T : S \to R$ transforms region S to region R, then:*

$$\iiint\limits_R f(x,y,z)\,\mathrm{d}V = \iiint\limits_S f(g(u,v,w),h(u,v,w),k(u,v,w))\left|\frac{\partial(x,y,z)}{\partial(u,v,w)}\right|\mathrm{d}\overline{V}$$

where $\mathrm{d}\overline{V} = \mathrm{d}u\,\mathrm{d}v\,\mathrm{d}w$ in the transformed integral.
Equivalently:

$$\mathrm{d}V = \left|\frac{\partial(x,y,z)}{\partial(u,v,w)}\right|\mathrm{d}\overline{V}$$

Example 4.18: Verify $\mathrm{d}V = \rho^2 \sin\varphi \,\mathrm{d}\rho\,\mathrm{d}\theta\,\mathrm{d}\varphi$ for spherical coordinates.
The spherical coordinate transformation is:

$$x = \rho \sin\varphi \cos\theta, \quad y = \rho \sin\varphi \sin\theta, \quad z = \rho \cos\varphi$$

Compute the Jacobian:

$$\frac{\partial(x, y, z)}{\partial(\rho, \theta, \varphi)} = \begin{vmatrix} \sin\varphi\cos\theta & -\rho\sin\varphi\sin\theta & \rho\cos\varphi\cos\theta \\ \sin\varphi\sin\theta & \rho\sin\varphi\cos\theta & \rho\cos\varphi\sin\theta \\ \cos\varphi & 0 & -\rho\sin\varphi \end{vmatrix}$$

Expanding along the third row:

$$\begin{aligned}
&= \cos\varphi \begin{vmatrix} -\rho\sin\varphi\sin\theta & \rho\cos\varphi\cos\theta \\ \rho\sin\varphi\cos\theta & \rho\cos\varphi\sin\theta \end{vmatrix} \\
&\quad - 0 + (-\rho\sin\varphi) \begin{vmatrix} \sin\varphi\cos\theta & -\rho\sin\varphi\sin\theta \\ \sin\varphi\sin\theta & \rho\sin\varphi\cos\theta \end{vmatrix} \\
&= \cos\varphi(-\rho^2\sin\varphi\cos\varphi\sin^2\theta - \rho^2\sin\varphi\cos\varphi\cos^2\theta) \\
&\quad + (-\rho\sin\varphi)(\rho\sin^2\varphi\cos^2\theta + \rho\sin^2\varphi\sin^2\theta) \\
&= -\rho^2\sin\varphi\cos^2\varphi - \rho^2\sin^3\varphi \\
&= -\rho^2\sin\varphi(\cos^2\varphi + \sin^2\varphi) \\
&= -\rho^2\sin\varphi
\end{aligned}$$

Therefore:

$$\mathrm{d}V = \left|-\rho^2\sin\varphi\right| \mathrm{d}\rho\,\mathrm{d}\theta\,\mathrm{d}\varphi = \rho^2\sin\varphi\,\mathrm{d}\rho\,\mathrm{d}\theta\,\mathrm{d}\varphi$$

since $\sin\varphi \geq 0$ for $0 \leq \varphi \leq \pi$.

4.9.1 Surface Area

Surface Area

The surface area S of a surface $z = f(x, y)$ over a region R in the xy-plane is given by:

$$S = \iint_R \sqrt{1 + \left(\frac{\partial f}{\partial x}\right)^2 + \left(\frac{\partial f}{\partial y}\right)^2}\,\mathrm{d}A$$

Example 4.19: Determine the surface area of the part of $z = xy$ that lies in the cylinder given by $x^2 + y^2 = 1$.
First, compute the partial derivatives:

$$\frac{\partial f}{\partial x} = y, \quad \frac{\partial f}{\partial y} = x$$

Given that D is a disk of radius 1, we do this integral in polar coordinates.

$$\begin{aligned}
S &= \iint\limits_D \sqrt{x^2+y^2+1}\,\mathrm{d}A \\
&= \int_0^{2\pi}\int_0^1 \sqrt{r^2+1}\cdot r\,\mathrm{d}r\,\mathrm{d}\theta \\
&= \int_0^{2\pi}\left[\frac{1}{3}(r^2+1)^{3/2}\right]_0^1 \mathrm{d}\theta \\
&= \int_0^{2\pi}\frac{1}{3}(2^{3/2}-1)\,\mathrm{d}\theta \\
&= \frac{2\pi}{3}(2^{3/2}-1)
\end{aligned}$$

5 Line Integrals

5.1 Vector Fields

Definition 5.1.1: Vector Field

A vector field on two or three dimensional space is a function $\vec{F}$ that assigns to each point (x, y) or (x, y, z) a two or three dimensional vector given by $\vec{F}(x, y)$ or $\vec{F}(x, y, z)$.

We can write vector fields in component form as follows. In two dimensions:

$$\vec{F}(x, y) = P(x, y)\,\vec{i} + Q(x, y)\,\vec{j}$$

In three dimensions:

$$\vec{F}(x, y, z) = P(x, y, z)\,\vec{i} + Q(x, y, z)\,\vec{j} + R(x, y, z)\,\vec{k}$$

where P, Q, and R are scalar functions called the component functions of the vector field.

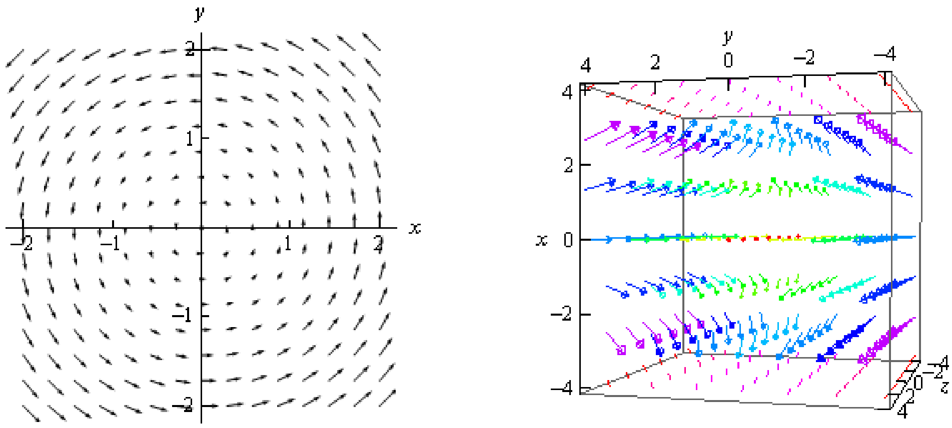
Definition 5.1.2: Gradient Vector Fields

If f is a scalar function, then its gradient ∇f is a vector field. We call ∇f a **gradient vector field**.
In two dimensions:

$$\nabla f = \langle f_x, f_y \rangle = \frac{\partial f}{\partial x}\,\vec{i} + \frac{\partial f}{\partial y}\,\vec{j}$$

In three dimensions:

$$\nabla f = \langle f_x, f_y, f_z \rangle = \frac{\partial f}{\partial x}\,\vec{i} + \frac{\partial f}{\partial y}\,\vec{j} + \frac{\partial f}{\partial z}\,\vec{k}$$



(a) $\vec{F}(x, y) = -y\hat{i} + x\hat{j}$ (b) $\vec{F}(x, y, z) = 2x\hat{i} - 2y\hat{j} - 2x\hat{k}$

Figure 5.1.1: Examples of vector fields in (a) two dimensions and (b) three dimensions.

Definition 5.1.3: Conservative Vector Field

A vector field $\vec{F}$ is called **conservative** if there exists a scalar function f such that $\vec{F} = \nabla f$. The function f is called a **potential function** for the vector field $\vec{F}$.
In other words, a vector field is conservative if it can be expressed as the gradient of some scalar function.

For example, the vector field $\vec{F} = y\hat{i} + x\hat{j}$ is a conservative vector field with a potential function of $f(x, y) = xy$ because $\nabla f = \langle y, x \rangle$.

5.2 Line Integrals wrt Arc Length

In this section we will look at line integrals with respect to arc length.

Definition 5.2.1: Smooth Curve

For a curve C defined by the parametric equations

$$x = x(t) \quad y = y(t) \quad a \le t \le b$$

$$\text{or } \vec{r}(t) = x(t)\hat{i} + y(t)\hat{j} \quad a \le t \le b$$

the curve is called **smooth** if $\vec{r}'(t)$ is continuous and $\vec{r}'(t) \neq 0$ for all t.

Definition 5.2.2: Line Integral

The **line integral** of $f(x, y)$ along a smooth curve C is given by

$$\int_C f(x, y)\,\mathrm{d}s$$

Here, $\mathrm{d}s$ is used to denote a small segment of arc length along the curve C. For this reason, this is sometimes called the **line integral of f with respect to arc length**.
Recall that $\mathrm{d}s = \|\vec{r}'(t)\|\,\mathrm{d}t$ where $\vec{r}(t) = \langle x(t), y(t)\rangle$ is the position vector for the curve. Therefore:

$$\mathrm{d}s = \|\vec{r}'(t)\|\,\mathrm{d}t = \sqrt{\left(\frac{\mathrm{d}x}{\mathrm{d}t}\right)^2 + \left(\frac{\mathrm{d}y}{\mathrm{d}t}\right)^2}\,\mathrm{d}t$$

Line Integral with Respect to Arc Length

The line integral of f along C with respect to arc length is:

$$\int_C f(x, y)\,\mathrm{d}s = \int_a^b f(x(t), y(t))\sqrt{\left(\frac{\mathrm{d}x}{\mathrm{d}t}\right)^2 + \left(\frac{\mathrm{d}y}{\mathrm{d}t}\right)^2}\,\mathrm{d}t$$

For three-dimensional curves given by $\vec{r}(t) = \langle x(t), y(t), z(t)\rangle$ where $a \le t \le b$:

$$\int_C f(x, y, z)\,\mathrm{d}s = \int_a^b f(x(t), y(t), z(t))\sqrt{\left(\frac{\mathrm{d}x}{\mathrm{d}t}\right)^2 + \left(\frac{\mathrm{d}y}{\mathrm{d}t}\right)^2 + \left(\frac{\mathrm{d}z}{\mathrm{d}t}\right)^2}\,\mathrm{d}t$$

Example 5.1: Evaluate $\int_C xyz\,\mathrm{d}s$ where C is the curve given by $x = t$, $y = t^2$, $z = t^3$ for $0 \le t \le 1$.

We have $x(t) = t$, $y(t) = t^2$, and $z(t) = t^3$.
First, compute $\mathrm{d}s$:

$$\frac{\mathrm{d}x}{\mathrm{d}t} = 1 \quad \frac{\mathrm{d}y}{\mathrm{d}t} = 2t \quad \frac{\mathrm{d}z}{\mathrm{d}t} = 3t^2$$

$$\therefore \mathrm{d}s = \sqrt{1 + 4t^2 + 9t^4}\,\mathrm{d}t$$

Now evaluate the integral:

$$\int_C xyz\,\mathrm{d}s = \int_0^1 (t)(t^2)(t^3)\sqrt{1 + 4t^2 + 9t^4}\,\mathrm{d}t = \int_0^1 t^6\sqrt{1 + 4t^2 + 9t^4}\,\mathrm{d}t$$

Notice that $1 + 4t^2 + 9t^4 = (1 + 3t^2)^2$, so:

$$\begin{aligned}
\int_C xyz\,\mathrm{d}s &= \int_0^1 t^6(1 + 3t^2)\,\mathrm{d}t \\
&= \int_0^1 (t^6 + 3t^8)\,\mathrm{d}t \\
&= \left.\frac{t^7}{7}\right|_0^1 + \left.\frac{3t^9}{9}\right|_0^1 \\
&= \frac{1}{7} + \frac{1}{3} = \frac{10}{21}
\end{aligned}$$

Definition 5.2.3: Piecewise Smooth Curves

If a curve C is not smooth, but can be broken into a finite number of smooth pieces $C_1, C_2, \dots, C_n$, then C is called **piecewise smooth**, and the line integral along C is defined as:

$$\int_C f\,\mathrm{d}s = \sum_{i=1}^{n} \int_{C_i} f\,\mathrm{d}s$$

Direction Independence

For line integrals with respect to arc length, changing the direction of the curve does **not** change the value of the integral. That is:

$$\int_C f\,\mathrm{d}s = \int_{-C} f\,\mathrm{d}s$$

where $-C$ denotes the curve C traversed in the opposite direction.

5.3 Line Integrals wrt x, y, z

Line Integral with Respect to x and y

For a two-dimensional curve C with parameterization $x = x(t)$, $y = y(t)$, $a \le t \le b$:
The **line integral of f with respect to x** is:

$$\int_C f(x, y)\,\mathrm{d}x = \int_a^b f(x(t), y(t))\frac{\mathrm{d}x}{\mathrm{d}t}\,\mathrm{d}t$$

The **line integral of f with respect to y** is:

$$\int_C f(x, y)\,\mathrm{d}y = \int_a^b f(x(t), y(t))\frac{\mathrm{d}y}{\mathrm{d}t}\,\mathrm{d}t$$

Note:-

These integrals often appear together, so we use the following shorthand notation:

$$\int_C P\,\mathrm{d}x + Q\,\mathrm{d}y = \int_C P(x, y)\,\mathrm{d}x + \int_C Q(x, y)\,\mathrm{d}y$$

Example 5.2: Evaluate $\int\limits_C \sin(\pi y)\,\mathrm{d}y + yx^2\,\mathrm{d}x$ where C is the line segment from $(0,2)$ to $(1,4)$.

First, parameterize the curve:

$$\vec{r}(t) = (1-t)\langle 0,2\rangle + t\langle 1,4\rangle = \langle t, 2+2t\rangle, \quad 0 \le t \le 1$$

So $x(t) = t$, $y(t) = 2+2t$, and:

$$\frac{\mathrm{d}x}{\mathrm{d}t} = 1 \quad \frac{\mathrm{d}y}{\mathrm{d}t} = 2$$

Now evaluate the line integral:

$$\begin{aligned}
\int\limits_C \sin(\pi y)\,\mathrm{d}y + yx^2\,\mathrm{d}x &= \int_0^1 \sin\big(\pi(2+2t)\big)(2)\,\mathrm{d}t + \int_0^1 (2+2t)(t)^2(1)\,\mathrm{d}t \\
&= \int_0^1 2\sin(2\pi+2\pi t)\,\mathrm{d}t + \int_0^1 (2t^2+2t^3)\,\mathrm{d}t \\
&= \left[-\frac{1}{\pi}\cos(2\pi+2\pi t)\right]_0^1 + \left[\frac{2t^3}{3} + \frac{t^4}{2}\right]_0^1 \\
&= -\frac{1}{\pi}(\cos(4\pi) - \cos(2\pi)) + \frac{2}{3} + \frac{1}{2} \\
&= -\frac{1}{\pi}(1-1) + \frac{4+3}{6} \\
&= 0 + \frac{7}{6} = \frac{7}{6}
\end{aligned}$$

Note:-

For three-dimensional curves C with parameterization $x = x(t)$, $y = y(t)$, $z = z(t)$, $a \le t \le b$:

$$\int\limits_C f(x,y,z)\,\mathrm{d}x = \int_a^b f(x(t),y(t),z(t))\frac{\mathrm{d}x}{\mathrm{d}t}\,\mathrm{d}t$$

$$\int\limits_C f(x,y,z)\,\mathrm{d}y = \int_a^b f(x(t),y(t),z(t))\frac{\mathrm{d}y}{\mathrm{d}t}\,\mathrm{d}t$$

$$\int\limits_C f(x,y,z)\,\mathrm{d}z = \int_a^b f(x(t),y(t),z(t))\frac{\mathrm{d}z}{\mathrm{d}t}\,\mathrm{d}t$$

Note:-

The shorthand notation for the combined form is:

$$\int\limits_C P\,\mathrm{d}x + Q\,\mathrm{d}y + R\,\mathrm{d}z = \int\limits_C P(x,y,z)\,\mathrm{d}x + \int\limits_C Q(x,y,z)\,\mathrm{d}y + \int\limits_C R(x,y,z)\,\mathrm{d}z$$

Example 5.3: Evaluate $\int\limits_C y\,\mathrm{d}x + x\,\mathrm{d}y + z\,\mathrm{d}z$ where C is given by $x = \cos t$, $y = \sin t$, $z = t^2$ for $0 \le t \le 2\pi$.

We have $x(t) = \cos t$, $y(t) = \sin t$, and $z(t) = t^2$.

Compute the derivatives:

$$\frac{\mathrm{d}x}{\mathrm{d}t} = -\sin t \quad \frac{\mathrm{d}y}{\mathrm{d}t} = \cos t \quad \frac{\mathrm{d}z}{\mathrm{d}t} = 2t$$

Evaluate the line integral:

$$\begin{aligned}
\int_C y\,\mathrm{d}x + x\,\mathrm{d}y + z\,\mathrm{d}z &= \int_0^{2\pi} \sin t(-\sin t)\,\mathrm{d}t + \int_0^{2\pi} \cos t(\cos t)\,\mathrm{d}t + \int_0^{2\pi} t^2(2t)\,\mathrm{d}t \\
&= \int_0^{2\pi} (-\sin^2 t + \cos^2 t + 2t^3)\,\mathrm{d}t \\
&= \int_0^{2\pi} \cos(2t)\,\mathrm{d}t + \int_0^{2\pi} 2t^3\,\mathrm{d}t \\
&= \frac{1}{2}\sin(2t)\Big|_0^{2\pi} + \frac{t^4}{2}\Big|_0^{2\pi} \\
&= 0 + \frac{(2\pi)^4}{2} = 8\pi^4
\end{aligned}$$

Direction Dependence

For line integrals with respect to x, y, and/or z, changing the direction of the curve **does** change the value of the integral by a factor of -1. That is:

$$\int_{-C} f(x,y)\,\mathrm{d}x = -\int_C f(x,y)\,\mathrm{d}x \quad \text{and} \quad \int_{-C} f(x,y)\,\mathrm{d}y = -\int_C f(x,y)\,\mathrm{d}y$$

For the combined form:

$$\int_{-C} P\,\mathrm{d}x + Q\,\mathrm{d}y = -\int_C P\,\mathrm{d}x + Q\,\mathrm{d}y$$

5.4 Line Integrals of Vector Fields

Let a three-dimensional vector field be

$$\vec{F}(x,y,z) = P(x,y,z)\,\vec{i} + Q(x,y,z)\,\vec{j} + R(x,y,z)\,\vec{k}$$

and let C a three-dimensional smooth curve be given by:

$$\vec{r}(t) = x(t)\,\vec{i} + y(t)\,\vec{j} + z(t)\,\vec{k}, \quad a \le t \le b$$

Definition 5.4.1: Line Integral of a Vector Field

The **line integral of $\vec{F}$ along** C is:

$$\int_C \vec{F}\cdot \mathrm{d}\vec{r} = \int_a^b \vec{F}(\vec{r}(t))\cdot \vec{r}'(t)\,\mathrm{d}t$$

Note:

$$\vec{F}(\vec{r}(t)) = \vec{F}(x(t), y(t), z(t))$$

Alternative Notation

We can also write line integrals of vector fields using the unit tangent vector $\vec{T}$:

$$\int\limits_C \vec{F} \cdot \mathrm{d}\vec{r} = \int\limits_C \vec{F} \cdot \vec{T}\,\mathrm{d}s$$

where $\vec{T}(t) = \dfrac{\vec{r}'(t)}{\|\vec{r}'(t)\|}$ is the unit tangent vector.

Relationship to Line Integrals with Respect to Coordinates

Line integrals of vector fields are related to line integrals with respect to x, y, and z:

$$\int\limits_C \vec{F} \cdot \mathrm{d}\vec{r} = \int\limits_C P\,\mathrm{d}x + Q\,\mathrm{d}y + R\,\mathrm{d}z$$

To see this, note that:

$$\begin{aligned}
\int\limits_C \vec{F} \cdot \mathrm{d}\vec{r} &= \int_a^b (P\,\vec{i} + Q\,\vec{j} + R\,\vec{k}) \cdot (x'\,\vec{i} + y'\,\vec{j} + z'\,\vec{k})\,\mathrm{d}t \\
&= \int_a^b (Px' + Qy' + Rz')\,\mathrm{d}t \\
&= \int_a^b Px'\,\mathrm{d}t + \int_a^b Qy'\,\mathrm{d}t + \int_a^b Rz'\,\mathrm{d}t \\
&= \int\limits_C P\,\mathrm{d}x + \int\limits_C Q\,\mathrm{d}y + \int\limits_C R\,\mathrm{d}z
\end{aligned}$$

Example 5.4: Evaluate $\int\limits_C \vec{F} \cdot \mathrm{d}\vec{r}$ where $\vec{F}(x,y,z) = 8x^2yz\,\vec{i} + 5z\,\vec{j} - 4xy\,\vec{k}$ and C is given by $\vec{r}(t) = t\,\vec{i} + t^2\,\vec{j} + t^3\,\vec{k}$ for $0 \le t \le 1$.

First, evaluate the vector field along the curve:

$$\vec{F}(\vec{r}(t)) = 8t^2(t^2)(t^3)\,\vec{i} + 5t^3\,\vec{j} - 4t(t^2)\,\vec{k} = 8t^7\,\vec{i} + 5t^3\,\vec{j} - 4t^3\,\vec{k}$$

Next, compute the derivative of the parameterization:

$$\vec{r}'(t) = \vec{i} + 2t\,\vec{j} + 3t^2\,\vec{k}$$

Now compute the dot product:

$$\begin{aligned}
\vec{F}(\vec{r}(t)) \cdot \vec{r}'(t) &= (8t^7)(1) + (5t^3)(2t) + (-4t^3)(3t^2) \\
&= 8t^7 + 10t^4 - 12t^5
\end{aligned}$$

Finally, evaluate the integral:

$$\begin{aligned}
\int\limits_C \vec{F} \cdot \mathrm{d}\vec{r} &= \int_0^1 (8t^7 + 10t^4 - 12t^5)\,\mathrm{d}t \\
&= \left[t^8 + 2t^5 - 2t^6\right]_0^1 \\
&= 1 + 2 - 2 = 1
\end{aligned}$$

Example 5.5: Evaluate $\int\limits_C \vec{F}\cdot \mathrm{d}\vec{r}$ where $\vec{F}(x,y,z) = xz\,\vec{i} - yz\,\vec{k}$ and C is the line segment from $(-1,2,0)$ to $(3,0,1)$.

First, parameterize the line segment:

$$\begin{aligned}\vec{r}(t) &= (1-t)\langle -1,2,0\rangle + t\langle 3,0,1\rangle \\ &= \langle 4t-1, 2-2t, t\rangle, \quad 0\le t\le 1\end{aligned}$$

Evaluate the vector field along the curve:

$$\begin{aligned}\vec{F}(\vec{r}(t)) &= (4t-1)(t)\,\vec{i} - (2-2t)(t)\,\vec{k} \\ &= (4t^2-t)\,\vec{i} - (2t-2t^2)\,\vec{k}\end{aligned}$$

Compute the derivative:

$$\vec{r}'(t) = \langle 4,-2,1\rangle$$

Compute the dot product:

$$\begin{aligned}\vec{F}(\vec{r}(t))\cdot\vec{r}'(t) &= (4t^2-t)(4) + 0(-2) + (-2t+2t^2)(1) \\ &= 16t^2 - 4t - 2t + 2t^2 \\ &= 18t^2 - 6t\end{aligned}$$

Evaluate the integral:

$$\begin{aligned}\int\limits_C \vec{F}\cdot\mathrm{d}\vec{r} &= \int_0^1 (18t^2-6t)\,\mathrm{d}t \\ &= \left[6t^3 - 3t^2\right]_0^1 \\ &= 6-3 = 3\end{aligned}$$

Work Interpretation

In physics, if $\vec{F}$ represents a force field and C is the path taken by an object, then $\int\limits_C \vec{F}\cdot\mathrm{d}\vec{r}$ represents the **work** done by the force field in moving the object along the path C.

Direction Dependence for Vector Fields

As with line integrals with respect to coordinates, reversing the direction of the curve changes the sign of the integral:

$$\int\limits_{-C} \vec{F}\cdot\mathrm{d}\vec{r} = -\int\limits_C \vec{F}\cdot\mathrm{d}\vec{r}$$

Note:-

We can write vector fields in component form as follows. In two dimensions:

$$\vec{F}(x,y) = P(x,y)\,\vec{i} + Q(x,y)\,\vec{j}$$

In three dimensions:

$$\vec{F}(x,y,z) = P(x,y,z)\,\vec{i} + Q(x,y,z)\,\vec{j} + R(x,y,z)\,\vec{k}$$

where P, Q, and R are scalar functions called the component functions of the vector field.

Example 5.6: Let $\vec{F}(x,y) = x^2\,\vec{i} + (2x+y)\,\vec{j}$. Evaluate $\vec{F}(1,2)$.

We simply substitute the point into the component functions:

$$\begin{aligned}\vec{F}(1,2) &= (1)^2\,\vec{i} + (2(1)+2)\,\vec{j} \\ &= \vec{i} + 4\,\vec{j}\end{aligned}$$

Gradient Vector Fields

If f is a scalar function, then its gradient ∇f is a vector field. We call ∇f a **gradient vector field**. In two dimensions:

$$\nabla f = \frac{\partial f}{\partial x}\,\vec{i} + \frac{\partial f}{\partial y}\,\vec{j}$$

In three dimensions:

$$\nabla f = \frac{\partial f}{\partial x}\,\vec{i} + \frac{\partial f}{\partial y}\,\vec{j} + \frac{\partial f}{\partial z}\,\vec{k}$$

5.5 Fundamental Theorem for Line Integrals

Theorem 5.5.1* (Fundamental Theorem for Line Integrals)*: *Let C be a smooth curve given by $\vec{r}(t), a <= t <= b$. Also, let f be a differentiable scalar function whose gradient vector field ∇f is continuous on C. Then:*

$$\int\limits_C \nabla f \cdot \mathrm{d}\vec{r} = f(\vec{r}(b)) - f(\vec{r}(a))$$

Proof:

By definition of the line integral of a vector field, we have:

$$\begin{aligned}\int\limits_C \nabla f \cdot \mathrm{d}\vec{r} &= \int_a^b \nabla f(\vec{r}(t)) \cdot \vec{r}'(t)\,\mathrm{d}t \\ &= \int_a^b \left(\frac{\partial f}{\partial x}\frac{\mathrm{d}x}{\mathrm{d}t} + \frac{\partial f}{\partial y}\frac{\mathrm{d}y}{\mathrm{d}t} + \frac{\partial f}{\partial z}\frac{\mathrm{d}z}{\mathrm{d}t}\right)\mathrm{d}t \\ &= \int_a^b \frac{\mathrm{d}}{\mathrm{d}t}[f(\vec{r}(t))]\,\mathrm{d}t \\ &= f(\vec{r}(b)) - f(\vec{r}(a)) \quad \square\end{aligned}$$

Example 5.7: Evaluate $\int\limits_C \nabla f \cdot \mathrm{d}\vec{r}$ where $f(x,y,z) = \cos(\pi x) + \sin(\pi y) - xyz$ and C is any path that starts at $\left(1, \frac{1}{2}, 2\right)$ and ends at $(2,1,-1)$.

By the Fundamental Theorem for Line Integrals, we have:

$$\int_C \nabla f \cdot d\vec{r} = f(2,1,-1) - f\left(1, \frac{1}{2}, 2\right)$$
$$= \left[\cos(2\pi) + \sin(\pi) - (2)(1)(-1)\right] - \left[\cos(\pi) + \sin\left(\frac{\pi}{2}\right) - (1)\left(\frac{1}{2}\right)(2)\right]$$
$$= [1 + 0 + 2] - [-1 + 1 - 1]$$
$$= 3 - (-1) = 4$$

Definition 5.5.2

Conservative Vector Field: A vector field $\vec{F}$ is called **conservative** if there exists a scalar function f such that $\vec{F} = \nabla f$. The function f is called a **potential function** for the vector field $\vec{F}$.

Path Independence: A line integral $\int_C \vec{F} \cdot d\vec{r}$ is called **path independent** if its value depends only on the endpoints of the curve C and not on the specific path taken between those points.

Closed Curve: A curve C is called **closed** if its initial point and terminal point are the same.

Simple Curve: A curve C in $\mathbb{R}^2$ or $\mathbb{R}^3$ is called **simple** if it does not intersect itself, except possibly at the endpoints if it is closed.

Open Region: A region D in $\mathbb{R}^2$ or $\mathbb{R}^3$ is called **open** if for every point in D, there exists a small neighborhood around that point which is entirely contained within D. Simply put, an open region does not include its boundary points.

Connected Region: A region D in $\mathbb{R}^2$ or $\mathbb{R}^3$ is called **connected** if any two points in D can be joined by a curve that lies entirely within D.

Simply Connected Region: A region D in $\mathbb{R}^2$ or $\mathbb{R}^3$ is called **simply connected** if any closed curve in D can be continuously contracted to a point without leaving D. Simply put, a simply connected region has no holes.

Note:-

1. $\int_C \nabla f \cdot d\vec{r}$ is independent of path.
2. If $\vec{F}$ is a conservative vector field, then $\int_C \vec{F} \cdot d\vec{r}$ is independent of path.
3. If $\vec{F}$ is a continuous vector field on an open connected region D and if $\int_C \vec{F} \cdot d\vec{r}$ is

independent of path (for any path in D), then $\vec{F}$ is a conservative vector field on D.

4. If $\displaystyle\int_C \vec{F} \cdot \mathrm{d}\vec{r}$ is independent of path, then $\displaystyle\int_C \vec{F} \cdot \mathrm{d}\vec{r} = 0$ for every closed path C.

5. If $\int_C \vec{F} \cdot \mathrm{d}\vec{r} = 0$ for every closed path C, then $\displaystyle\int_C \vec{F} \cdot \mathrm{d}\vec{r}$ is independent of path.

5.6 Conservative Vector Fields

Theorem 5.6.1* (Characterization of Conservative Vector Fields in $\mathbb{R}^2$)*: *Let $\vec{F}(x, y) = P(x, y)\,\vec{i} + Q(x, y)\,\vec{j}$ be a vector field defined on an **open** and **simply connected** region D of $\mathbb{R}^2$, where P and Q have continuous first partial derivatives in D. Then $\vec{F}$ is a conservative vector field if and only if:*

$$\frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x}$$

Finding Potential Functions in $\mathbb{R}^2$

Now that we have a way to determine if a vector field is conservative, we can find its potential function. We can say that

$$\nabla f = \frac{\partial f}{\partial x}\hat{i} + \frac{\partial f}{\partial y}\hat{j} = P\hat{i} + Q\hat{j} = \vec{F}$$

Equating components, we have:

$$\frac{\partial f}{\partial x} = P(x, y) \quad \text{and} \quad \frac{\partial f}{\partial y} = Q(x, y)$$

Then, to find f, we can integrate P with respect to x and Q with respect to y.

$$f(x, y) = \int P(x, y)\,\mathrm{d}x \quad \text{and} \quad f(x, y) = \int Q(x, y)\,\mathrm{d}y$$

Finding Potential Functions in $\mathbb{R}^3$

Similarly, for a vector field in $\mathbb{R}^3$:

$$\nabla f = \frac{\partial f}{\partial x}\hat{i} + \frac{\partial f}{\partial y}\hat{j} + \frac{\partial f}{\partial z}\hat{k} = P\hat{i} + Q\hat{j} + R\hat{k} = \vec{F}$$

Equating components, we have:

$$\frac{\partial f}{\partial x} = P(x, y, z) \quad \frac{\partial f}{\partial y} = Q(x, y, z) \quad \text{and} \quad \frac{\partial f}{\partial z} = R(x, y, z)$$

Then, to find f, we can integrate P with respect to x, Q with respect to y, and R with respect to z.

$$f(x, y, z) = \int P(x, y, z)\,\mathrm{d}x \quad f(x, y, z) = \int Q(x, y, z)\,\mathrm{d}y \quad f(x, y, z) = \int R(x, y, z)\,\mathrm{d}z$$

5.7 Green's Theorem

Green's Theorem establishes a relationship between certain line integrals on closed paths and double integrals over the region enclosed by the path.

Definition 5.7.1: Positively Oriented Curve

A simple closed curve C is **positively oriented** if it is traced out in a counter-clockwise direction. Equivalently, as we traverse the path following the positive orientation, the region D enclosed by C is always on the left.

***Theorem 5.7.2* (Green's Theorem)**: *Let C be a positively oriented, piecewise smooth, simple closed curve, and let D be the region bounded by C. If P and Q have continuous first order partial derivatives on an open region that contains D, then:*

$$\oint_C P\,\mathrm{d}x + Q\,\mathrm{d}y = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right)\mathrm{d}A$$

Notation for Closed Curves

When working with a line integral in which the path satisfies the conditions of Green's Theorem, we often denote the line integral as:

$$\oint_C P\,\mathrm{d}x + Q\,\mathrm{d}y$$

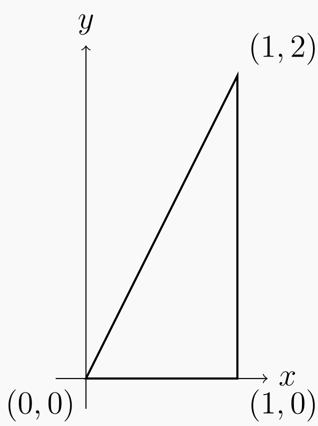
The circle on the integral sign indicates that C is a closed curve. Sometimes the curve C is viewed as the boundary of region D and is denoted as ∂D.

Example 5.8: Use Green's Theorem to evaluate $\oint_C xy\,\mathrm{d}x + x^2y^3\,\mathrm{d}y$ where C is the triangle with vertices $(0,0)$, $(1,0)$, $(1,2)$ with positive orientation.

First, sketch the region to verify that the conditions of Green's Theorem are met. The triangle is a simple closed curve traversed counter-clockwise, so it has positive orientation.

The region D can be described by:

$$0 \le x \le 1, \quad 0 \le y \le 2x$$



Next, identify P and Q:

$$P(x,y) = xy \qquad Q(x,y) = x^2y^3$$

Compute the partial derivatives:

$$\frac{\partial Q}{\partial x} = 2xy^3 \qquad \frac{\partial P}{\partial y} = x$$

Apply Green's Theorem:

$$\oint_C xy\,\mathrm{d}x + x^2y^3\,\mathrm{d}y = \iint_D (2xy^3 - x)\,\mathrm{d}A$$
$$= \int_0^1 \int_0^{2x} (2xy^3 - x)\,\mathrm{d}y\,\mathrm{d}x$$
$$= \int_0^1 \left[\frac{1}{2}xy^4 - xy\right]_0^{2x} \mathrm{d}x$$
$$= \int_0^1 \left(\frac{1}{2}x(2x)^4 - x(2x)\right) \mathrm{d}x$$
$$= \int_0^1 (8x^5 - 2x^2)\,\mathrm{d}x$$
$$= \left[\frac{4x^6}{3} - \frac{2x^3}{3}\right]_0^1$$
$$= \frac{4}{3} - \frac{2}{3} = \frac{2}{3}$$

Example 5.9: Evaluate $\oint_C y^3\,\mathrm{d}x - x^3\,\mathrm{d}y$ where C is the positively oriented circle of radius 2 centered at the origin.

Identify P and Q:

$$P(x,y) = y^3 \qquad Q(x,y) = -x^3$$

Be careful with the minus sign on Q!
Compute the partial derivatives:

$$\frac{\partial Q}{\partial x} = -3x^2 \qquad \frac{\partial P}{\partial y} = 3y^2$$

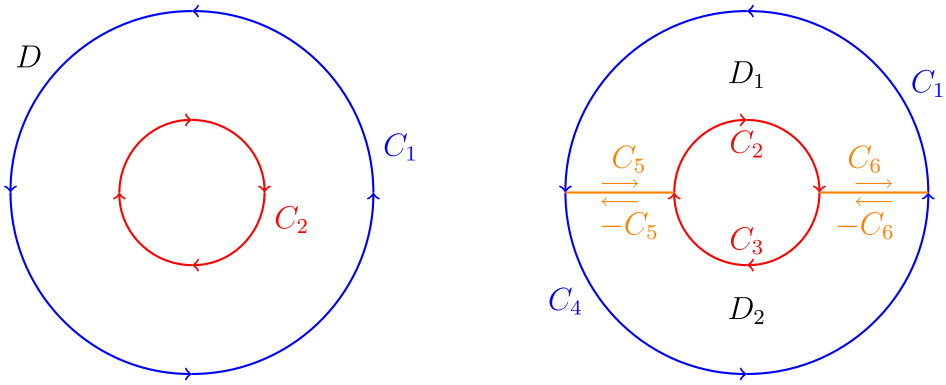
Apply Green's Theorem where D is the disk of radius 2:

$$\oint_C y^3\,\mathrm{d}x - x^3\,\mathrm{d}y = \iint_D (-3x^2 - 3y^2)\,\mathrm{d}A$$
$$= -3\iint_D (x^2 + y^2)\,\mathrm{d}A$$

Since D is a disk, use polar coordinates where $x^2 + y^2 = r^2$:

$$\oint_C y^3\,\mathrm{d}x - x^3\,\mathrm{d}y = -3\int_0^{2\pi}\int_0^2 r^2 \cdot r\,\mathrm{d}r\,\mathrm{d}\theta$$
$$= -3\int_0^{2\pi}\int_0^2 r^3\,\mathrm{d}r\,\mathrm{d}\theta$$
$$= -3\int_0^{2\pi} \left[\frac{r^4}{4}\right]_0^2 \mathrm{d}\theta$$
$$= -3\int_0^{2\pi} 4\,\mathrm{d}\theta = -12\theta\Big|_0^{2\pi} = -24\pi$$

So far, the region D was assumed to have no holes. However, many regions do have holes in them, and Green's Theorem can be extended to regions with holes.
Consider a region D that looks like a ring (annulus) as shown in Figure 5.7.1a. Let C_1 be the outer boundary and C_2 be the inner boundary (the hole). For Green's Theorem to work, we need both boundaries to be positively oriented relative to D.



(a) (b)

Figure 5.7.1: (a) Region D with outer boundary C_1 and inner boundary C_2. (b) Region D cut into two sub-regions D_1 and D_2 along cut lines C_5 and C_6.

Now, since the region D has a hole in it, we can't directly apply Green's Theorem. However, we can cut the disk in half and rename all the various portions of the curve as shown in Figure 5.7.1b. Thus we have two regions D_1, which is bounded by $C_1 \cup C_2 \cup C_5 \cup C_6$ and D_2, which is bounded by $C_1 \cup C_2 \cup (-C_5) \cup (-C_6)$.
Now we can apply Green's Theorem to each of these two regions since they have no holes.

$$\begin{aligned}\iint\limits_D (Q_x - P_y)\,\mathrm{d}A &= \iint\limits_{D_1} (Q_x - P_y)\,\mathrm{d}A + \iint\limits_{D_2} (Q_x - P_y)\,\mathrm{d}A \\ &= \oint\limits_{C_1\cup C_2\cup C_5\cup C_6} P\,\mathrm{d}x + Q\,\mathrm{d}y + \oint\limits_{C_1\cup C_2\cup(-C_5)\cup(-C_6)} P\,\mathrm{d}x + Q\,\mathrm{d}y\end{aligned}$$

Now we can break up the line integrals. Also note that the integrals over C_5 and $-C_5$ cancel out, as do the integrals over C_6 and $-C_6$:

$$\begin{aligned}\iint\limits_D (Q_x - P_y)\,\mathrm{d}A &= \oint\limits_{C_1} P\,\mathrm{d}x + Q\,\mathrm{d}y + \oint\limits_{C_2} P\,\mathrm{d}x + Q\,\mathrm{d}y + \oint\limits_{C_3} P\,\mathrm{d}x + Q\,\mathrm{d}y + \oint\limits_{C_4} P\,\mathrm{d}x + Q\,\mathrm{d}y \\ &= \oint\limits_{C_1\cup C_2\cup C_3\cup C_4} P\,\mathrm{d}x + Q\,\mathrm{d}y \\ &= \oint\limits_{C} P\,\mathrm{d}x + Q\,\mathrm{d}y\end{aligned}$$

Green's Theorem for Regions with Holes

For a region with multiple holes, if C consists of an outer boundary C_1 and inner boundaries $C_2, C_3, \ldots, C_n$, (where C_1 is counter-clockwise and all other C_i are clockwise) then:

$$\sum_{i=1}^{n} \oint\limits_{C_i} P\,\mathrm{d}x + Q\,\mathrm{d}y = \iint\limits_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right)\mathrm{d}A$$

Example 5.10: Evaluate $\oint\limits_C y^3\,\mathrm{d}x - x^3\,\mathrm{d}y$ where C consists of two circles of radius 2 and radius 1 centered at the origin, both with positive orientation.

This is the same integrand as the previous example, but now the region D is an annulus (ring) between the two circles.
From before, we have:

$$P(x, y) = y^3, \quad Q(x, y) = -x^3, \quad \frac{\partial Q}{\partial x} = -3x^2, \quad \frac{\partial P}{\partial y} = 3y^2$$

Apply Green's Theorem:

$$\oint\limits_C y^3\,\mathrm{d}x - x^3\,\mathrm{d}y = -3 \iint\limits_D (x^2 + y^2)\,\mathrm{d}A$$

In polar coordinates, the region is $1 \le r \le 2$, $0 \le \theta \le 2\pi$:

$$\begin{aligned}
\oint\limits_C y^3\,\mathrm{d}x - x^3\,\mathrm{d}y &= -3 \int_0^{2\pi} \int_1^2 r^3\,\mathrm{d}r\,\mathrm{d}\theta \\
&= -3 \int_0^{2\pi} \left[\frac{r^4}{4}\right]_1^2 \mathrm{d}\theta \\
&= -3 \int_0^{2\pi} \left(\frac{16}{4} - \frac{1}{4}\right) \mathrm{d}\theta \\
&= -3 \int_0^{2\pi} \frac{15}{4}\,\mathrm{d}\theta \\
&= -\frac{45}{4} [\theta]_0^{2\pi} \\
&= -\frac{45\pi}{2}
\end{aligned}$$

Area Formula Using Green's Theorem

The area of a region D can be computed using Green's Theorem. Since we want:

$$A = \iint\limits_D \mathrm{d}A$$

we need $\dfrac{\partial Q}{\partial x} - \dfrac{\partial P}{\partial y} = 1$. Several choices of P and Q work:

- $P = 0, Q = x$ gives $A = \oint\limits_C x\,\mathrm{d}y$
- $P = -y, Q = 0$ gives $A = -\oint\limits_C y\,\mathrm{d}x$
- $P = -\frac{y}{2}, Q = \frac{x}{2}$ gives $A = \dfrac{1}{2} \oint\limits_C x\,\mathrm{d}y - y\,\mathrm{d}x$

where C is the boundary of D with positive orientation.

Example 5.11: Use Green's Theorem to find the area of a disk of radius a.

We'll use the formula:

$$A = \frac{1}{2} \oint_C x\,\mathrm{d}y - y\,\mathrm{d}x$$

where C is the circle of radius a centered at the origin.
Parameterize C as:

$$x = a\cos t, \quad y = a\sin t, \quad 0 \le t \le 2\pi$$

Then:

$$\mathrm{d}x = -a\sin t\,\mathrm{d}t\,, \quad \mathrm{d}y = a\cos t\,\mathrm{d}t$$

Compute the area:

$$\begin{aligned}
A &= \frac{1}{2} \oint_C x\,\mathrm{d}y - y\,\mathrm{d}x \\
&= \frac{1}{2} \int_0^{2\pi} \big[(a\cos t)(a\cos t) - (a\sin t)(-a\sin t)\big]\,\mathrm{d}t \\
&= \frac{1}{2} \int_0^{2\pi} \big(a^2\cos^2 t + a^2\sin^2 t\big)\,\mathrm{d}t \\
&= \frac{1}{2} \int_0^{2\pi} a^2\,\mathrm{d}t = \frac{a^2}{2} \cdot t\Big|_0^{2\pi} \\
&= \frac{a^2}{2}(2\pi) = \pi a^2
\end{aligned}$$

This confirms the familiar formula for the area of a circle.

6 Surface Integrals

6.1 Curl and Divergence

Definition 6.1.1: Curl

Given a vector field $\vec{F} = P\hat{\imath} + Q\hat{\jmath} + R\hat{k}$ the curl of $\vec{F}$ is defined to be,

$$\text{curl } \vec{F} = \nabla \times \vec{F} = \begin{vmatrix} \hat{\imath} & \hat{\jmath} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ P & Q & R \end{vmatrix}$$

where ∇ is the **del** operator and is defined as:

$$\nabla = \frac{\partial}{\partial x}\hat{\imath} + \frac{\partial}{\partial y}\hat{\jmath} + \frac{\partial}{\partial z}\hat{k}$$

Physical Interpretation of Curl

If we think of $\vec{F}$ as the velocity field of a flowing fluid then curl $\vec{F}$ represents the tendency of the fluid to rotate about the point (x, y, z). If curl $\vec{F} = \vec{0}$ then there is no tendency to rotate and the fluid is said to be **irrotational**.

Note:-

1. If $f(x, y, z)$ has continuous second order partial derivatives then $\text{curl}(\nabla f) = \vec{0}$.
2. If $\vec{F}$ is a conservative vector field then curl $\vec{F} = \vec{0}$.
3. If $\vec{F}$ is defined on all of $\mathbb{R}^3$ whose components have continuous first order partial derivative and curl $\vec{F} = \vec{0}$ then $\vec{F}$ is a conservative vector field.

Example 6.1: Determine if $\vec{F} = x^2y\hat{\imath} + xyz\hat{\jmath} - x^2y^2\hat{k}$ is a conservative vector field.

We need to do compute the curl and see if we get the zero vector or not.

$$\begin{aligned} \text{curl } \vec{F} &= \begin{vmatrix} \hat{\imath} & \hat{\jmath} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ x^2y & xyz & -x^2y^2 \end{vmatrix} \\ &= -2x^2y\hat{\imath} + yz\hat{k} - \left(-2xy^2\hat{\jmath}\right) - xy\hat{\imath} - x^2\hat{k} \\ &= -\left(2x^2y + xy\right)\hat{\imath} + 2xy^2\hat{\jmath} + \left(yz - x^2\right)\hat{k} \\ &\neq \vec{0} \end{aligned}$$

So, the curl isn't the zero vector and so this vector field is not conservative.

Definition 6.1.2: Divergence

Given a vector field $\vec{F} = P\hat{\imath} + Q\hat{\jmath} + R\hat{k}$ the divergence of $\vec{F}$ is defined to be,

$$\text{div } \vec{F} = \nabla \cdot \vec{F} = \frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} + \frac{\partial R}{\partial z}$$

Physical Interpretation of Divergence

If we think of $\vec{F}$ as the velocity field of a flowing fluid then div $\vec{F}$ represents the net rate of change of the mass of the fluid flowing from the point (x, y, z) per unit volume. This can also be thought of as the tendency of a fluid to diverge from a point. If div $\vec{F} = 0$ then the $\vec{F}$ is called **incompressible**.

Note:-

For any vector field $\vec{F}$ with continuous second order partial derivatives we have the following fact,

$$\text{div}\left(\text{curl } \vec{F}\right) = 0$$

The physical interpretation of this fact is that the tendency of a fluid to rotate about a point does not contribute to the net rate of change of the mass of the fluid flowing from that point per unit volume.

Example 6.2: Verify the above fact for the vector field $\vec{F} = yz^2\hat{\imath} + xy\hat{\jmath} + yz\hat{k}$.

Let's first compute the curl.

$$\begin{aligned}
\text{curl } \vec{F} &= \begin{vmatrix} \hat{\imath} & \hat{\jmath} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ yz^2 & xy & yz \end{vmatrix} \\
&= z\hat{\imath} + 2yz\hat{\jmath} + y\hat{k} - z^2\hat{k} \\
&= z\hat{\imath} + 2yz\hat{\jmath} + \left(y - z^2\right)\hat{k}
\end{aligned}$$

Now compute the divergence of this.

$$\text{div}\left(\text{curl } \vec{F}\right) = \frac{\partial}{\partial x}(z) + \frac{\partial}{\partial y}(2yz) + \frac{\partial}{\partial z}\left(y - z^2\right) = 2z - 2z = 0$$

Definition 6.1.3: Laplace Operator

The Laplace operator, denoted by ∇^2, is defined as the divergence of the gradient of a function. That is, for a scalar function $f(x, y, z)$,

$$\text{div}(\nabla f) = \nabla \cdot \nabla f = \nabla^2 f = \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} + \frac{\partial^2 f}{\partial z^2}$$

Here, ∇^2 is also called the **Laplacian**:

$$\nabla^2 = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}$$

Vector Forms of Green's Theorem

***Theorem 6.1.4* (Vector Form of Green's Theorem):** *Let $\vec{F} = P\hat{\imath} + Q\hat{\jmath}$ be a vector field in the plane with P and Q having continuous partial derivatives on an open region that contains a simply closed curve C and its interior D. Then,*

$$\oint_C \vec{F} \cdot d\vec{r} = \iint_D (\mathit{curl}\ \vec{F}) \cdot \hat{k}\, dA$$

where $\mathit{curl}\ \vec{F} = \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right)\hat{k}$.

For the other form, let the curve C be parameterized by

$$\vec{r}(t) = x(t)\hat{\imath} + y(t)\hat{\jmath}, \quad a \le t \le b$$

then the outward unit normal vector to the curve C is given by,

$$\vec{n} = \frac{y'(t)}{\|\vec{r}'(t)\|}\hat{\imath} - \frac{x'(t)}{\|\vec{r}'(t)\|}\hat{\jmath}$$

Figure 6.1.1 illustrates this concept.

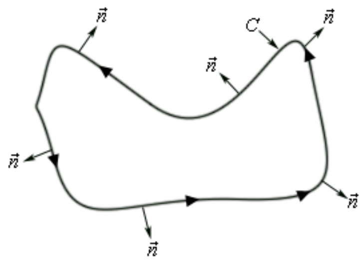


Figure 6.1.1: Outward Unit Normal Vector to Curve C

***Theorem 6.1.5* (Another Vector Form of Green's Theorem):** *Let $\vec{F} = P\hat{\imath} + Q\hat{\jmath}$ be a vector field in the plane with P and Q having continuous partial derivatives on an open region that contains a simply closed curve C and its interior D.*
Then,

$$\oint_C \vec{F} \cdot \vec{n}\, ds = \iint_D (\mathit{div}\ \vec{F})\, dA$$

where $\mathit{div}\ \vec{F} = \dfrac{\partial P}{\partial x} + \dfrac{\partial Q}{\partial y}$ *and $\vec{n}$ is the outward unit normal vector to the curve C.*

6.2 Parametric Surfaces

Parametric Equations

The **parametric representation** of a **parametric surface** is given by the vector function

$$\vec{r}(u, v) = x(u, v)\hat{\imath} + y(u, v)\hat{\jmath} + z(u, v)\hat{k}$$

The parametric equations for a surface are:

$$x = x(u,v) \qquad y = y(u,v) \qquad z = z(u,v)$$

Common Forms

$$\begin{aligned} z = f(x,y) &\implies \vec{r}(x,y) = x\hat{i} + y\hat{j} + f(x,y)\hat{k} \\ x = f(y,z) &\implies \vec{r}(y,z) = f(y,z)\hat{i} + y\hat{j} + z\hat{k} \\ y = f(x,z) &\implies \vec{r}(x,z) = x\hat{i} + f(x,z)\hat{j} + z\hat{k} \end{aligned}$$

Example 6.3: Identify the surface $\vec{r}(u,v) = u\hat{i} + u\cos v\hat{j} + u\sin v\hat{k}$.
Parametric equations:

$$x = u \qquad y = u\cos v \qquad z = u\sin v$$

Squaring and adding:

$$y^2 + z^2 = u^2\cos^2 v + u^2\sin^2 v = u^2 = x^2$$

Surface equation: $x^2 = y^2 + z^2$ (cone opening along x-axis)

Example 6.4: Parameterize the sphere $x^2 + y^2 + z^2 = 30$.
In spherical coordinates: $\rho = \sqrt{30}$
Using $x = \rho\sin\varphi\cos\theta$, $y = \rho\sin\varphi\sin\theta$, $z = \rho\cos\varphi$:

$$\vec{r}(\theta,\varphi) = \sqrt{30}\sin\varphi\cos\theta\hat{i} + \sqrt{30}\sin\varphi\sin\theta\hat{j} + \sqrt{30}\cos\varphi\hat{k}$$

with $0 \le \theta \le 2\pi$ and $0 \le \varphi \le \pi$.

Example 6.5: Parameterize the cylinder $y^2 + z^2 = 25$ (centered on x-axis).
In cylindrical coordinates: $r = 5$
Using $x = x$, $y = r\sin\theta$, $z = r\cos\theta$:

$$\vec{r}(x,\theta) = x\hat{i} + 5\sin\theta\hat{j} + 5\cos\theta\hat{k}$$

with $0 \le \theta \le 2\pi$ (no restriction on x).

Tangent Planes

Tangent Vectors

For the parametric surface $\vec{r}(u,v) = x(u,v)\hat{i} + y(u,v)\hat{j} + z(u,v)\hat{k}$, the tangent vectors are:

$$\begin{aligned} \vec{r}_u &= \frac{\partial x}{\partial u}\hat{i} + \frac{\partial y}{\partial u}\hat{j} + \frac{\partial z}{\partial u}\hat{k} \\ \vec{r}_v &= \frac{\partial x}{\partial v}\hat{i} + \frac{\partial y}{\partial v}\hat{j} + \frac{\partial z}{\partial v}\hat{k} \end{aligned}$$

$\vec{r}_u$ is tangent to curves where v is constant.
$\vec{r}_v$ is tangent to curves where u is constant.

Normal Vector

Provided $\vec{r}_u \times \vec{r}_v \neq \vec{0}$, the cross product $\vec{r}_u \times \vec{r}_v$ is normal to the surface and can be used for the tangent plane equation.

Example 6.6: Find the tangent plane to $\vec{r}(u,v) = u\hat{\imath} + 2v^2\hat{\jmath} + (u^2+v)\hat{k}$ at $(2,2,3)$.
Compute partial derivatives:

$$\vec{r}_u = \hat{\imath} + 2u\hat{k} \qquad \vec{r}_v = 4v\hat{\jmath} + \hat{k}$$

Normal vector:

$$\vec{n} = \vec{r}_u \times \vec{r}_v = \begin{vmatrix} \hat{\imath} & \hat{\jmath} & \hat{k} \\ 1 & 0 & 2u \\ 0 & 4v & 1 \end{vmatrix} = -8uv\hat{\imath} - \hat{\jmath} + 4v\hat{k}$$

Find u, v for point $(2,2,3)$:

$$u = 2, \quad 2v^2 = 2 \implies v = \pm 1, \quad u^2 + v = 3 \implies 4 + v = 3 \implies v = -1$$

At $(u,v) = (2,-1)$:

$$\vec{n} = 16\hat{\imath} - \hat{\jmath} - 4\hat{k}$$

Tangent plane:

$$16(x-2) - (y-2) - 4(z-3) = 0 \implies 16x - y - 4z = 18$$

Surface Area

Surface Area Formula

For parametric surface $\vec{r}(u,v)$ over region D:

$$A = \iint_D \|\vec{r}_u \times \vec{r}_v\| \, dA$$

Example 6.7: Find the surface area of the portion of sphere $x^2 + y^2 + z^2 = 16$ inside cylinder $x^2 + y^2 = 12$ and above the xy-plane.
Sphere: $\vec{r}(\theta,\varphi) = 4\sin\varphi\cos\theta\hat{\imath} + 4\sin\varphi\sin\theta\hat{\jmath} + 4\cos\varphi\hat{k}$
Ranges: $0 \leq \theta \leq 2\pi$
For φ: Intersection of sphere and cylinder gives $12 + z^2 = 16 \implies z = 2$
Using $z = 4\cos\varphi$: $\cos\varphi = \frac{1}{2} \implies \varphi = \frac{\pi}{3}$
So $0 \leq \varphi \leq \frac{\pi}{3}$.
Compute partial derivatives:

$$\begin{aligned} \vec{r}_\theta &= -4\sin\varphi\sin\theta\hat{\imath} + 4\sin\varphi\cos\theta\hat{\jmath} \\ \vec{r}_\varphi &= 4\cos\varphi\cos\theta\hat{\imath} + 4\cos\varphi\sin\theta\hat{\jmath} - 4\sin\varphi\hat{k} \end{aligned}$$

Cross product:

$$\vec{r}_\theta \times \vec{r}_\varphi = -16\sin^2\varphi\cos\theta\hat{\imath} - 16\sin^2\varphi\sin\theta\hat{\jmath} - 16\sin\varphi\cos\varphi\hat{k}$$

Magnitude:

$$\begin{aligned}\|\vec{r}_\theta \times \vec{r}_\varphi\| &= \sqrt{256\sin^4\varphi + 256\sin^2\varphi\cos^2\varphi} \\ &= \sqrt{256\sin^2\varphi(\sin^2\varphi + \cos^2\varphi)} = 16\sin\varphi\end{aligned}$$

Surface area:

$$\begin{aligned}A &= \int_0^{2\pi}\int_0^{\pi/3} 16\sin\varphi\, d\varphi\, d\theta \\ &= \int_0^{2\pi} [-16\cos\varphi]_0^{\pi/3}\, d\theta \\ &= \int_0^{2\pi} (-8+16)\, d\theta = 16\pi\end{aligned}$$

6.3 Surface Integrals

Surface integrals extend line integrals to two dimensions, allowing us to integrate over surfaces in three-dimensional space.

Surface Integral - Explicit Form

For a surface S given by $z = g(x,y)$ over region D in the xy-plane:

$$\iint_S f(x,y,z)\,\mathrm{d}S = \iint_D f\left(x,y,g(x,y)\right)\sqrt{1+\left(\frac{\partial g}{\partial x}\right)^2+\left(\frac{\partial g}{\partial y}\right)^2}\,\mathrm{d}A$$

Similarly, for $x = g(y,z)$ over D in yz-plane:

$$\iint_S f(x,y,z)\,\mathrm{d}S = \iint_D f\left(g(y,z),y,z\right)\sqrt{1+\left(\frac{\partial g}{\partial y}\right)^2+\left(\frac{\partial g}{\partial z}\right)^2}\,\mathrm{d}A$$

For $y = g(x,z)$ over D in xz-plane:

$$\iint_S f(x,y,z)\,\mathrm{d}S = \iint_D f\left(x,g(x,z),z\right)\sqrt{1+\left(\frac{\partial g}{\partial x}\right)^2+\left(\frac{\partial g}{\partial z}\right)^2}\,\mathrm{d}A$$

Surface Integral - Parametric Form

For a surface S parameterized by $\vec{r}(u,v) = x(u,v)\hat{\imath} + y(u,v)\hat{\jmath} + z(u,v)\hat{k}$:

$$\iint_S f(x,y,z)\,\mathrm{d}S = \iint_D f(\vec{r}(u,v))\|\vec{r}_u \times \vec{r}_v\|\,\mathrm{d}A$$

where D is the parameter domain.

Example 6.8: Evaluate $\iint\limits_S 6xy\,\mathrm{d}S$ where S is the portion of the plane $x+y+z=1$ in the first octant, in front of the yz-plane.

Write the surface as $x = g(y,z) = 1-y-z$.
Region D in yz-plane: Setting $x=0$ gives $y+z=1$, so $0 \le y \le 1$ and $0 \le z \le 1-y$.
Compute:

$$\frac{\partial g}{\partial y} = -1 \qquad \frac{\partial g}{\partial z} = -1$$

Surface integral:

$$\begin{aligned}
\iint\limits_S 6xy\,\mathrm{d}S &= \iint\limits_D 6(1-y-z)y\sqrt{1+1+1}\,\mathrm{d}A \\
&= \sqrt{3}\int_0^1 \int_0^{1-y} 6y(1-y-z)\,\mathrm{d}z\,\mathrm{d}y \\
&= 6\sqrt{3}\int_0^1 \int_0^{1-y} (y-y^2-yz)\,\mathrm{d}z\,\mathrm{d}y \\
&= 6\sqrt{3}\int_0^1 \left[yz - y^2z - \frac{1}{2}yz^2\right]_0^{1-y}\mathrm{d}y \\
&= 6\sqrt{3}\int_0^1 \left(y(1-y) - y^2(1-y) - \frac{1}{2}y(1-y)^2\right)\mathrm{d}y \\
&= 6\sqrt{3}\int_0^1 \left(\frac{1}{2}y - y^2 + \frac{1}{2}y^3\right)\mathrm{d}y \\
&= 6\sqrt{3}\left[\frac{1}{4}y^2 - \frac{1}{3}y^3 + \frac{1}{8}y^4\right]_0^1 \\
&= 6\sqrt{3}\left(\frac{1}{4} - \frac{1}{3} + \frac{1}{8}\right) = 6\sqrt{3}\cdot\frac{1}{24} = \frac{\sqrt{3}}{4}
\end{aligned}$$

Example 6.9: Evaluate $\iint\limits_S z\,\mathrm{d}S$ where S is the upper hemisphere of radius 2.

Use spherical parameterization:

$$\vec{r}(\theta,\varphi) = 2\sin\varphi\cos\theta\hat{i} + 2\sin\varphi\sin\theta\hat{j} + 2\cos\varphi\hat{k}$$

Ranges: $0 \le \theta \le 2\pi$, $0 \le \varphi \le \frac{\pi}{2}$ (upper hemisphere).
Compute tangent vectors:

$$\begin{aligned}
\vec{r}_\theta &= -2\sin\varphi\sin\theta\hat{i} + 2\sin\varphi\cos\theta\hat{j} \\
\vec{r}_\varphi &= 2\cos\varphi\cos\theta\hat{i} + 2\cos\varphi\sin\theta\hat{j} - 2\sin\varphi\hat{k}
\end{aligned}$$

Cross product:

$$\vec{r}_\theta \times \vec{r}_\varphi = -4\sin^2\varphi\cos\theta\hat{i} - 4\sin^2\varphi\sin\theta\hat{j} - 4\sin\varphi\cos\varphi\hat{k}$$

Magnitude:

$$\begin{aligned}
\|\vec{r}_\theta \times \vec{r}_\varphi\| &= \sqrt{16\sin^4\varphi + 16\sin^2\varphi\cos^2\varphi} \\
&= 4\sin\varphi\sqrt{\sin^2\varphi + \cos^2\varphi} = 4\sin\varphi
\end{aligned}$$

Surface integral (using $z = 2\cos\varphi$):

$$\iint_S z\,\mathrm{d}S = \int_0^{2\pi}\int_0^{\pi/2} 2\cos\varphi \cdot 4\sin\varphi\,\mathrm{d}\varphi\,\mathrm{d}\theta$$
$$= \int_0^{2\pi}\int_0^{\pi/2} 4\sin(2\varphi)\,\mathrm{d}\varphi\,\mathrm{d}\theta$$
$$= \int_0^{2\pi} \big[-2\cos(2\varphi)\big]_0^{\pi/2}\,\mathrm{d}\theta$$
$$= \int_0^{2\pi} (-2(-1) + 2(1))\,\mathrm{d}\theta = \int_0^{2\pi} 4\,\mathrm{d}\theta = 8\pi$$

Example 6.10: Evaluate $\iint_S (y+z)\,\mathrm{d}S$ where S consists of: the cylinder wall $x^2+y^2 = 3$ with $0 \le z \le 4-y$, the top cap (plane $z = 4-y$), and bottom disk (in xy-plane).

Split into three parts: $S = S_1 \cup S_2 \cup S_3$.

S_1: Cylinder wall

Parameterization: $\vec{r}(z,\theta) = \sqrt{3}\cos\theta\hat{i} + \sqrt{3}\sin\theta\hat{j} + z\hat{k}$

Ranges: $0 \le \theta \le 2\pi$, $0 \le z \le 4 - y = 4 - \sqrt{3}\sin\theta$

From previous example: $\|\vec{r}_z \times \vec{r}_\theta\| = \sqrt{3}$

$$\iint_{S_1} (y+z)\,\mathrm{d}S = \sqrt{3}\int_0^{2\pi}\int_0^{4-\sqrt{3}\sin\theta} (\sqrt{3}\sin\theta + z)\,\mathrm{d}z\,\mathrm{d}\theta$$
$$= \sqrt{3}\int_0^{2\pi}\left[\sqrt{3}\sin\theta\cdot z + \frac{z^2}{2}\right]_0^{4-\sqrt{3}\sin\theta}\mathrm{d}\theta$$
$$= \sqrt{3}\int_0^{2\pi}\left(\sqrt{3}\sin\theta(4-\sqrt{3}\sin\theta) + \frac{(4-\sqrt{3}\sin\theta)^2}{2}\right)\mathrm{d}\theta$$
$$= \sqrt{3}\int_0^{2\pi}\left(8 - \frac{3}{2}\sin^2\theta\right)\mathrm{d}\theta$$
$$= \sqrt{3}\int_0^{2\pi}\left(8 - \frac{3}{4}(1-\cos(2\theta))\right)\mathrm{d}\theta \qquad \left[\sin^2\theta = \frac{1-\cos(2\theta)}{2}\right]$$
$$= \sqrt{3}\left[\frac{29}{4}\theta + \frac{3}{8}\sin(2\theta)\right]_0^{2\pi} = \frac{29\sqrt{3}\pi}{2}$$

S_2: Top cap (plane $z = 4-y$)

Surface: $z = g(x,y) = 4-y$ over disk $D : x^2+y^2 \le 3$

$$\iint_{S_2} (y+z)\,\mathrm{d}S = \iint_D (y+4-y)\sqrt{0^2+(-1)^2+1}\,\mathrm{d}A$$
$$= \sqrt{2}\iint_D 4\,\mathrm{d}A = 4\sqrt{2}\cdot\pi(\sqrt{3})^2 = 12\sqrt{2}\pi$$

S_3: Bottom disk ($z=0$)
Surface: $z=0$ over disk $D: x^2+y^2 \le 3$

$$\iint_{S_3} (y+z)\,\mathrm{d}S = \iint_D y\cdot 1\,\mathrm{d}A = \int_0^{2\pi}\int_0^{\sqrt{3}} r\sin\theta\cdot r\,\mathrm{d}r\,\mathrm{d}\theta$$
$$= \int_0^{2\pi}\left[\frac{r^3}{3}\sin\theta\right]_0^{\sqrt{3}}\mathrm{d}\theta$$
$$= \int_0^{2\pi}\sqrt{3}\sin\theta\,\mathrm{d}\theta = \left[-\sqrt{3}\cos\theta\right]_0^{2\pi} = 0$$

Total:

$$\iint_S (y+z)\,\mathrm{d}S = \frac{29\sqrt{3}\pi}{2} + 12\sqrt{2}\pi + 0 = \frac{\pi}{2}(29\sqrt{3}+24\sqrt{2})$$

6.4 Surface Integrals of Vector Fields

Surface integrals of vector fields extend the concept to integrating a vector field over an oriented surface.

Oriented Surfaces

Definition 6.4.1: Oriented Surface

A surface S is **oriented** if we choose one of two possible unit normal vector fields $\vec{n}$ on S. For surfaces with two sides, at each point we have $\vec{n}_1$ and $\vec{n}_2 = -\vec{n}_1$.

Orientation for Closed Surfaces

For a closed surface S (the boundary of a solid region E), the **positive orientation** is defined by choosing the unit normal vectors that point outward from E. The **negative orientation** points inward toward E.

Finding Unit Normal Vectors

For $z = g(x,y)$ (Explicit Form)

Define $f(x,y,z) = z - g(x,y)$, where $z = g(x,y)$. Hence we get $f(x,y,z) = 0$. Then the unit normal vector is:

$$\vec{n} = \frac{\nabla f}{\|\nabla f\|} = \frac{-g_x\hat{i} - g_y\hat{j} + \hat{k}}{\sqrt{g_x^2+g_y^2+1}}$$

This points generally upward (positive z-component). Use $-\vec{n}$ for downward orientation. Similar formulas hold for $x = g(y,z)$ or $y = g(x,z)$.

For Parametric Surfaces

Given $\vec{r}(u, v) = x(u, v)\hat{i} + y(u, v)\hat{j} + z(u, v)\hat{k}$:

$$\vec{n} = \frac{\vec{r}_u \times \vec{r}_v}{\|\vec{r}_u \times \vec{r}_v\|}$$

Check the direction and use $-\vec{n}$ if needed to match orientation.

Surface Integral of Vector Field

Definition 6.4.2: Flux

The surface integral of vector field $\vec{F}$ over oriented surface S is:

$$\iint_S \vec{F} \cdot \mathrm{d}\vec{S} = \iint_S \vec{F} \cdot \vec{n} \, \mathrm{d}S$$

This is called the **flux** of $\vec{F}$ across S.

Physical Interpretation

If $\vec{v}$ is the velocity field of a fluid, then $\iint\limits_S \vec{v} \cdot \mathrm{d}\vec{S}$ represents the volume of fluid flowing through S per unit time (flux).

This formula can be simplified depending on how the surface is defined.
If the surface is given by $z = g(x, y)$ over region D in the xy-plane, and $\vec{F} = P\hat{i} + Q\hat{j} + R\hat{k}$, and the orientation is upward, then using the unit normal vector from above:

$$\begin{aligned}
\iint_S \vec{F} \cdot \mathrm{d}\vec{S} = \iint_S \vec{F} \cdot \vec{n} \, \mathrm{d}S \\
&= \iint_D (P\hat{i} + Q\hat{j} + R\hat{k}) \cdot \left(\frac{-g_x\hat{i} - g_y\hat{j} + \hat{k}}{\sqrt{g_x^2 + g_y^2 + 1}} \right) \sqrt{g_x^2 + g_y^2 + 1} \, \mathrm{d}A \\
&= \iint_D (P\hat{i} + Q\hat{j} + R\hat{k}) \cdot (-g_x\hat{i} - g_y\hat{j} + \hat{k}) \, \mathrm{d}A \\
&= \iint_D (-Pg_x - Qg_y + R) \, \mathrm{d}A
\end{aligned}$$

Formula for $z = g(x, y)$ (Upward Orientation)

For $\vec{F} = P\hat{i} + Q\hat{j} + R\hat{k}$ and $z = g(x, y)$ over region D:

$$\iint_S \vec{F} \cdot \mathrm{d}\vec{S} = \iint_D (-Pg_x - Qg_y + R) \, \mathrm{d}A$$

For downward orientation, change signs.

If the surface is parameterized by $\vec{r}(u,v)$ over region D in the uv-plane, then:

$$\begin{aligned}
\iint\limits_S \vec{F}\cdot \mathrm{d}\vec{S} &= \iint\limits_S \vec{F}\cdot\vec{n}\,\mathrm{d}S \\
&= \iint\limits_D \vec{F}(\vec{r}(u,v))\cdot\left(\frac{\vec{r}_u\times\vec{r}_v}{\|\vec{r}_u\times\vec{r}_v\|}\right)\|\vec{r}_u\times\vec{r}_v\|\,\mathrm{d}A \\
&= \iint\limits_D \vec{F}(\vec{r}(u,v))\cdot(\vec{r}_u\times\vec{r}_v)\,\mathrm{d}A
\end{aligned}$$

Formula for Parametric Surfaces

For $\vec{r}(u,v)$ over parameter domain D:

$$\iint\limits_S \vec{F}\cdot\mathrm{d}\vec{S} = \iint\limits_D \vec{F}\cdot(\vec{r}_u\times\vec{r}_v)\cdot\mathrm{d}A$$

Adjust sign of cross product to match orientation.

Example 6.11: Evaluate $\iint\limits_S \vec{F}\cdot\mathrm{d}\vec{S}$ where $\vec{F} = y\hat{j} - z\hat{k}$ and S is the paraboloid $y = x^2+z^2$ for $0\le y\le 1$ with the disk $x^2+z^2\le 1$ at $y=1$, positive orientation.

Split: $S = S_1\cup S_2$ where S_1 is paraboloid, S_2 is disk cap.

For positive orientation (pointing out), normal on S_1 points in negative y direction.

S_1: Paraboloid

Define $f(x,y,z) = y - x^2 - z^2$. Then $\nabla f = \langle -2x, 1, -2z\rangle$.

For outward pointing normal: $\vec{n} = \frac{\langle 2x,-1,2z\rangle}{\|\nabla f\|}$

Surface integral over S_1:

$$\begin{aligned}
\iint\limits_{S_1} \vec{F}\cdot\mathrm{d}\vec{S} &= \iint\limits_D (y\hat{j} - z\hat{k})\cdot\langle 2x,-1,2z\rangle\,\mathrm{d}A \\
&= \iint\limits_D (-y-2z^2)\,\mathrm{d}A = \iint\limits_D (-(x^2+z^2)-2z^2)\,\mathrm{d}A \\
&= -\iint\limits_D (x^2+3z^2)\,\mathrm{d}A
\end{aligned}$$

Using polar: $x = r\cos\theta$, $z = r\sin\theta$, $0\le r\le 1$, $0\le\theta\le 2\pi$:

$$\begin{aligned}
&= -\int_0^{2\pi}\int_0^1 (r^2\cos^2\theta + 3r^2\sin^2\theta)r\,\mathrm{d}r\,\mathrm{d}\theta \\
&= -\int_0^{2\pi}\int_0^1 (\cos^2\theta + 3\sin^2\theta)r^3\,\mathrm{d}r\,\mathrm{d}\theta \\
&= -\int_0^{2\pi} (\cos^2\theta + 3\sin^2\theta)\frac{1}{4}\,\mathrm{d}\theta \\
&= -\frac{1}{4}\int_0^{2\pi}\left(\frac{1+\cos(2\theta)}{2} + 3\frac{1-\cos(2\theta)}{2}\right)\mathrm{d}\theta \\
&= -\frac{1}{8}\int_0^{2\pi}(4 - 2\cos(2\theta))\,\mathrm{d}\theta = -\frac{1}{8}[4\theta - \sin(2\theta)]_0^{2\pi} = -\pi
\end{aligned}$$

S_2**: Disk cap at** $y = 1$
Surface: plane $y = 1$. Outward normal: $\vec{n} = \hat{j}$

$$\begin{aligned}
\iint\limits_{S_2} \vec{F} \cdot \mathrm{d}\vec{S} &= \iint\limits_{S_2} (y\hat{j} - z\hat{k}) \cdot \hat{j} \, \mathrm{d}S \\
&= \iint\limits_{S_2} y \, \mathrm{d}S = \iint\limits_{D} 1 \cdot \sqrt{0^2 + 1 + 0^2} \, \mathrm{d}A \\
&= \iint\limits_{D} \mathrm{d}A = \pi(1)^2 = \pi
\end{aligned}$$

Total:

$$\iint\limits_{S} \vec{F} \cdot \mathrm{d}\vec{S} = -\pi + \pi = 0$$

Example 6.12: Evaluate $\iint\limits_{S} \vec{F} \cdot \mathrm{d}\vec{S}$ **where** $\vec{F} = x\hat{i} + y\hat{j} + z^4\hat{k}$ **and** S **is the upper hemisphere** $x^2 + y^2 + z^2 = 9$ **with** $z \geq 0$ **and the disk** $x^2 + y^2 \leq 9$ **at** $z = 0$**, positive orientation.**

Split: $S = S_1 \cup S_2$ where S_1 is hemisphere, S_2 is disk base.
S_1**: Hemisphere**
Parameterize:

$$\vec{r}(\theta, \varphi) = 3 \sin\varphi \cos\theta \hat{i} + 3 \sin\varphi \sin\theta \hat{j} + 3 \cos\varphi \hat{k}$$

with $0 \leq \theta \leq 2\pi$, $0 \leq \varphi \leq \frac{\pi}{2}$.
Tangent vectors:

$$\begin{aligned}
\vec{r}_\theta &= -3 \sin\varphi \sin\theta \hat{i} + 3 \sin\varphi \cos\theta \hat{j} \\
\vec{r}_\varphi &= 3 \cos\varphi \cos\theta \hat{i} + 3 \cos\varphi \sin\theta \hat{j} - 3 \sin\varphi \hat{k}
\end{aligned}$$

Cross product:

$$\vec{r}_\theta \times \vec{r}_\varphi = -9 \sin^2\varphi \cos\theta \hat{i} - 9 \sin^2\varphi \sin\theta \hat{j} - 9 \sin\varphi \cos\varphi \hat{k}$$

For outward pointing (positive z-component), use $-(\vec{r}_\theta \times \vec{r}_\varphi)$.
Evaluate $\vec{F}$ on surface:

$$\vec{F}(\vec{r}(\theta, \varphi)) = 3 \sin\varphi \cos\theta \hat{i} + 3 \sin\varphi \sin\theta \hat{j} + 81 \cos^4\varphi \hat{k}$$

Dot product with $-(\vec{r}_\theta \times \vec{r}_\varphi)$:

$$\begin{aligned}
\vec{F} \cdot (-\vec{r}_\theta \times \vec{r}_\varphi) &= 27 \sin^3\varphi \cos^2\theta + 27 \sin^3\varphi \sin^2\theta + 729 \sin\varphi \cos^5\varphi \\
&= 27 \sin^3\varphi + 729 \sin\varphi \cos^5\varphi
\end{aligned}$$

Surface integral:

$$\begin{aligned}
\iint_{S_1} \vec{F} \cdot \mathrm{d}\vec{S} &= \int_0^{2\pi} \int_0^{\pi/2} (27 \sin^3 \varphi + 729 \sin \varphi \cos^5 \varphi) \, \mathrm{d}\varphi \, \mathrm{d}\theta \\
&= \int_0^{2\pi} \int_0^{\pi/2} 27 \sin \varphi (1 - \cos^2 \varphi) + 729 \sin \varphi \cos^5 \varphi \, \mathrm{d}\varphi \, \mathrm{d}\theta \\
&= \int_0^{2\pi} \left[-27 \cos \varphi + 9 \cos^3 \varphi - \frac{729}{6} \cos^6 \varphi \right]_0^{\pi/2} \mathrm{d}\theta \\
&= \int_0^{2\pi} \frac{279}{2} \, \mathrm{d}\theta = 279\pi
\end{aligned}$$

S_2**: Disk at** $z = 0$
Surface: plane $z = 0$. Outward normal: $\vec{n} = -\hat{k}$

$$\begin{aligned}
\iint_{S_2} \vec{F} \cdot \mathrm{d}\vec{S} &= \iint_{S_2} (x\hat{i} + y\hat{j} + z^4\hat{k}) \cdot (-\hat{k}) \, \mathrm{d}S \\
&= \iint_{S_2} -z^4 \, \mathrm{d}S = \iint_{S_2} 0 \, \mathrm{d}S = 0
\end{aligned}$$

Total:

$$\iint_S \vec{F} \cdot \mathrm{d}\vec{S} = 279\pi + 0 = 279\pi$$

6.5 Stokes' Theorem

Stokes' Theorem relates a line integral around a boundary curve to a surface integral over the enclosed surface.

Boundary Curve and Orientation

For an oriented surface S with boundary curve C:

- The **boundary curve** C is the edge of surface S
- **Positive orientation of** C: Walking along C with head pointing in direction of normal vectors, surface is on the left

***Theorem 6.5.1* (Stokes' Theorem)**: *Let S be an oriented smooth surface bounded by simple, closed, smooth boundary curve C with positive orientation. For vector field $\vec{F}$:*

$$\int_C \vec{F} \cdot d\vec{r} = \iint_S curl\ \vec{F} \cdot d\vec{S}$$

Note:-

The surface S can be *any* surface with boundary curve C. Choose the simplest surface for easier computation.

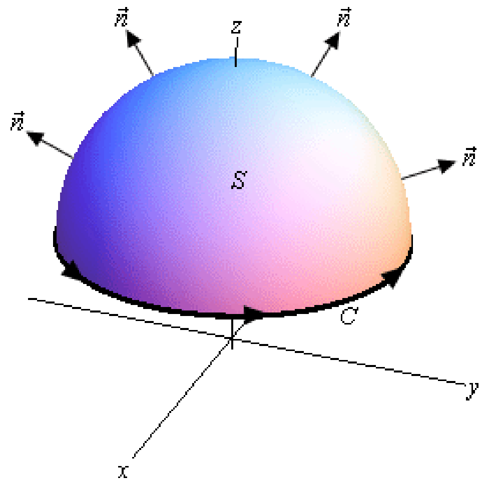


Figure 6.5.1: Orientation of surface and boundary curve

Example 6.13: Evaluate $\iint\limits_S$ curl $\vec{F} \cdot d\vec{S}$ where $\vec{F} = z^2\hat{\imath} - 3xy\hat{\jmath} + x^3y^3\hat{k}$ and S is part of $z = 5 - x^2 - y^2$ above $z = 1$, oriented upward.

The boundary curve is where $z = 1$ intersects the paraboloid:

$$\begin{aligned} 1 &= 5 - x^2 - y^2 \\ x^2 + y^2 &= 4 \text{ at } z = 1 \end{aligned}$$

Parameterize C (circle of radius 2 at $z = 1$):

$$\vec{r}(t) = 2\cos t\,\hat{\imath} + 2\sin t\,\hat{\jmath} + \hat{k}, \quad 0 \le t \le 2\pi$$

By Stokes' Theorem:

$$\iint\limits_S \text{curl } \vec{F} \cdot d\vec{S} = \int_C \vec{F} \cdot d\vec{r}$$

Evaluate $\vec{F}$ on curve:

$$\begin{aligned} \vec{F}(\vec{r}(t)) &= (1)^2\hat{\imath} - 3(2\cos t)(2\sin t)\hat{\jmath} + (2\cos t)^3(2\sin t)^3\hat{k} \\ &= \hat{\imath} - 12\cos t\sin t\,\hat{\jmath} + 64\cos^3 t\sin^3 t\,\hat{k} \end{aligned}$$

Compute derivative and dot product:

$$\begin{aligned} \vec{r}'(t) &= -2\sin t\,\hat{\imath} + 2\cos t\,\hat{\jmath} \\ \vec{F}(\vec{r}(t)) \cdot \vec{r}'(t) &= -2\sin t - 24\sin t\cos^2 t \end{aligned}$$

Evaluate integral:

$$\begin{aligned} \iint\limits_S \text{curl } \vec{F} \cdot d\vec{S} &= \int_0^{2\pi} (-2\sin t - 24\sin t\cos^2 t)\,\mathrm{d}t \\ &= \left[2\cos t + 8\cos^3 t\right]_0^{2\pi} \\ &= 0 \end{aligned}$$

Example 6.14: Evaluate $\int_C \vec{F}\cdot d\vec{r}$ where $\vec{F} = z^2\hat{i} + y^2\hat{j} + x\hat{k}$ and C is the triangle with vertices $(1,0,0)$, $(0,1,0)$, $(0,0,1)$ with counterclockwise orientation.

First compute curl:

$$\text{curl } \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ z^2 & y^2 & x \end{vmatrix} = (2z-1)\hat{j}$$

The triangle lies on plane $x+y+z=1$, so $z = 1-x-y$. Orient plane upward, inducing positive direction on C. For the surface: $f(x,y,z) = z-(1-x-y) = z-1+x+y$

$$\nabla f = \hat{i} + \hat{j} + \hat{k}$$

Region D in xy-plane: $0 \le x \le 1$, $0 \le y \le 1-x$

By Stokes' Theorem:

$$\begin{aligned}
\int_C \vec{F}\cdot d\vec{r} &= \iint\limits_S \text{curl } \vec{F}\cdot d\vec{S} \\
&= \iint\limits_D (2z-1)\hat{j}\cdot(\hat{i}+\hat{j}+\hat{k})\,\mathrm{d}A \\
&= \iint\limits_D (2z-1)\,\mathrm{d}A \\
&= \int_0^1 \int_0^{1-x} [2(1-x-y)-1]\,\mathrm{d}y\,\mathrm{d}x \qquad [z = 1-x-y] \\
&= \int_0^1 \int_0^{1-x} (1-2x-2y)\,\mathrm{d}y\,\mathrm{d}x \\
&= \int_0^1 \left[y - 2xy - y^2\right]_0^{1-x}\,\mathrm{d}x \\
&= \int_0^1 [(1-x) - 2x(1-x) - (1-x)^2]\,\mathrm{d}x \\
&= \int_0^1 (x^2 - x)\,\mathrm{d}x \\
&= \left[\frac{x^3}{3} - \frac{x^2}{2}\right]_0^1 \\
&= \frac{1}{3} - \frac{1}{2} = -\frac{1}{6}
\end{aligned}$$

6.6 Divergence Theorem

***Theorem 6.6.1* (Divergence Theorem):** *Let E be a simple solid region and S be the boundary surface of E with positive orientation (outward normal). For vector field $\vec{F}$ with continuous first order partial derivatives:*

$$\iint\limits_S \vec{F} \cdot \mathrm{d}\vec{S} = \iiint\limits_E \textit{div } \vec{F} \, \mathrm{d}V$$

Example 6.15: Use the divergence theorem to evaluate $\iint\limits_S \vec{F} \cdot \mathrm{d}\vec{S}$ **where** $\vec{F} = xy\hat{\imath} - \frac{1}{2}y^2\hat{\jmath} + z\hat{k}$ **and the surface consists of the three surfaces,** $z = 4 - 3x^2 - 3y^2, 1 \le z \le 4$ **on the top,** $x^2 + y^2 = 1, 0 \le z \le 1$ **on the sides and** $z = 0$ **on the bottom**

First compute divergence:

$$\text{div } \vec{F} = \frac{\partial}{\partial x}(xy) + \frac{\partial}{\partial y}\left(-\frac{1}{2}y^2\right) + \frac{\partial}{\partial z}(z) = y - y + 1 = 1$$

The solid region E is bounded above by the paraboloid $z = 4 - 3x^2 - 3y^2$ and below by the disk $x^2 + y^2 \le 1$ at $z = 0$.
Using cylindrical coordinates: $x = r\cos\theta$, $y = r\sin\theta$, $z = z$, with $0 \le r \le 1$, $0 \le \theta \le 2\pi$, and $0 \le z \le 4 - 3r^2$.
Volume integral:

$$\begin{aligned}
\iiint\limits_E \text{div } \vec{F} \, \mathrm{d}V = \iiint\limits_E 1 \, \mathrm{d}V &= \int_0^{2\pi} \int_0^1 \int_0^{4-3r^2} r \, \mathrm{d}z \, \mathrm{d}r \, \mathrm{d}\theta \\
&= \int_0^{2\pi} \int_0^1 r(4 - 3r^2) \, \mathrm{d}r \, \mathrm{d}\theta \\
&= \int_0^{2\pi} \left[2r^2 - \frac{3}{4}r^4\right]_0^1 \mathrm{d}\theta \\
&= \int_0^{2\pi} \left(2 - \frac{3}{4}\right) \mathrm{d}\theta = \int_0^{2\pi} \frac{5}{4} \, \mathrm{d}\theta = \frac{5\pi}{2}
\end{aligned}$$

Therefore:

$$\iint\limits_S \vec{F} \cdot \mathrm{d}\vec{S} = \frac{5\pi}{2}$$